LECTURE 76

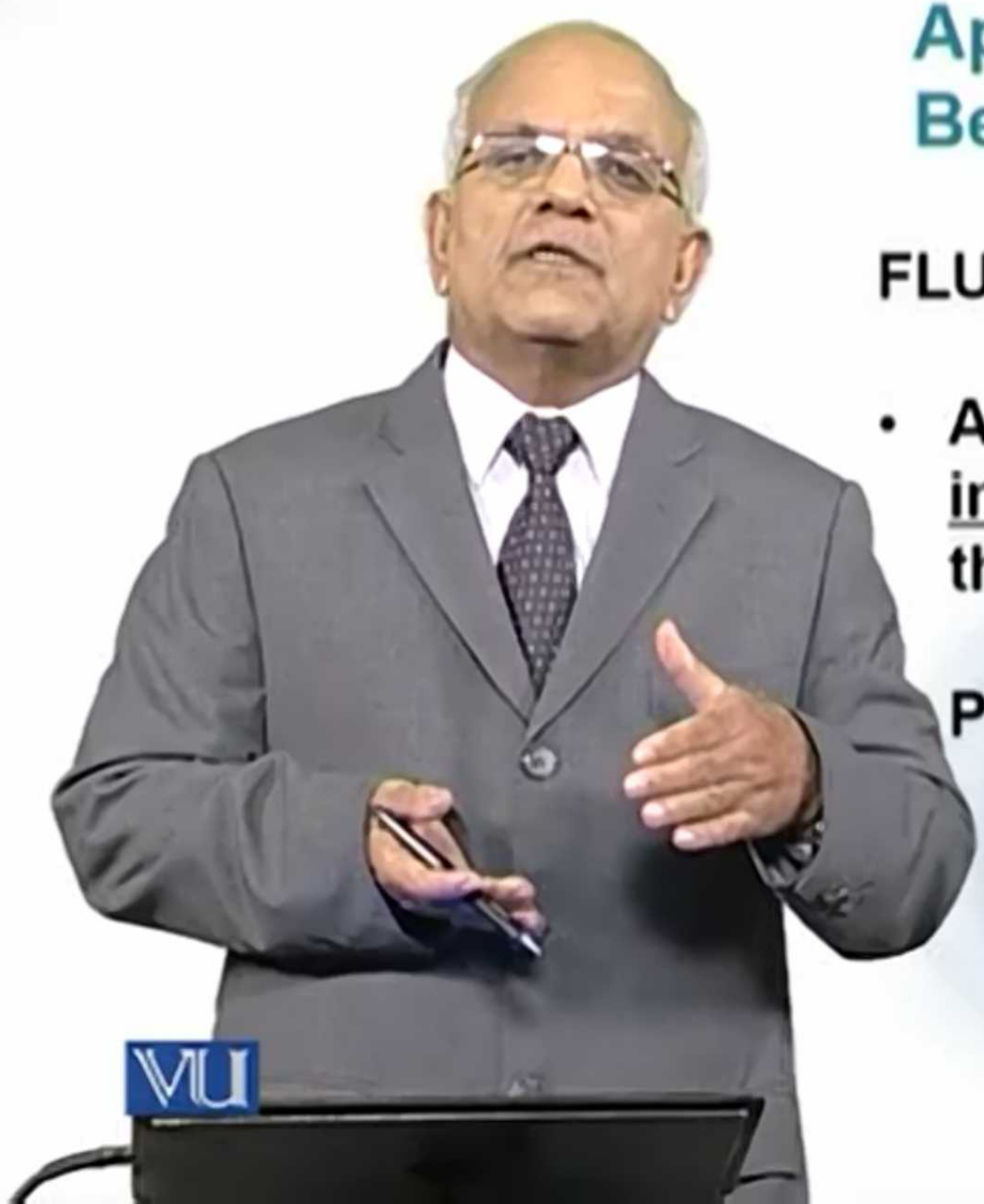
MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

- How can a plane fly?
- How does a perfume spray work?
- What is the venturi effect?
- Why does a cricket ball swing or a baseball curve?

Advanced Fluid Dynamics



Applications of Bernoulli's Principle

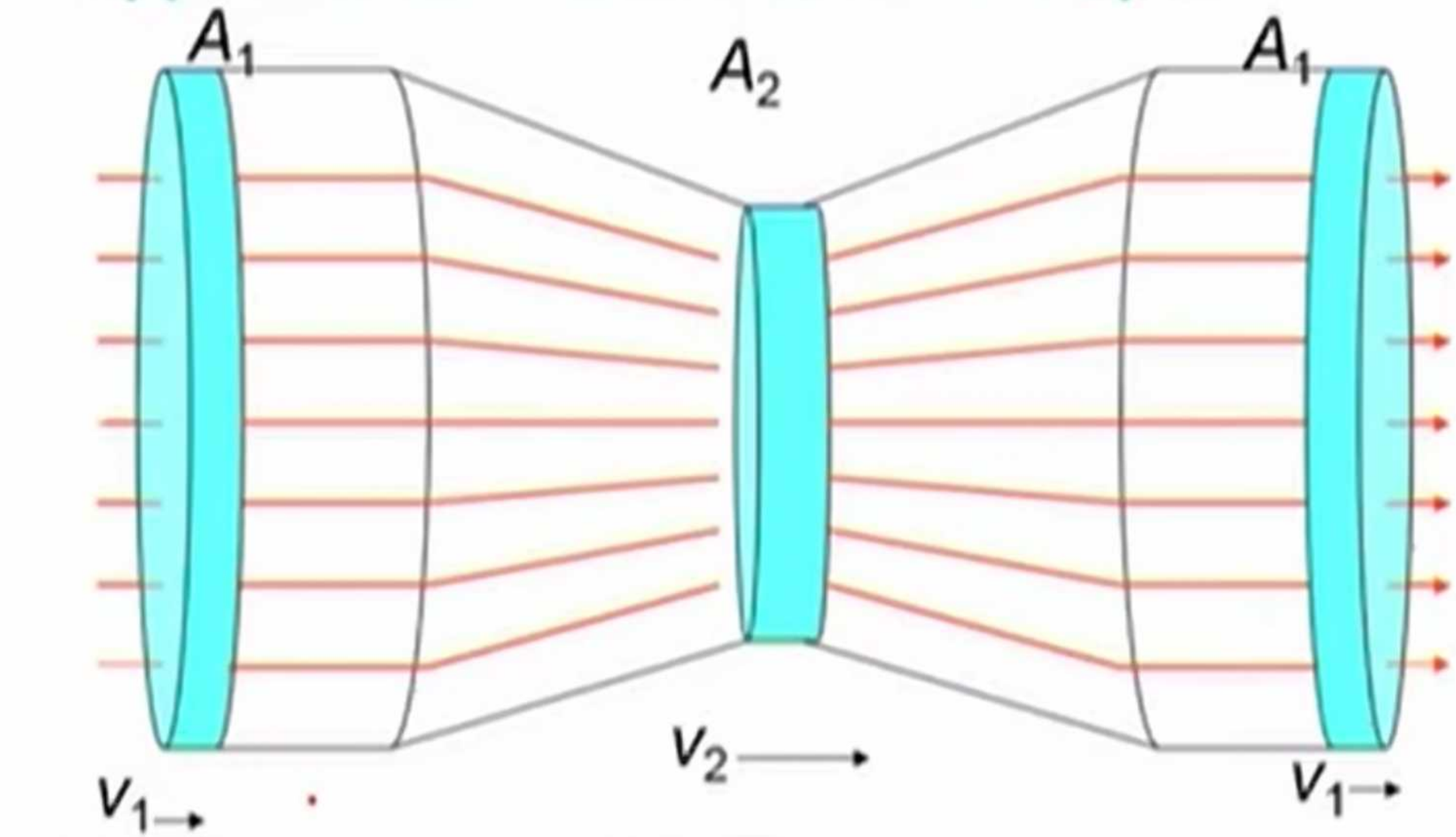
FLUID FLOW : IDEAL FLUID

- **As the speed of a fluid increases, the pressure in the fluid decreases.**

P α 1/v

Advanced Fluid Dynamics

Applications of Bernoulli's Principle



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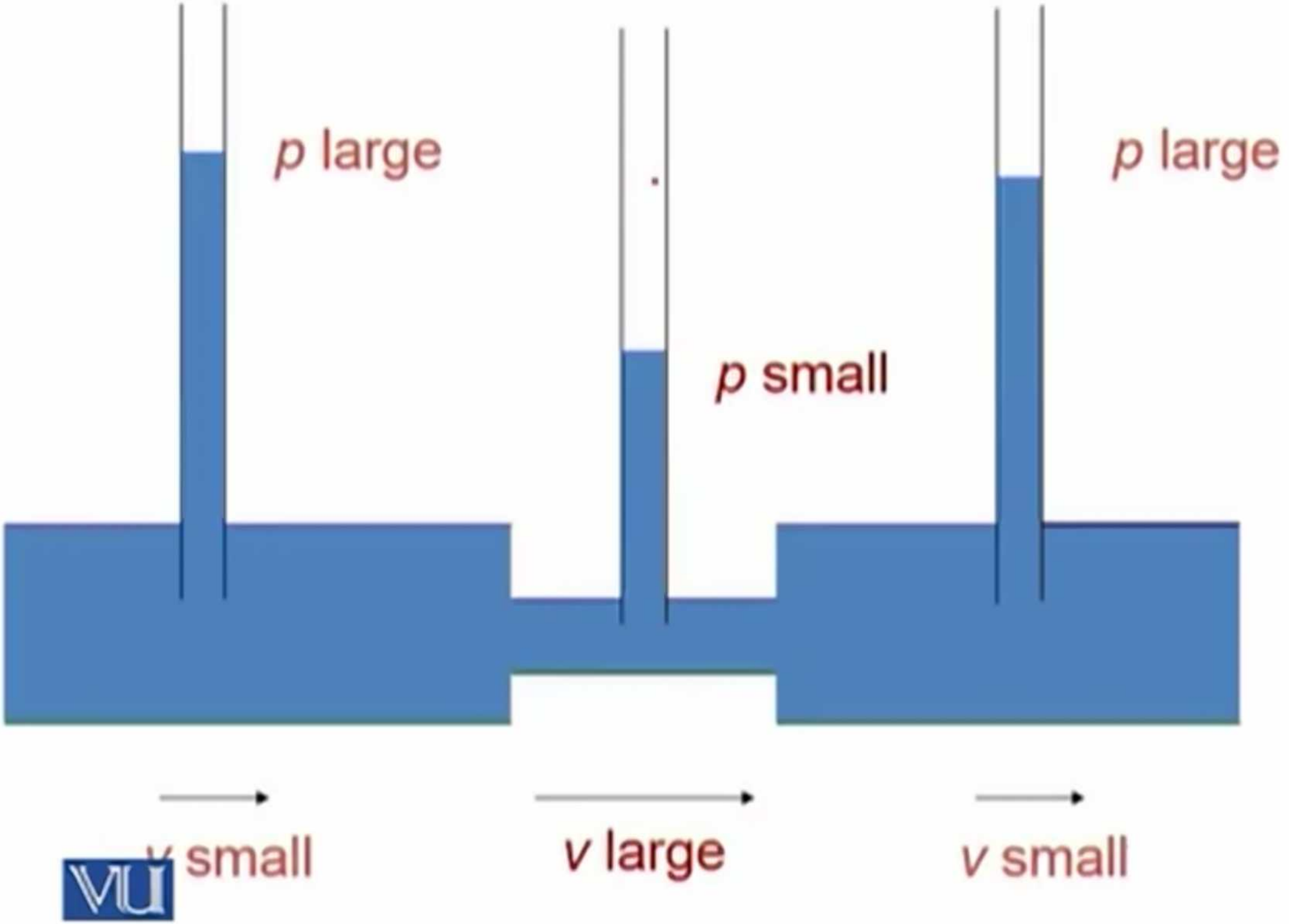
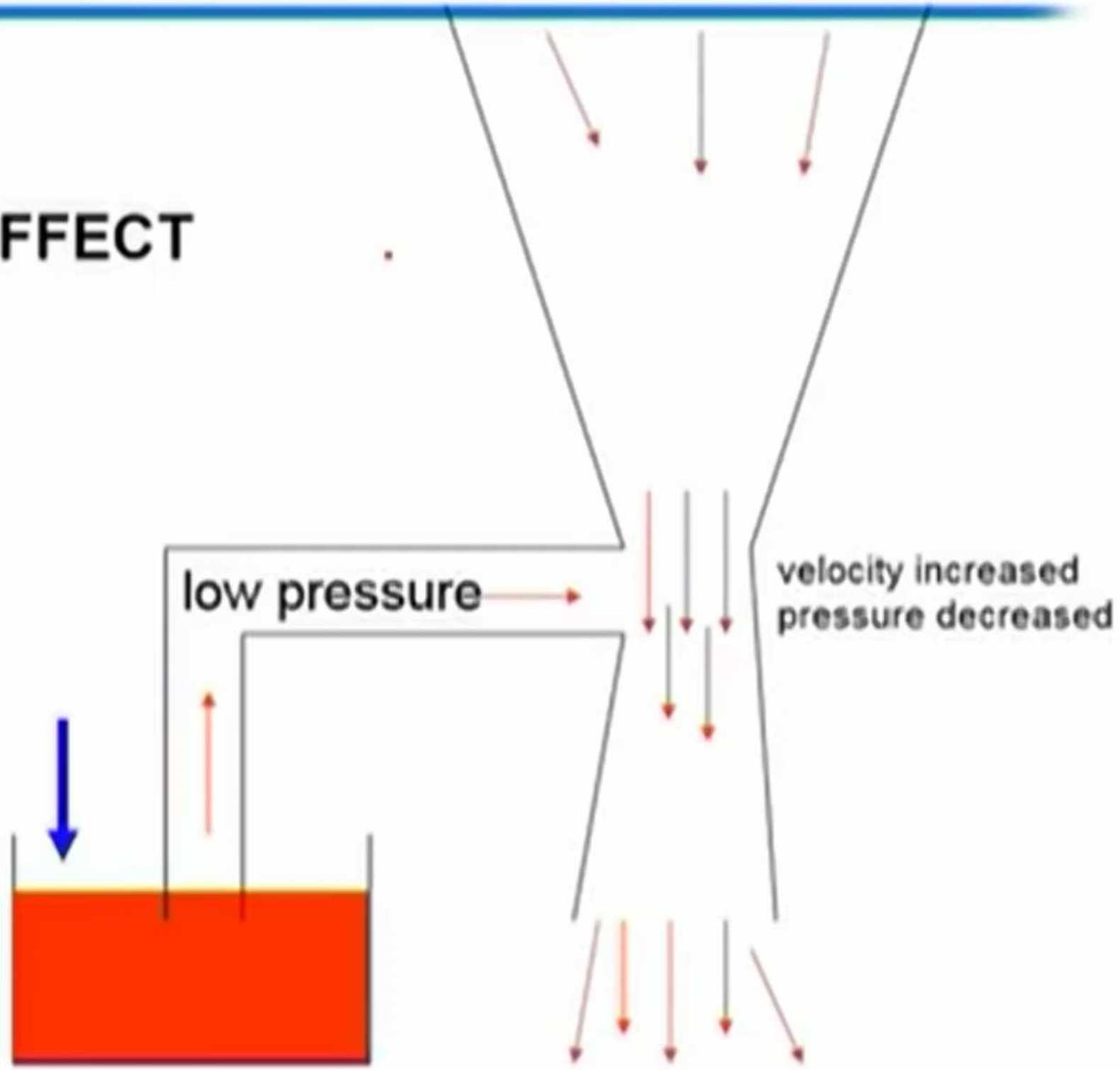


Advanced Fluid Dynamics

VENTURI EFFECT

high pressure (p_{atm})



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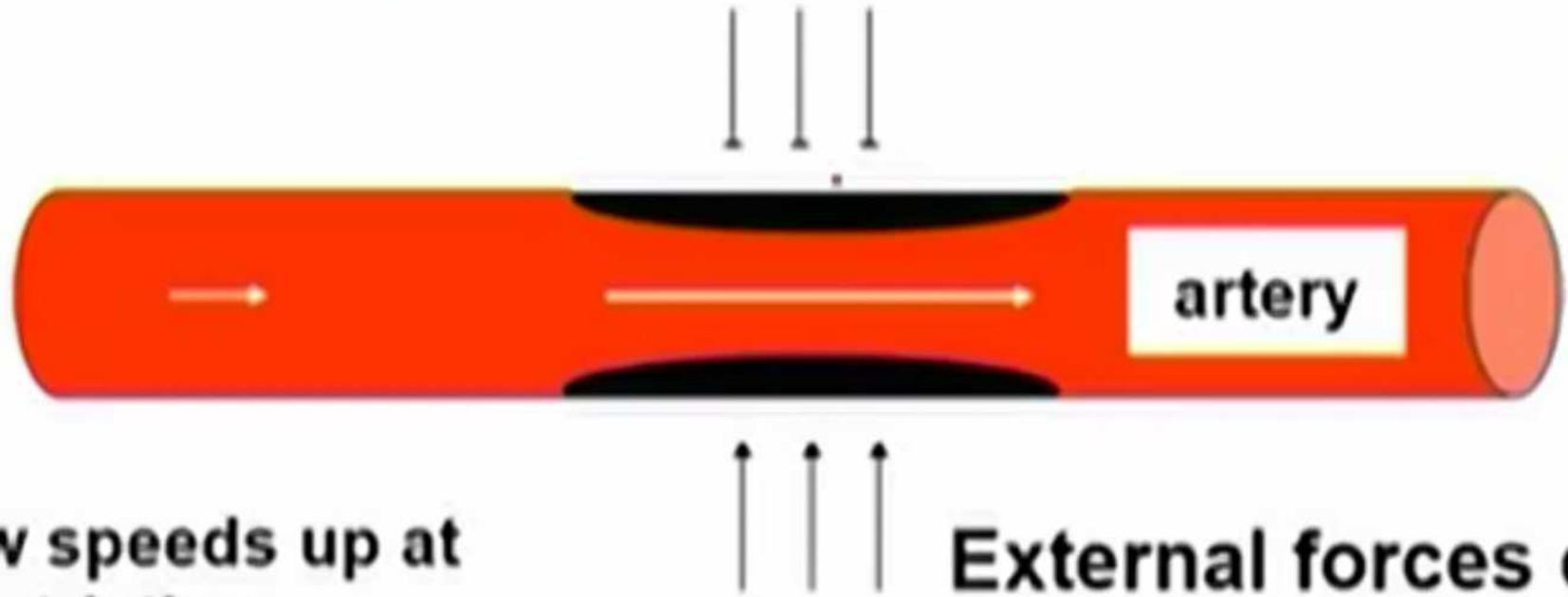
Applications of Bernoulli's Principle



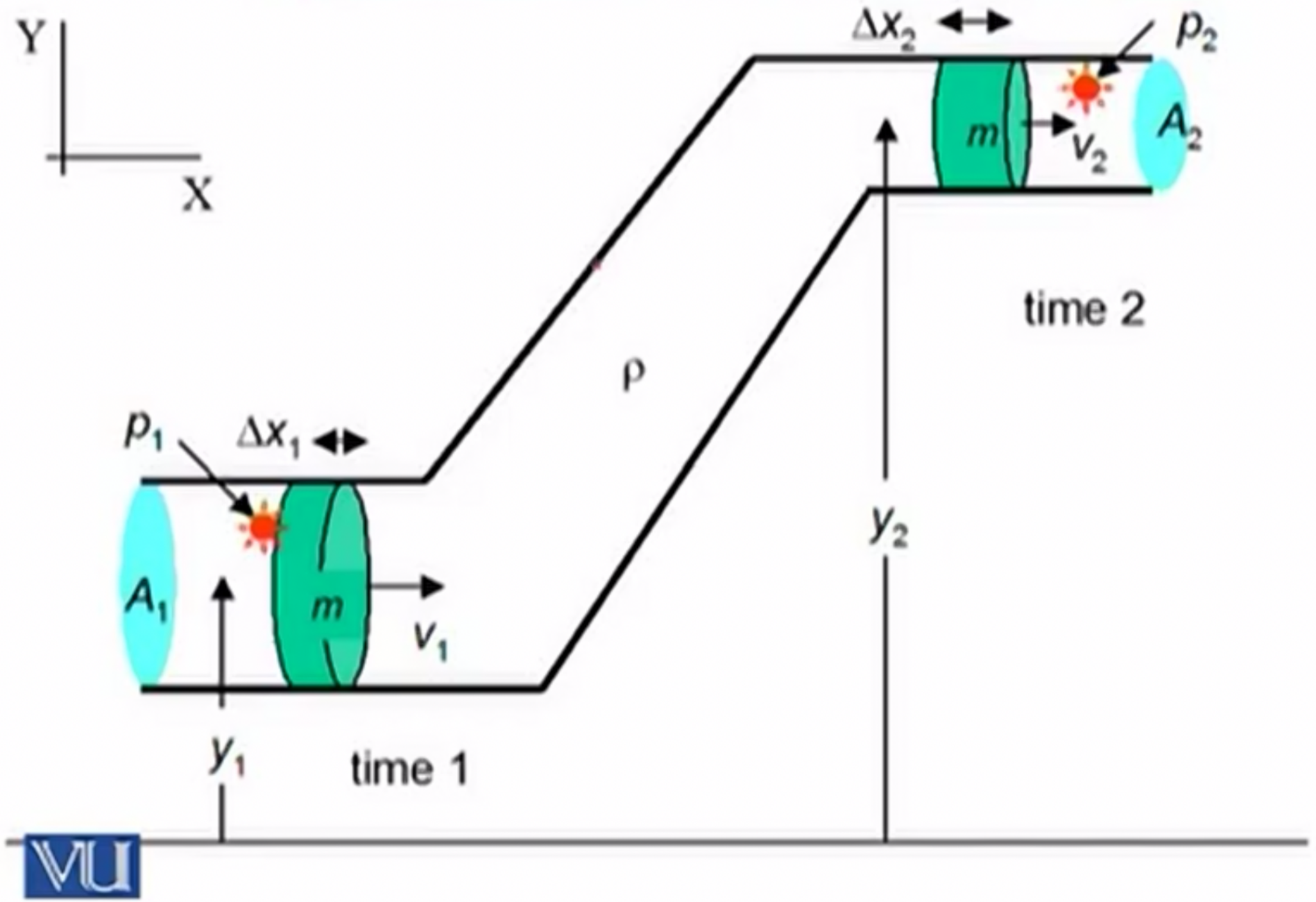
Flow speeds up at constriction
Pressure is lower
Internal force acting on artery wall is reduced

External forces causes artery to collapse

Arteriosclerosis and vascular flutter

Advanced Fluid Dynamics

Applications of Bernoulli's Principle



Advanced Fluid Dynamics

Bernoulli's Equation :for any point along a flow tube or streamline

$p + \frac{1}{2} \rho v^2 + g y = \text{constant}$

Dimensions

ρ [Pa] = [N.m-2] = [N.m.m-3] = [J.m-3]

$\frac{1}{2} \rho v^2$ [kg.m-3.m2.s-2] = [kg.m-1.s-2] = [N.m.m-3] = [J.m-3]

$\rho g h$ [kg.m-3 m.s-2. m] = [kg.m.s-2.m.m-3] = [N.m.m-3] = [J.m-3]

Each term has the dimensions of energy / volume or energy density.

$\frac{1}{2} \rho v^2$ **KE of bulk motion of fluid**

$\rho g h$ **GPE for location of fluid**

p **Pressure energy density arising from internal forces within moving fluid (similar to energy stored in a spring)**

Advanced Fluid Dynamics

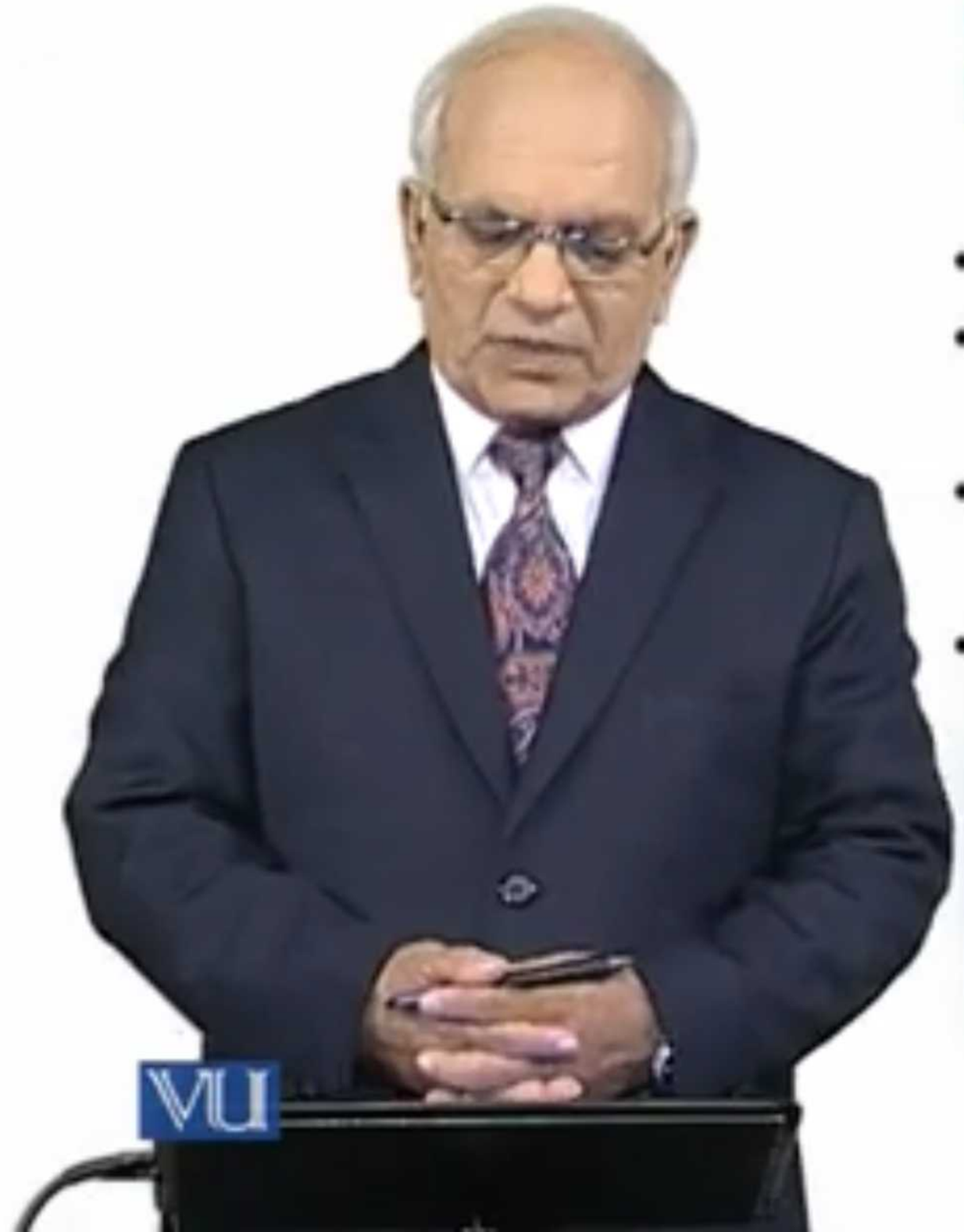
Applications of Bernoulli's Principle

- **Bernoulli's Principle is what allows birds and planes to fly.**
- **The secret behind flight is 'under the wings.'**

LECTURE 77

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics



Applications of Bernoulli's Principle

- How can a plane fly?
- How does a perfume spray work?
- What is the venturi effect?
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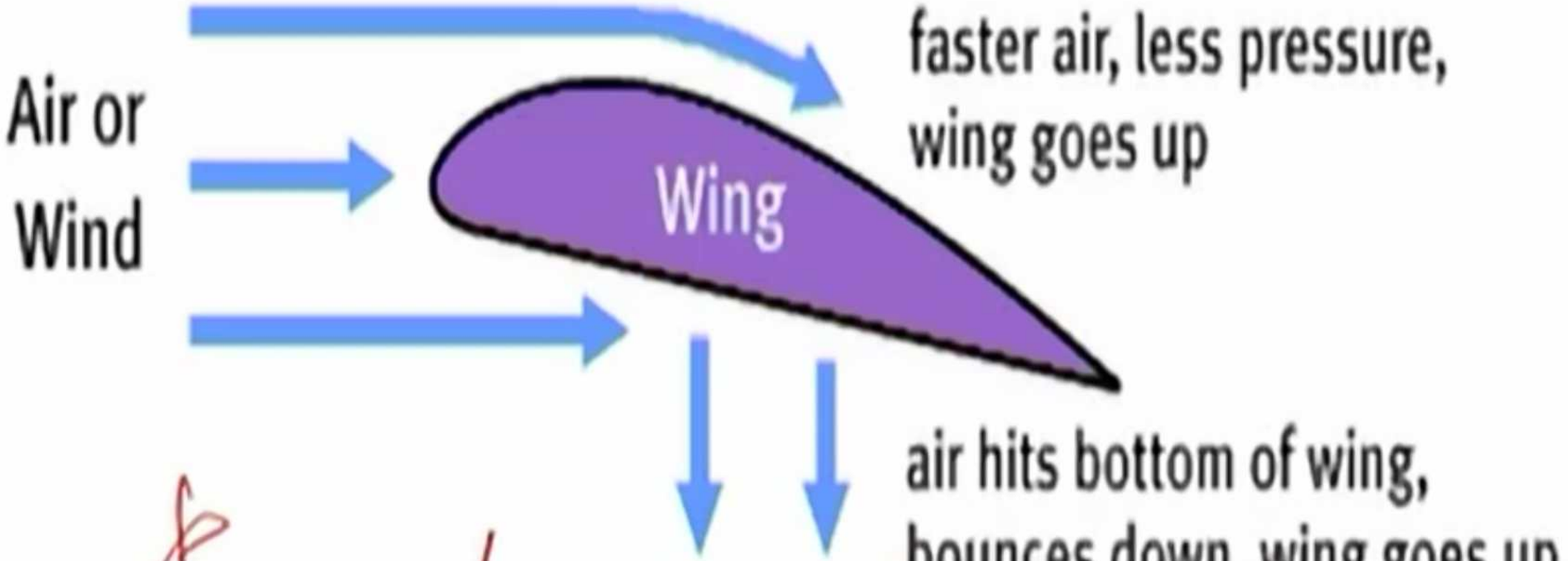
Advanced Fluid Dynamics

Applications of Bernoulli's Principle



$$\frac{p}{\rho} + \frac{1}{2}v^2 + gh = \text{constant}$$

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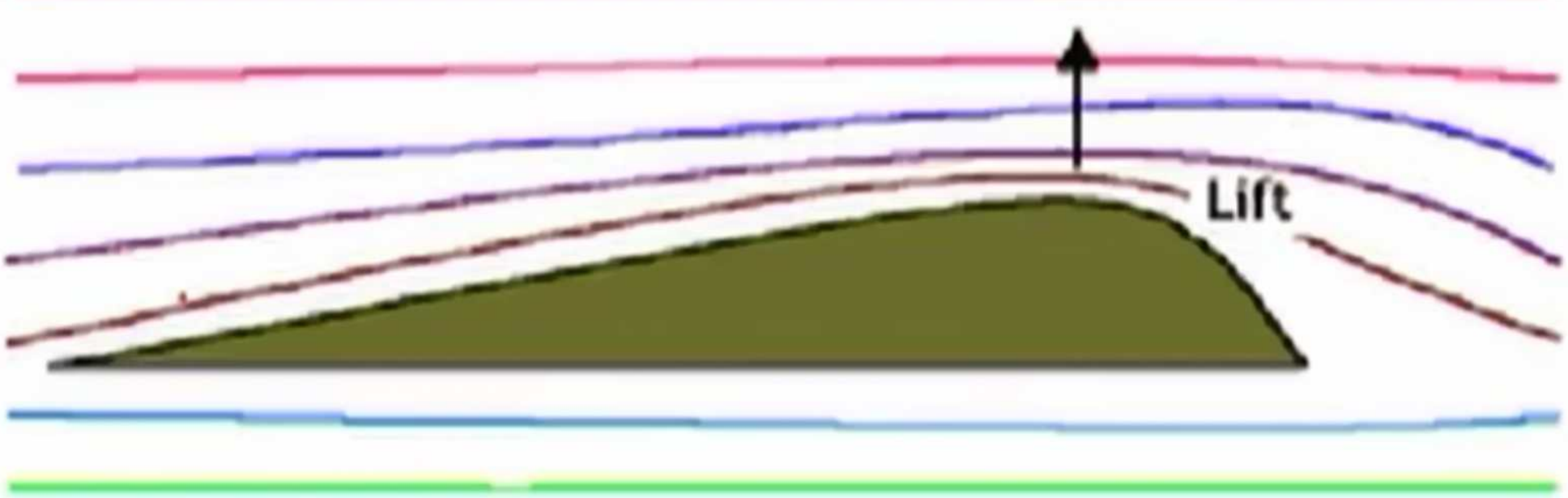
Applications of Bernoulli's Principle

AIRFOIL

On top: greater air speed and less air pressure



On bottom: less air speed and more air pressure

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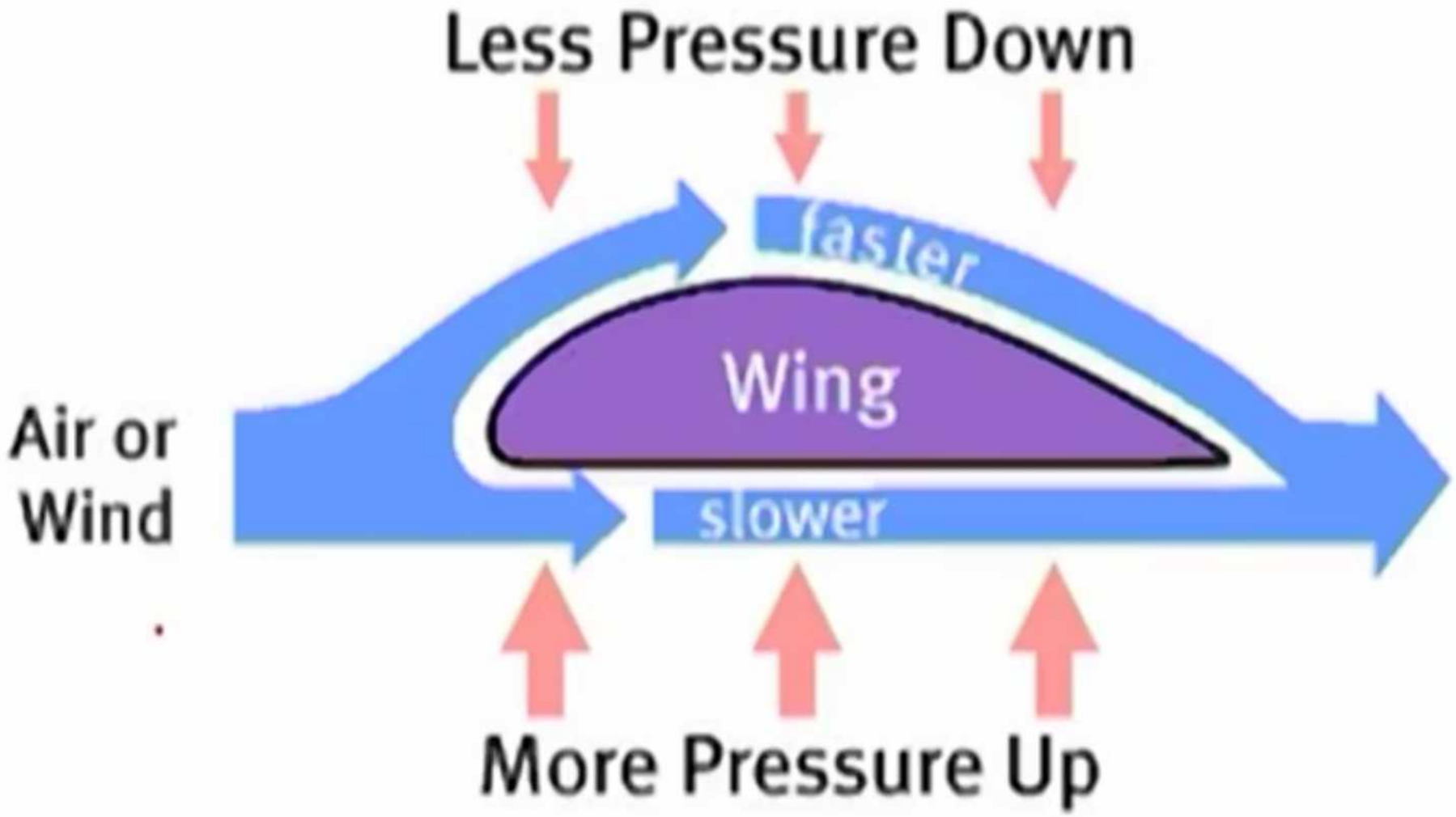
Applications of Bernoulli's Principle



Net force on wing? upwards

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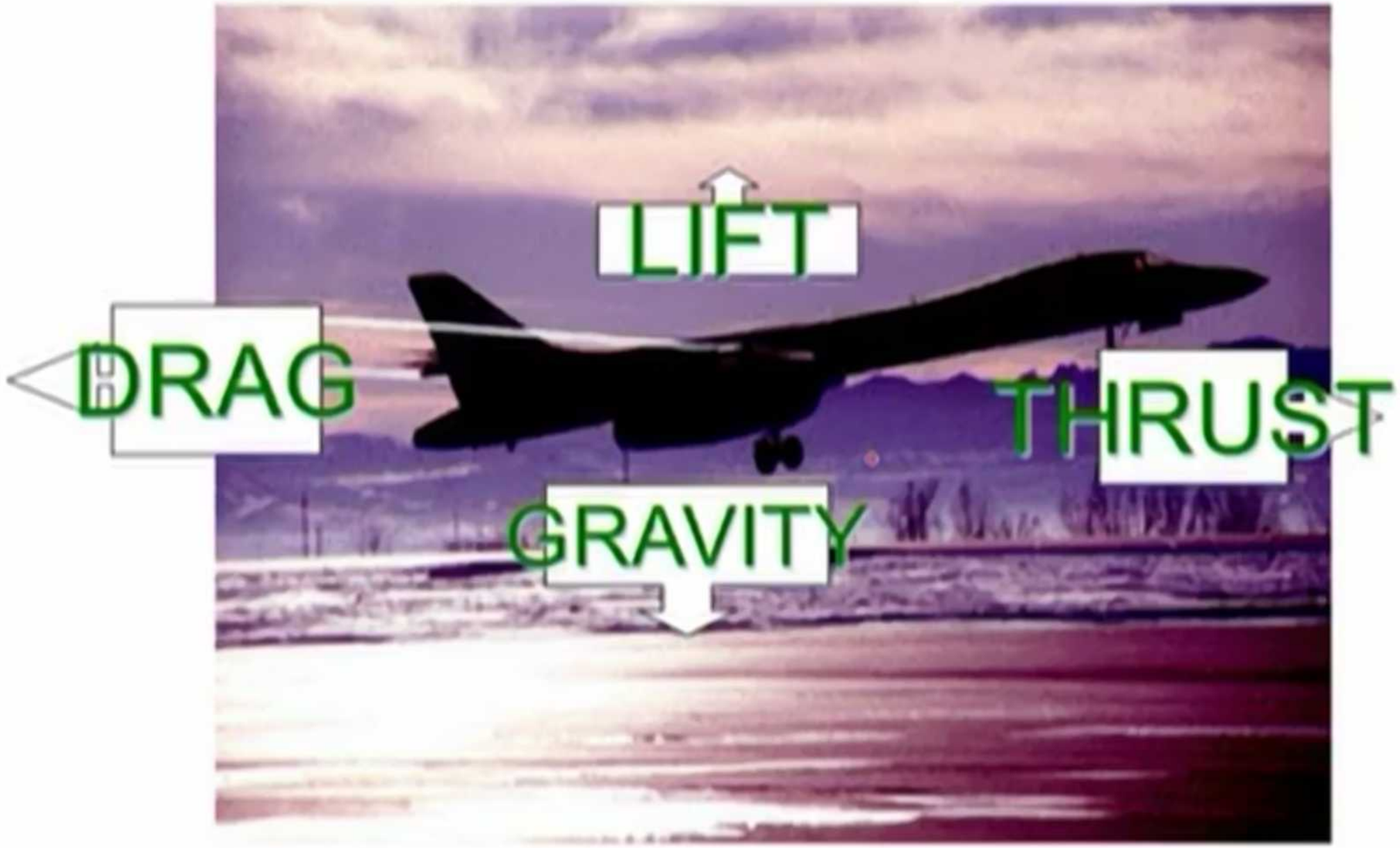
Applications of Bernoulli's Principle



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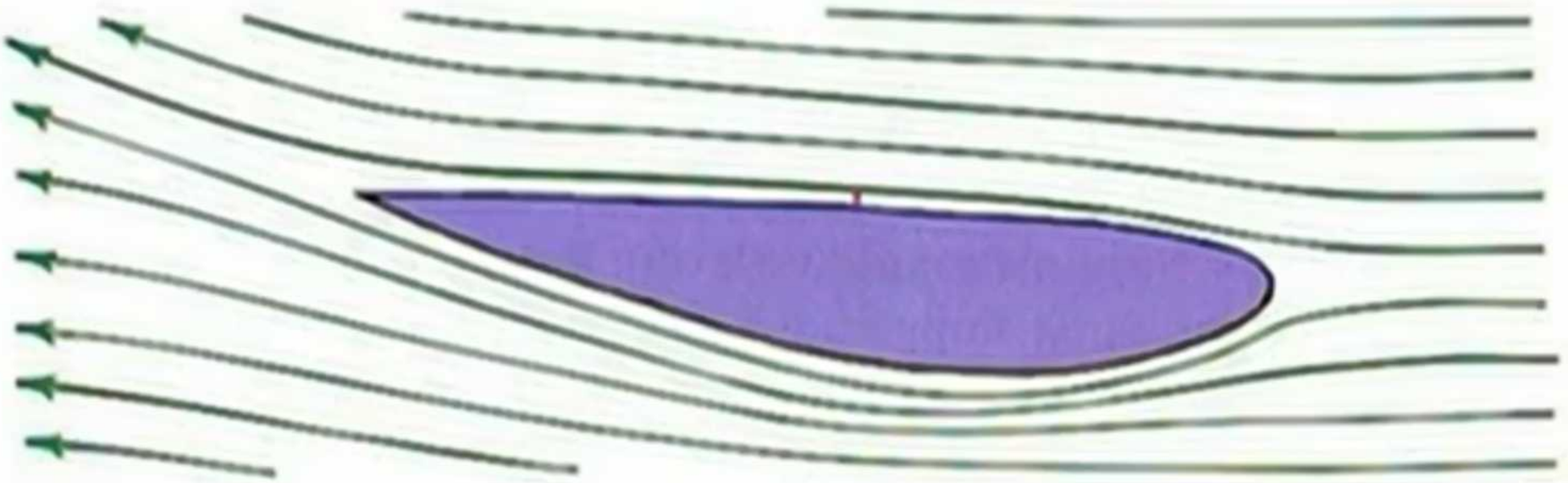
Applications of Bernoulli's Principle

Spoiler – airfoil reversed



LECTURE 78

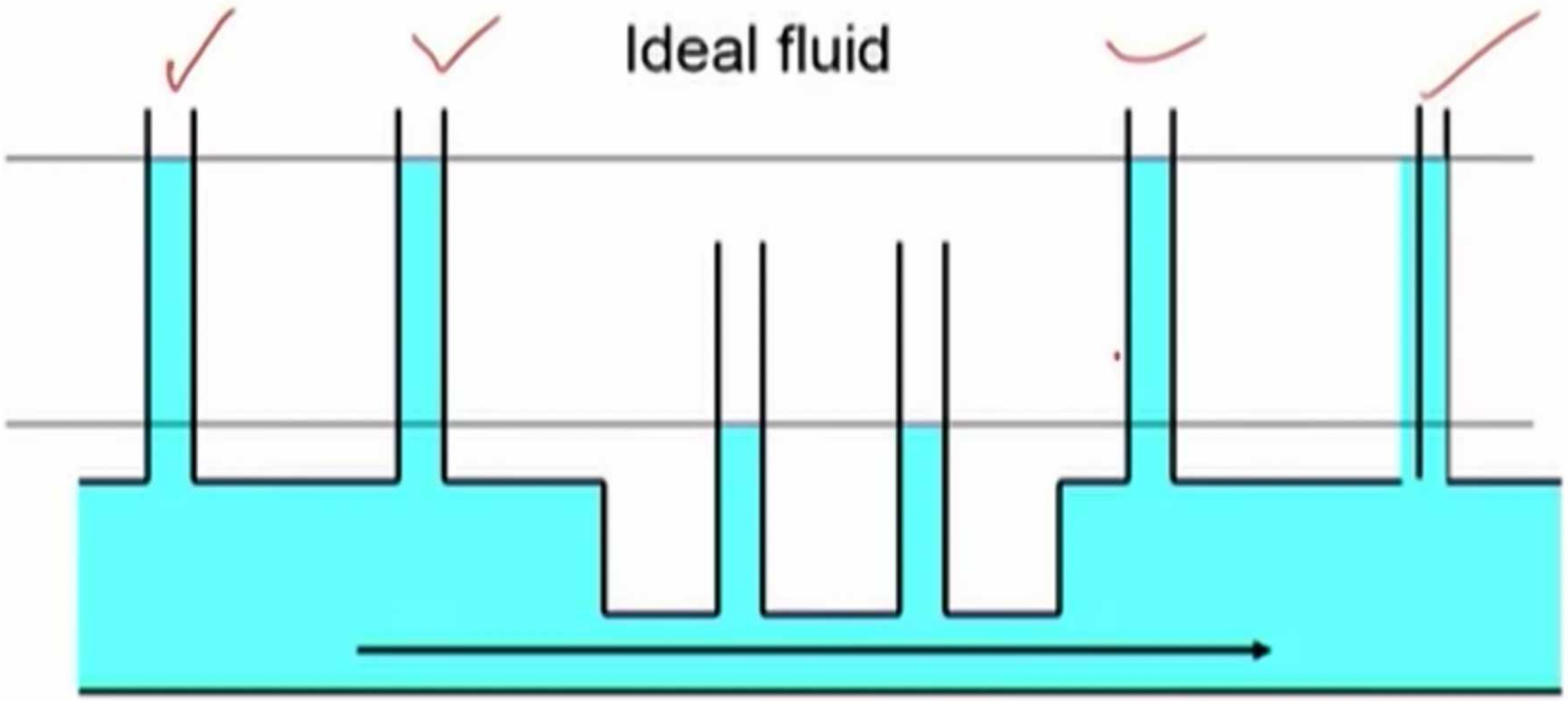
MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Advanced Fluid Dynamics

Applications of Bernoulli's Principle



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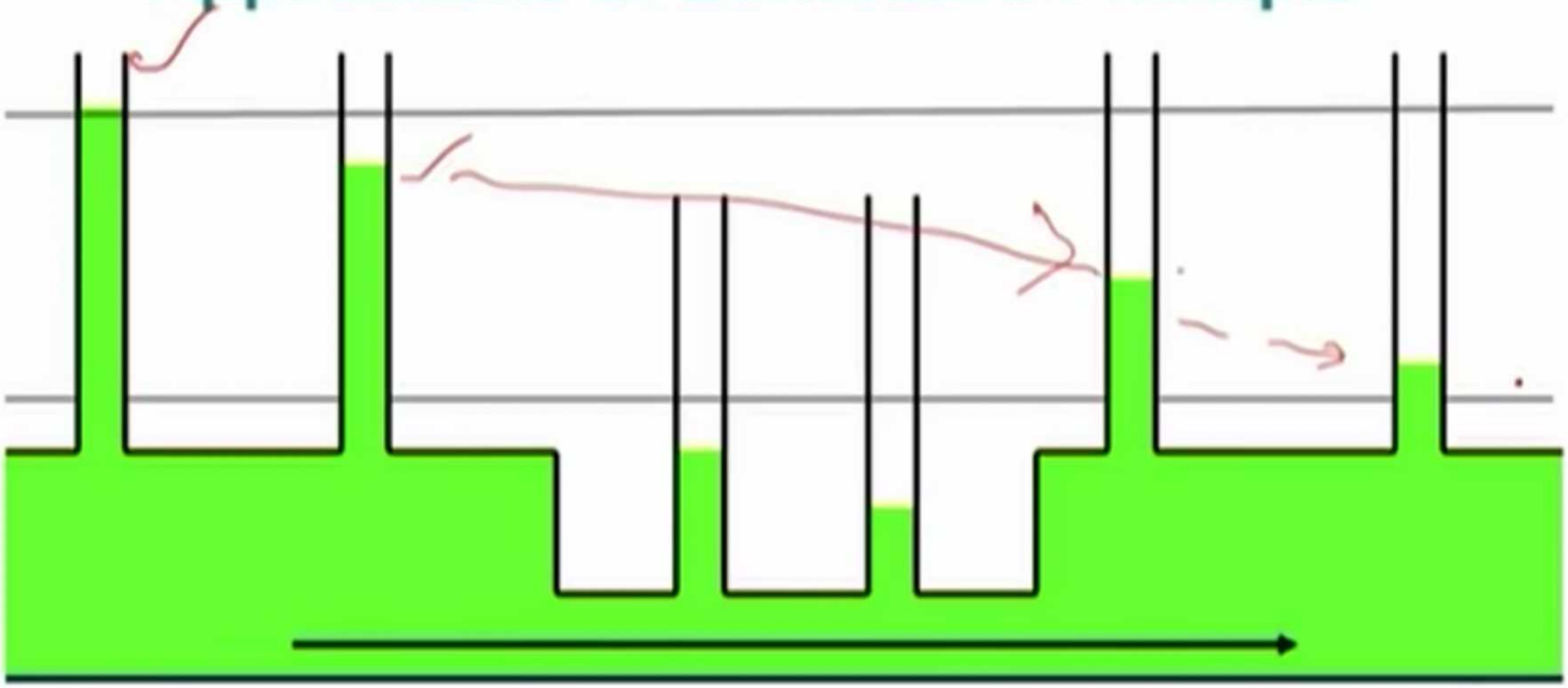
Applications of Bernoulli's Principle



For the viscous fluid there is a gradual decrease in pressure along the pipe.

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Applications of Bernoulli's Principle

- Viscosity ⇒ frictional forces ⇒ energy lost by heating fluid ⇒ increase in internal energy of fluid ⇒ increase in temperature of fluid. Where does the energy come from?

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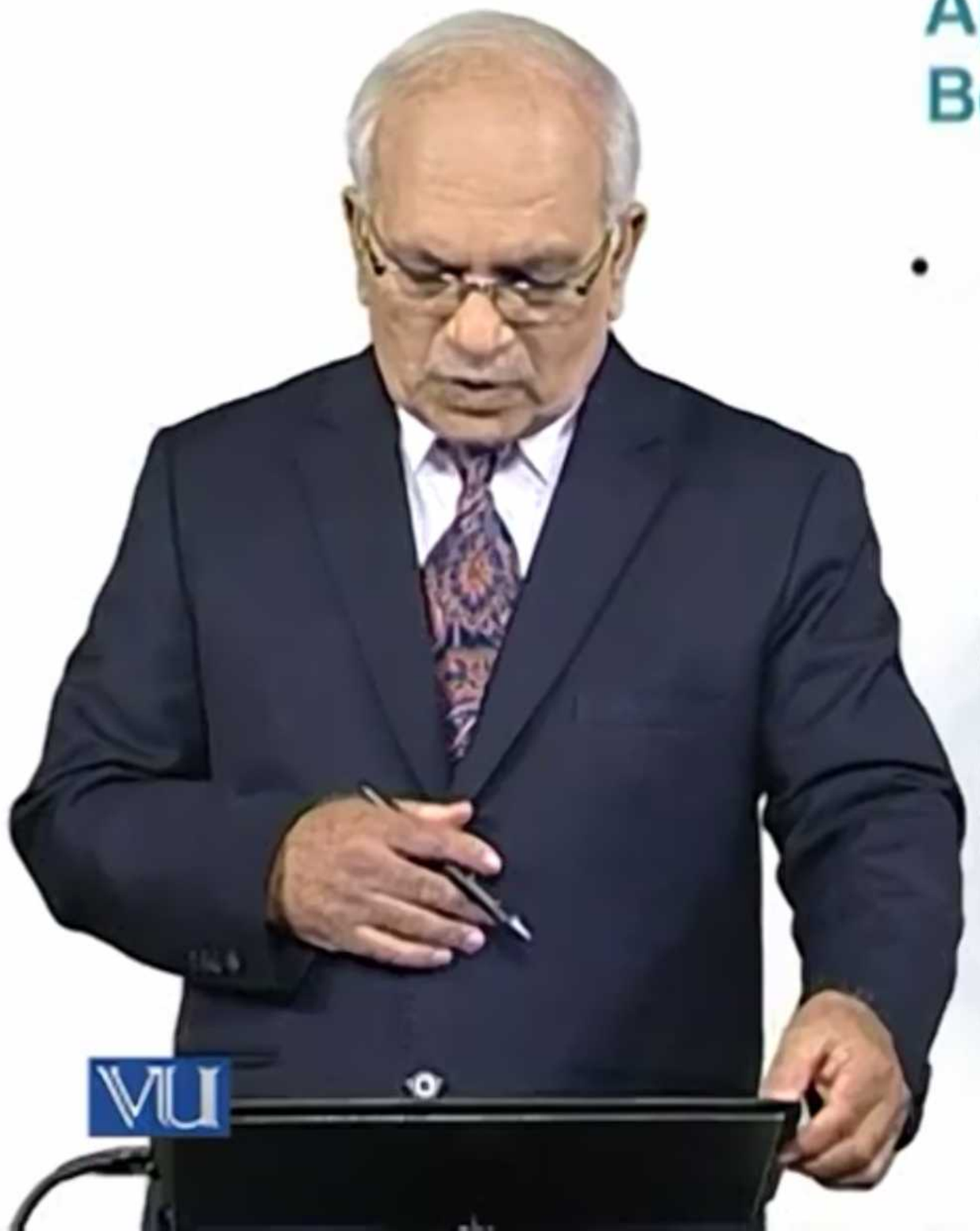
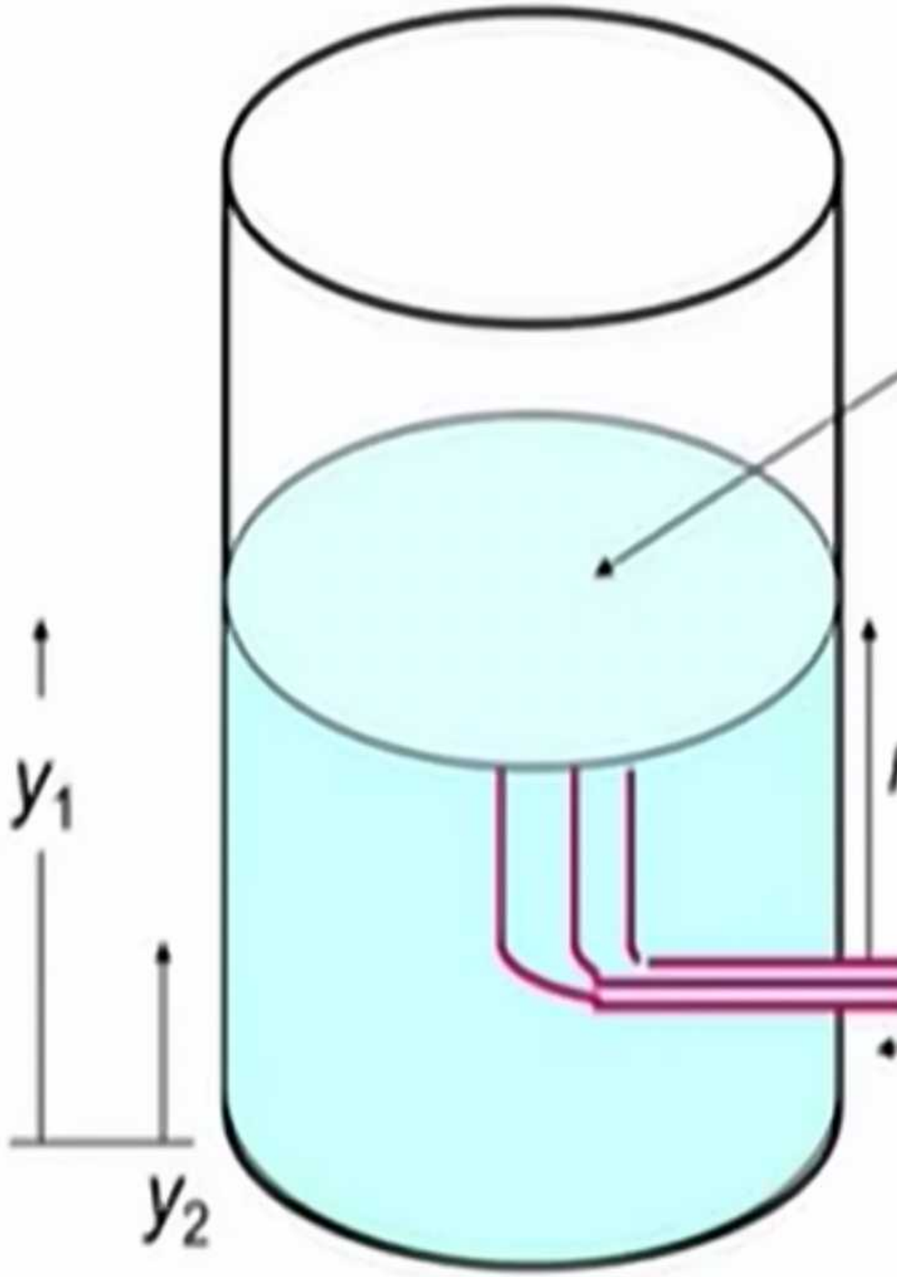


Applications of Bernoulli's Principle

- **Equation of continuity still applies ⇒ energy can not come from KE of fluid ⇒ energy must be provided from the energy of the fluid ⇒ pressure drop along tube.**

Advanced Fluid Dynamics



$$\frac{p_1}{\rho} + \frac{1}{2}v_1^2 + g y_1 = \frac{p_2}{\rho} + \frac{1}{2}v_2^2 + g y_2 .$$

What is the speed with which liquid flows from a hole at the bottom of a tank?

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Assume liquid behaves as an ideal fluid and that Bernoulli's equation can be applied

$$p_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

A small hole is at level (2) and the water level at (1) drops slowly $\Rightarrow v_1 = 0$

$$p_1 = p_{atm} \qquad p_2 = p_{atm}$$

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

$$\rho g y_1 = \tfrac{1}{2} \rho v_2^2 + \rho g y_2$$

$$v_2^2 = 2 g (y_1 - y_2) = 2 g h \qquad h = (y_1 - y_2)$$

$$v_2 = \sqrt{(2 g h)}$$ Torricelli formula (1608 – 1647)

This is the same velocity as a particle falling freely through a height h

LECTURE 79

MTH642 FLUID MECHANICS

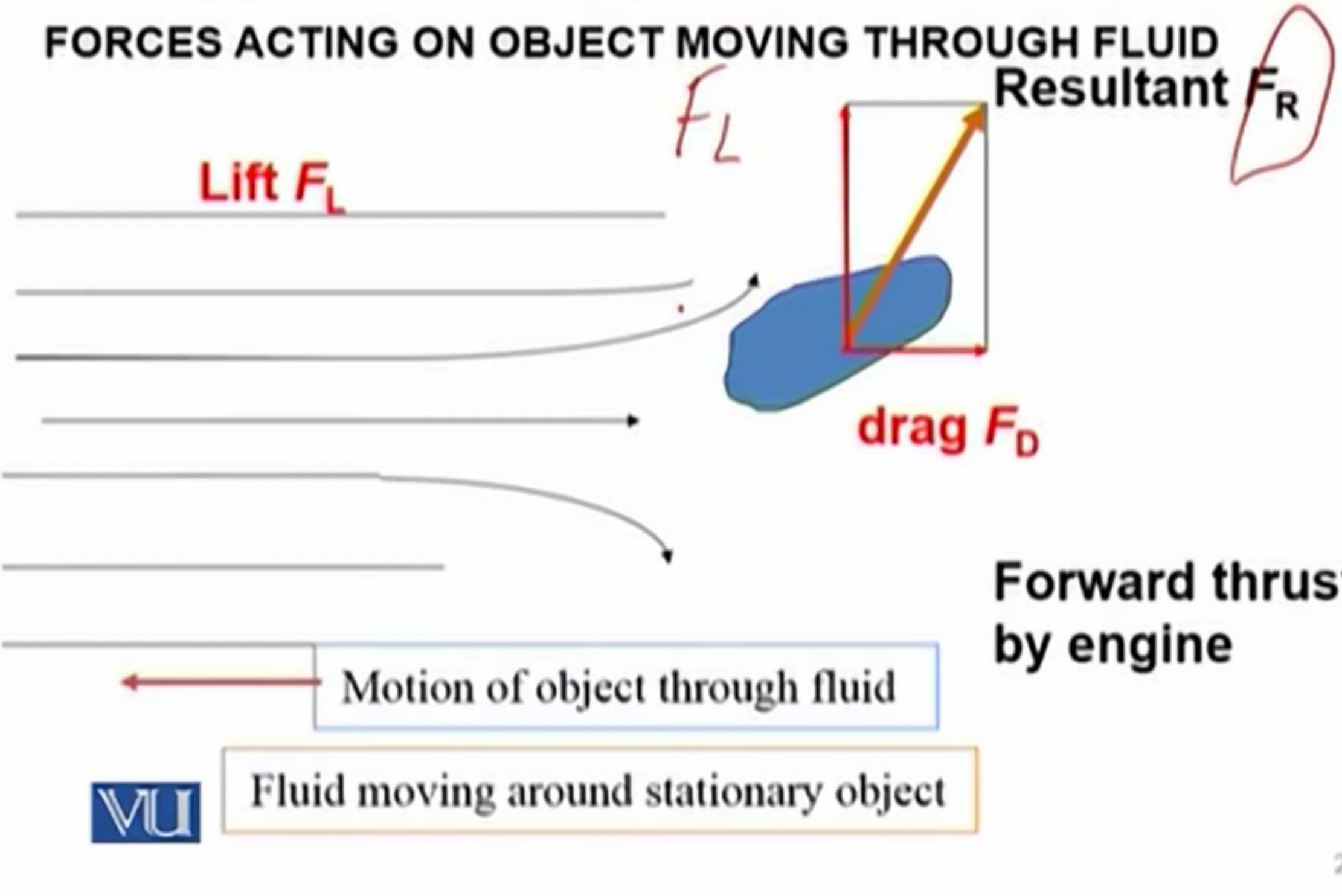
Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

FORCES ACTING ON OBJECT MOVING THROUGH FLUID



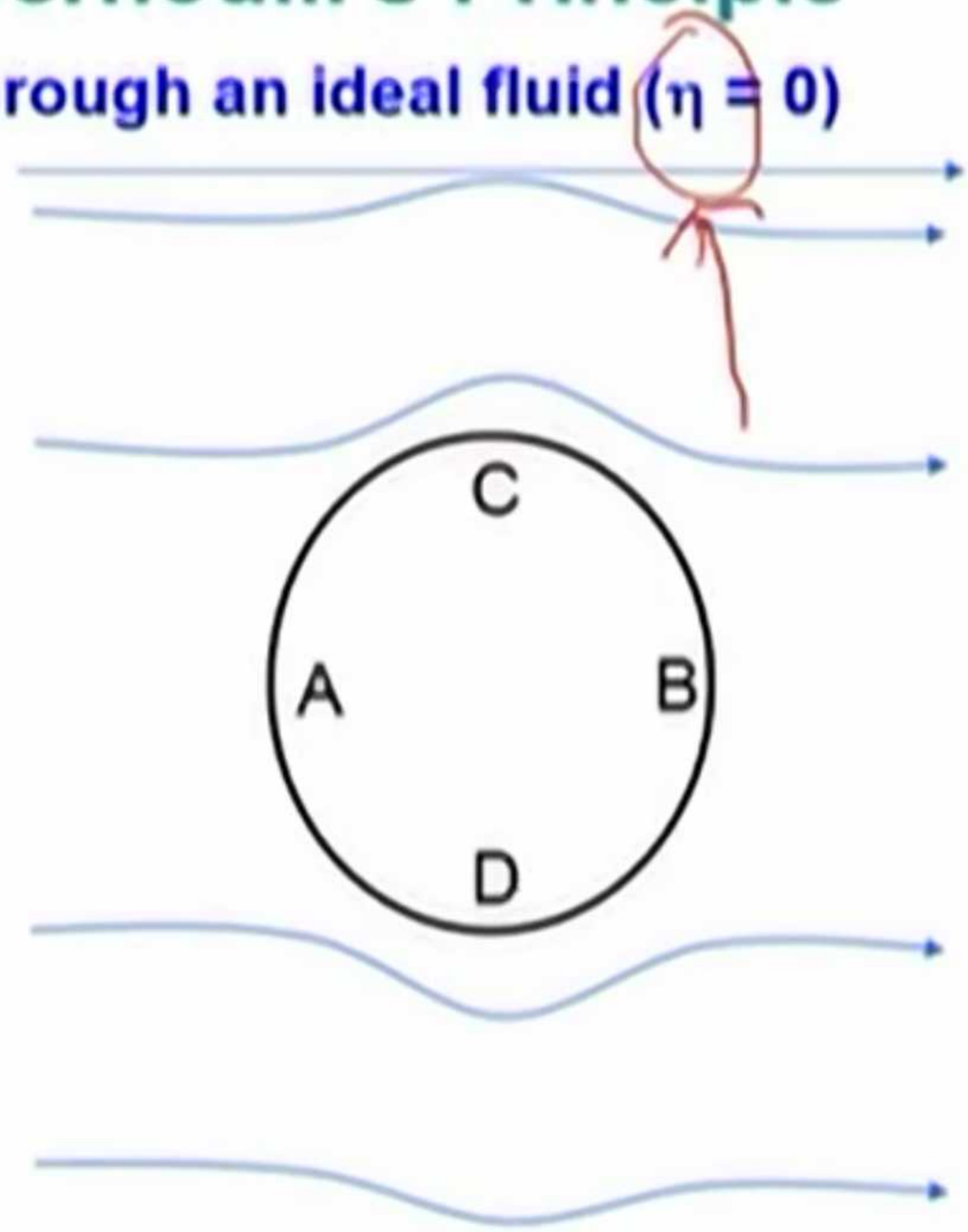
Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Uniform motion of an object through an ideal fluid ($\eta = 0$)

The pattern is symmetrical

$$\Rightarrow F_R = 0$$



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Applications of Bernoulli's Principle

Why does a cricket ball swing or a baseball curve?

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Applications of Bernoulli's Principle

Drag force

frictional drag (viscosity)

pressure drag (eddies – lower pressure)

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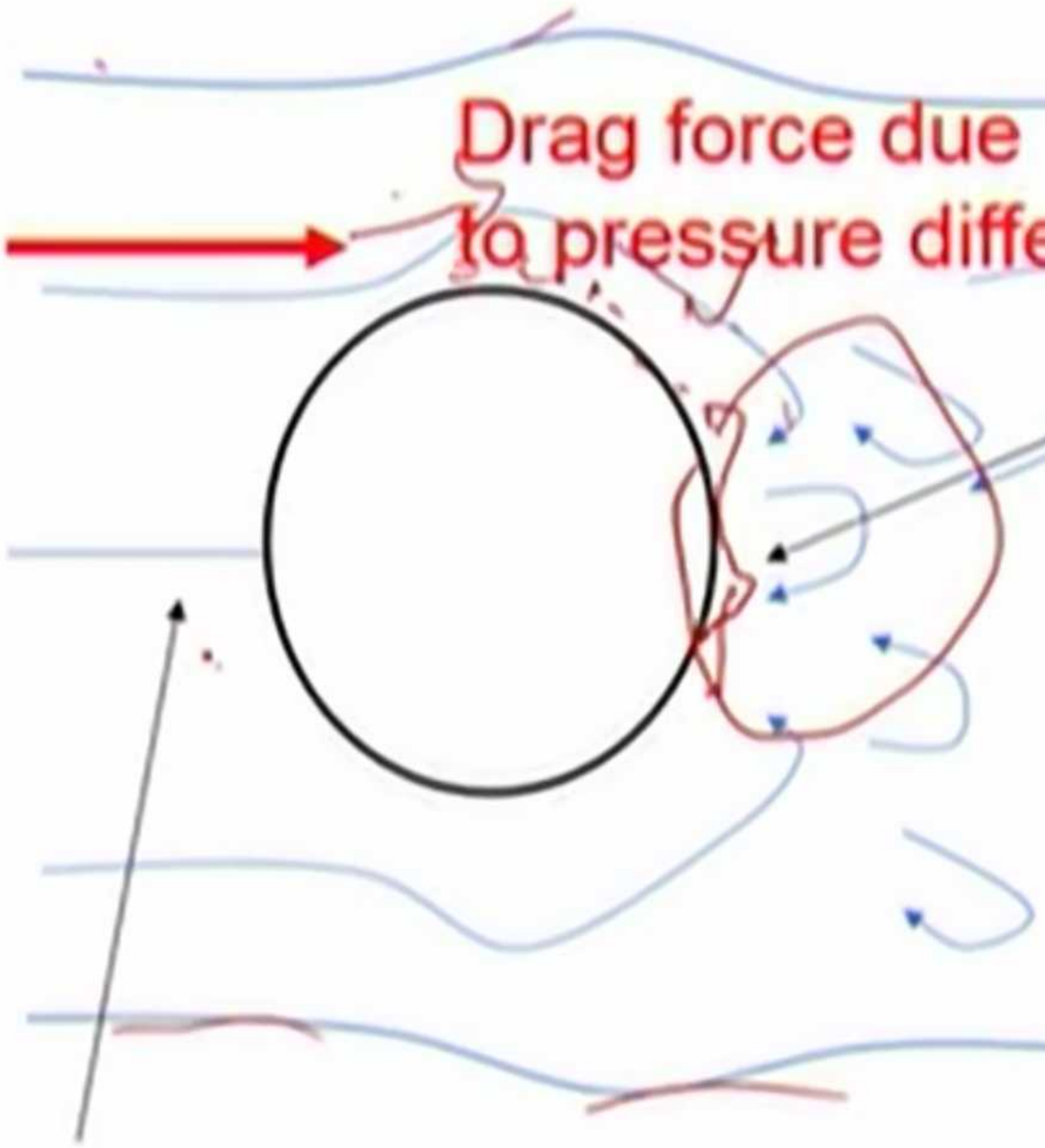

No CURVE
Drag force is opposite to the direction of motion

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ball: curved flight

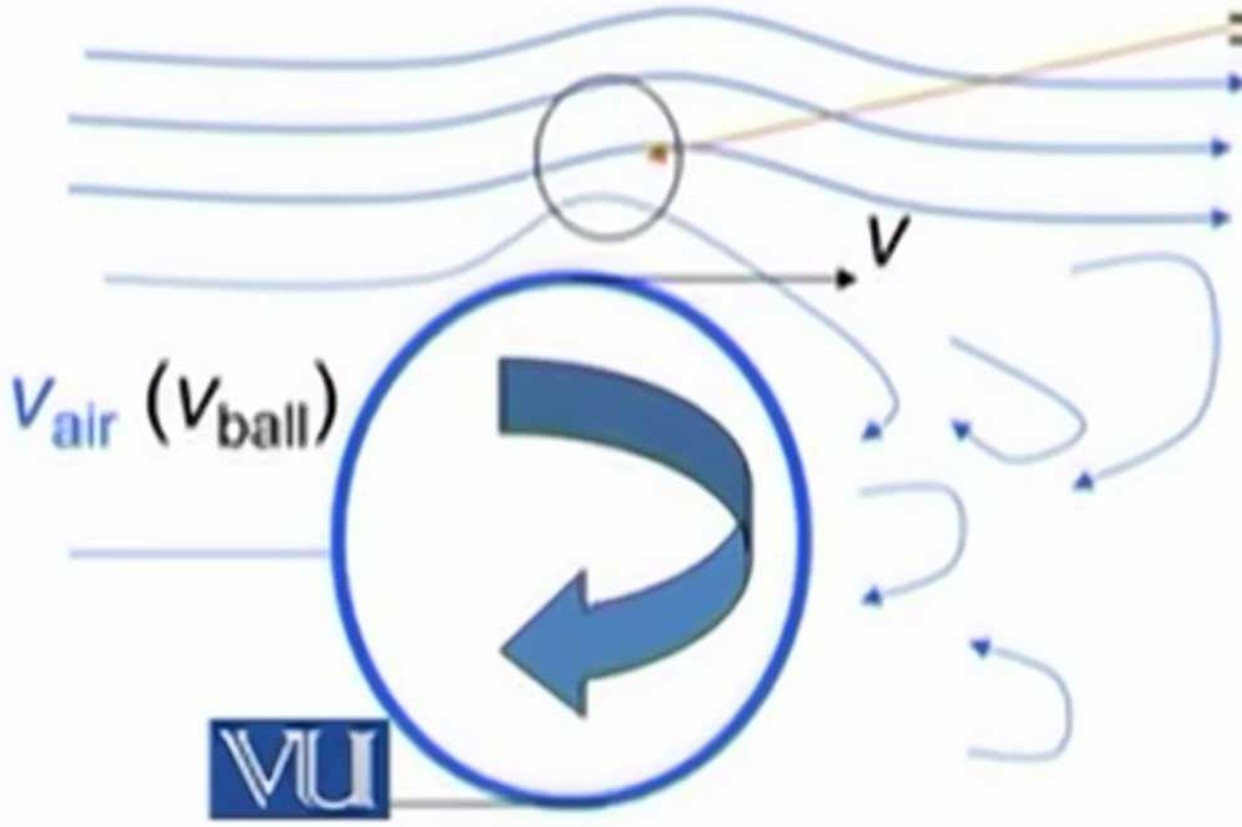
motion of air

motion of object

Drag force due
to pressure difference

flow speed (high) $v_{air} + v$

⇒ reduced pressure

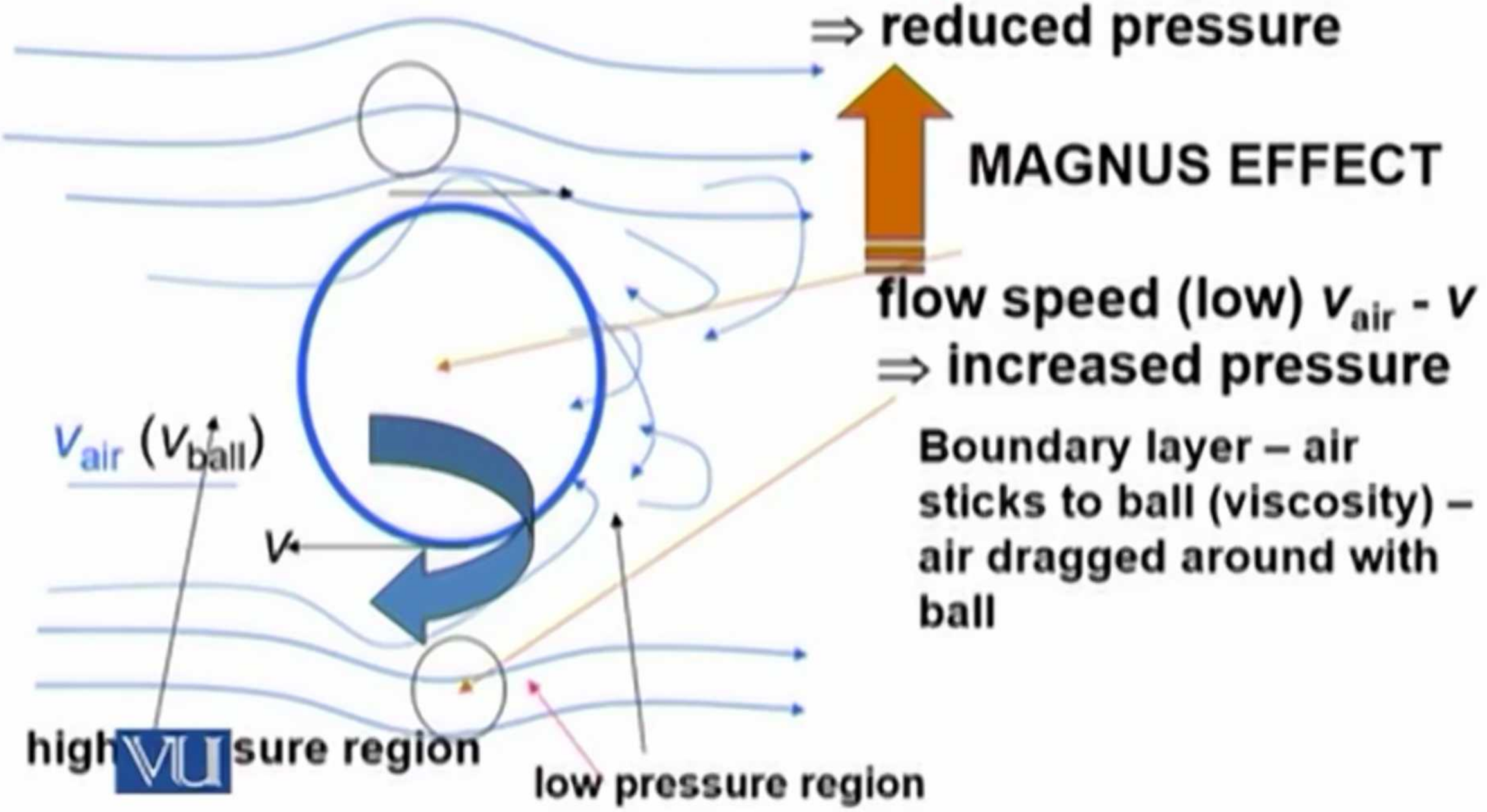


MAGNUS EFFECT

flow speed (low) $v_{air} - v$

⇒ increased pressure

Advanced Fluid Dynamics



LECTURE 80

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

$$p + \tfrac{1}{2} \rho v^2 + \rho g y = \text{constant}$$

Dimensions

p [Pa] = [N.m^{-2}] = [N.m.m^{-3}] = [J.m^{-3}]

$\tfrac{1}{2} \rho v^2$ [kg.m^{-3}.m^2.s^{-2}] = [kg.m^{-1}.s^{-2}] = [N.m.m^{-3}] = [J.m^{-3}]

$\rho g h$ [kg.m^{-3} m.s^{-2}. m] = [kg.m.s^{-2}.m.m^{-3}] = [N.m.m^{-3}] = [J.m^{-3}]

VU

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Each term has the dimensions of energy / volume or energy density.

$\frac{1}{2} \rho v^2$ KE of bulk motion of fluid

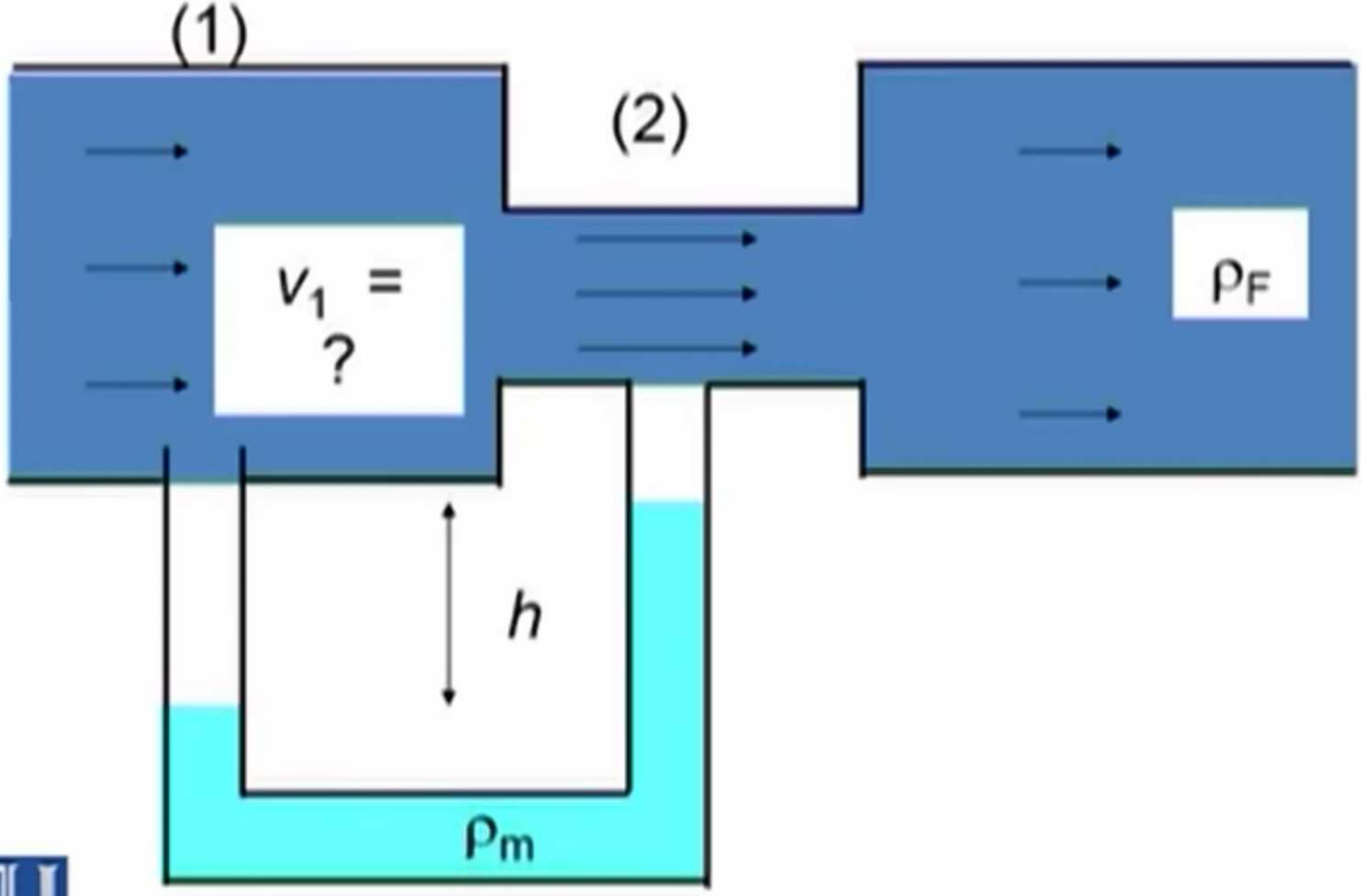
$\rho g h$ PE for location of fluid

p **pressure energy density arising from internal forces within moving fluid (similar to energy stored in a spring)**

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

How do you measure the speed of flow for a fluid?



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Applications of Bernoulli's Principle

One method of measuring it is to use a manometer type venturi meter.

Assume liquid behaves as an ideal fluid and that Bernoulli's equation can be applied for the flow along a streamline

$$p_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

$$y_1 = y_2$$

$$p_1 - p_2 = \tfrac{1}{2}\rho_F (v_2^2 - v_1^2)$$

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Applications of Bernoulli's Principle

$$p_1 - p_2 = \rho_m g h$$

$$A_1 v_1 = A_2 v_2 \Rightarrow v_2 = v_1 (A_1 / A_2)$$

$$\rho_m g h = \tfrac{1}{2} \rho_F \{ v_1^2 (A_1 / A_2)^2 - v_1^2 \}$$

$$= \tfrac{1}{2} \rho_F v_1^2 \{(A_1 / A_2)^2 - 1\}$$

$$v_1 = \sqrt{\frac{2 g h \rho_m}{\rho_F \left\{ \left(\frac{A_1}{A_2} \right)^2 - 1 \right\}}}$$

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LECTURE 81

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

A large artery in a body has an inner radius of 4.00×10^{-3} m. Blood flows through the artery at the rate of 1.00×10^{-6} $m^3.s^{-1}$. The blood has a viscosity of 2.084×10^{-3} Pa.s and a density of 1.06×10^{3} $kg.m^{-3}$.

Calculate:

a. The average blood velocity in the artery.
b. The pressure drop in a 0.100 m segment of the artery.
c. The Reynolds number for the blood flow.

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Briefly discuss each of the following:

(iv) The velocity profile across the artery (diagram may be helpful).

(v) The pressure drop along the segment of the artery.

(vi) The significance of the value of the Reynolds number calculated in part (iii).

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Solution

radius $R = 4.00 \times 10^{-3}$ m

volume flow rate $Q = 1.00 \times 10^{-6}$ $m^3.s^{-1}$

viscosity of blood $\eta = 2.084 \times 10^{-3}$ Pa.s

density of blood $\rho = 1.060 \times 10^{-3}$ $kg.m^{-3}$

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Advanced Fluid Dynamics

Applications of Bernoulli's Principle

(i) **Equation of continuity:** $Q = A\,v$

$$A = \pi R^2 = \pi\,(4.00\times10^{-3})^2 = 5.03\times10^{-5}\ \text{m}^2$$

$$v = Q / A = 1.00\times10^{-6} / 5.03\times10^{-5}\ \text{m.s}^{-1}$$
$$= 1.99\times10^{-2}\ \text{m.s}^{-1}$$

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

(iii) Reynolds Number

$R_e = \rho v L / \eta$ where $L = 2R$ (diameter of artery)

$R_e = (1.060\times10^{3})(1.99\times10^{-2})(2)(4.00\times10^{-3}) / (2.084\times10^{-3})$

$Re = 81$

use diameter not length

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

(i) **Equation of continuity:** $Q = A\,v$

$$A = \pi R^2 = \pi\,(4.00\times10^{-3})^2 = 5.03\times10^{-5}\ \text{m}^2$$

$$v = Q / A = 1.00\times10^{-6} / 5.03\times10^{-5}\ \text{m.s}^{-1}$$
$$= 1.99\times10^{-2}\ \text{m.s}^{-1}$$

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MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

(ii) Poiseuille's Equation

$$Q = \Delta P\, \pi\, R^4 / (8\, \eta\, L) \qquad L = 0.100 \text{ m}$$

$$\Delta P = 8\, \eta\, L\, Q / (\pi\, R^4)$$

$$\Delta P = (8)(2.084\times10^{-3})(0.1)(1.00\times10^{-6}) / \{(\pi)(4.00\times10^{-3})^4\} \text{ Pa}$$

$$\Delta P = 2.07 \text{ Pa}$$

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Applications of Bernoulli's Principle

(iii) Reynolds Number

$R_e = \rho v L / \eta$ where $L = 2R$ (diameter of artery)

$R_e = (1.060\times10^{3})(1.99\times10^{-2})(2)(4.00\times10^{-3}) / (2.084\times10^{-3})$

$Re = 81$

use diameter not length

Advanced Fluid Dynamics

Applications of Bernoulli's Principle

(iv) **Parabolic velocity profile: velocity of blood zero at sides of artery**

(v) **Viscosity ⇒ internal friction ⇒ energy dissipated as thermal energy ⇒ pressure drop along artery**

(vi) **Re very small ⇒ laminar flow ($R_\theta < 2000$)**

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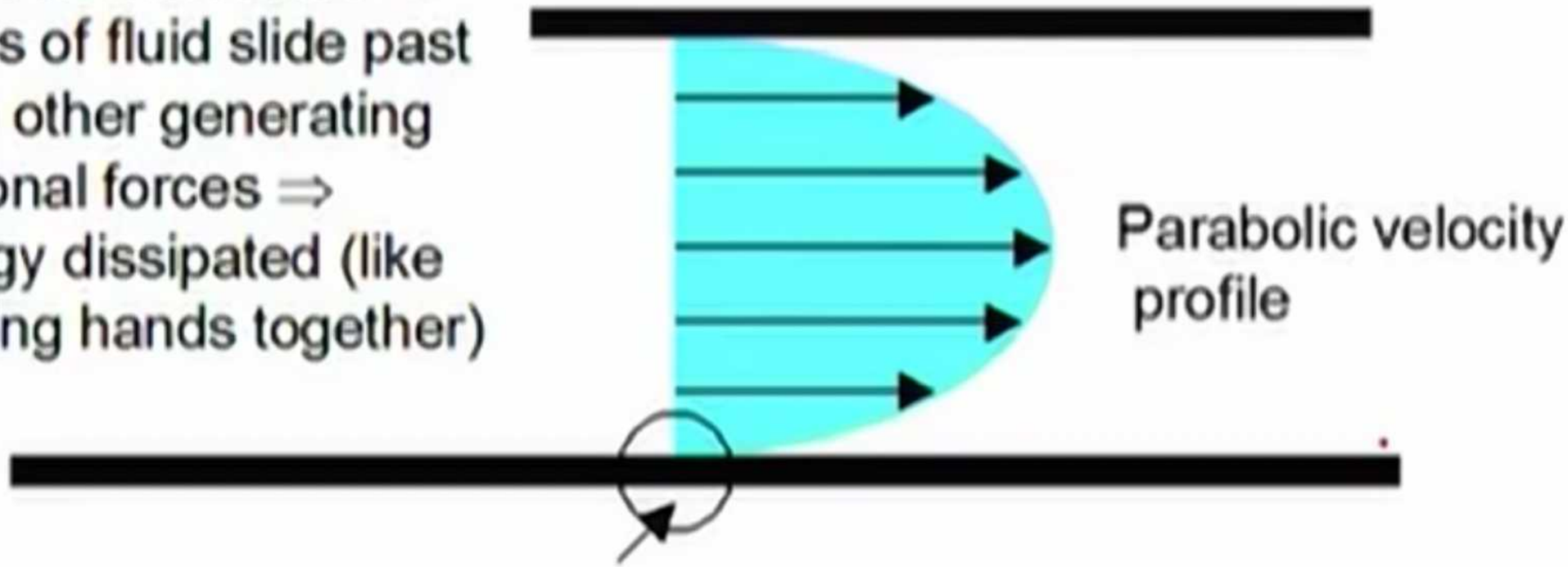
Advanced Fluid Dynamics

Flow of a viscous newtonain fluid through a pipe
Velocity Profile

Cohesive forces between molecules ⇒ layers of fluid slide past each other generating frictional forces ⇒ energy dissipated (like rubbing hands together)



Adhesive forces between fluid and surface ⇒ fluid stationary at surface

LECTURE 83

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Newton's Law's of Motion

Objectives

- **Identify the various kinds of forces and moments acting on a control volume**
- **Use control volume analysis to determine the forces associated with fluid flow**
- **Use control volume analysis to determine the moments caused by fluid flow and the torque transmitted**

LECTURE 84

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Newton's Law's of Motion

- Newton's laws:
- Relations between motions of bodies and the forces acting on them.
- Newton's first law: A body at rest remains at rest, and a body in motion remains in motion at the same velocity in a straight path when the net force acting on it is zero.

Advanced Fluid Dynamics

Newton's Law's of Motion

- Therefore, a body tends to preserve its state of inertia.
- Newton's second law: The acceleration of a body is proportional to the net force acting on it and is inversely proportional to its mass.

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Newton's Law's of Motion

Newton's third law: When a body exerts a force on a second body, the second body exerts an equal and opposite force on the first.

- **Therefore, the direction of an exposed reaction force depends on the body taken as the system.**

Advanced Fluid Dynamics

Newton's Law's of Motion

Newton's second law:

$$\vec{F} = m\vec{a} = m\frac{d\vec{V}}{dt} = \frac{d(m\vec{V})}{dt}$$

Therefore, the direction of an exposed reaction force depends on the body taken as the system

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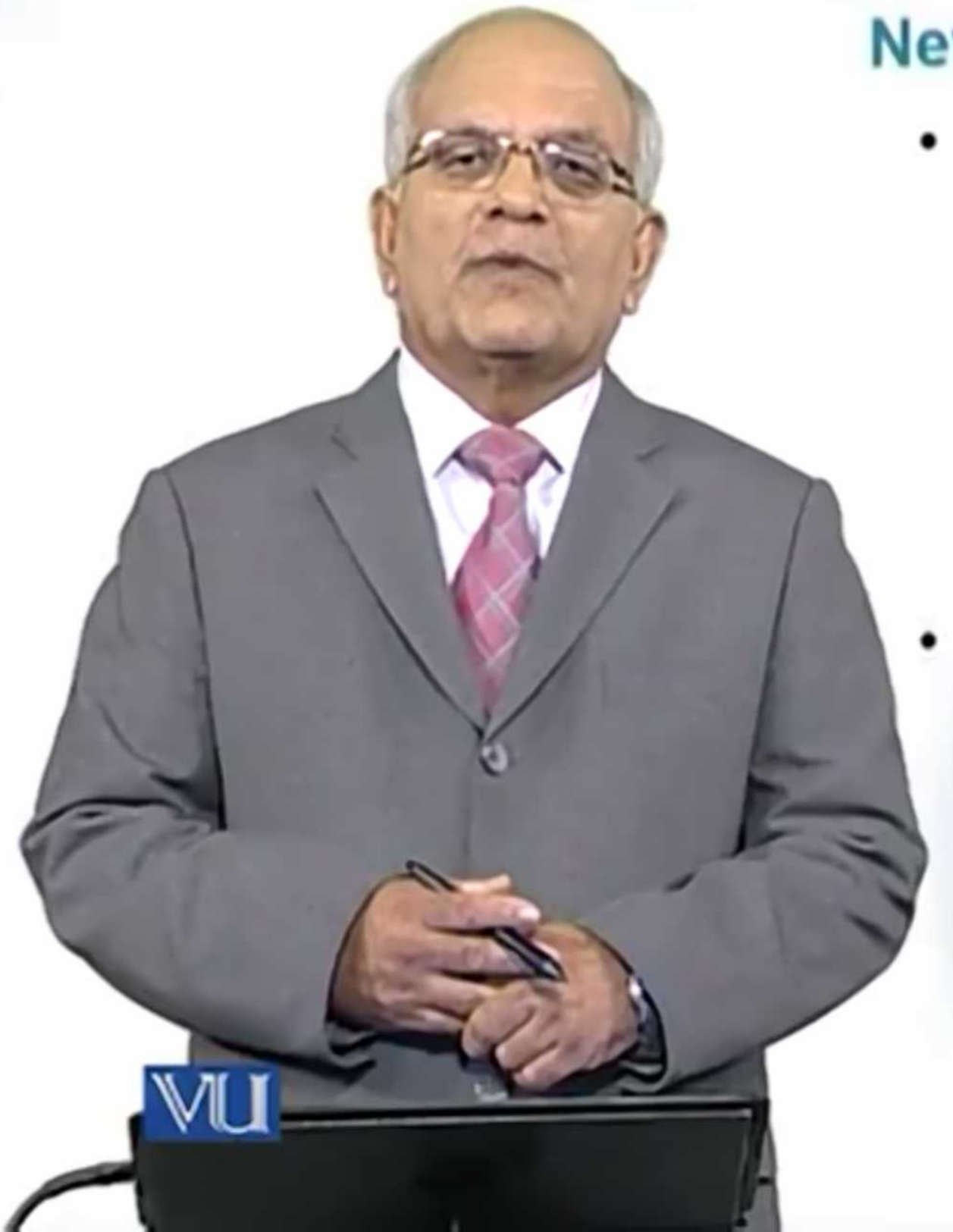


Newton's Law's of Motion

- ***Linear momentum*** **or just the *momentum* of the body: The product of the mass and the velocity of a body.**

- **Newton's second law is usually referred to as the *linear momentum equation*.**

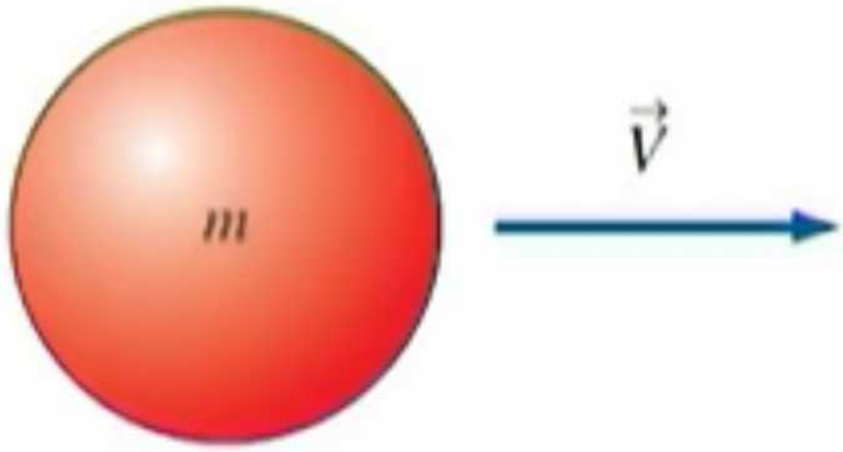
LECTURE 85

MTH642 FLUID MECHANICS

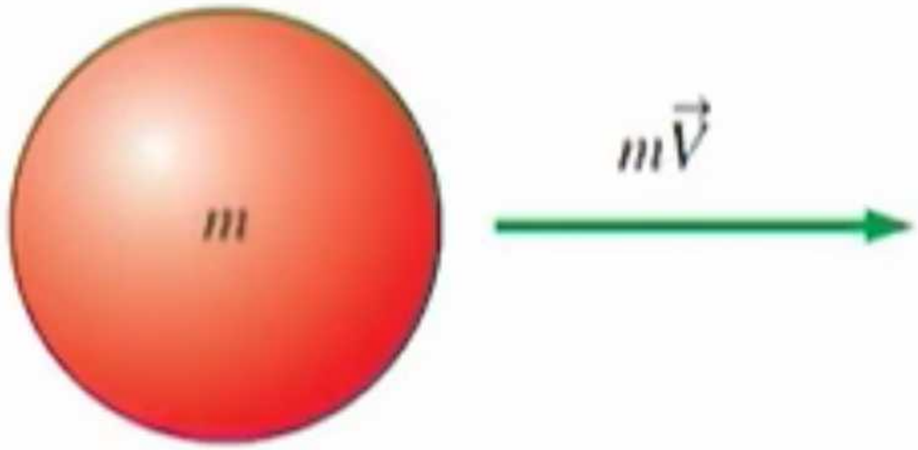
Advanced Fluid Dynamics

Newton's Law's of Motion

Advanced Fluid Dynamics

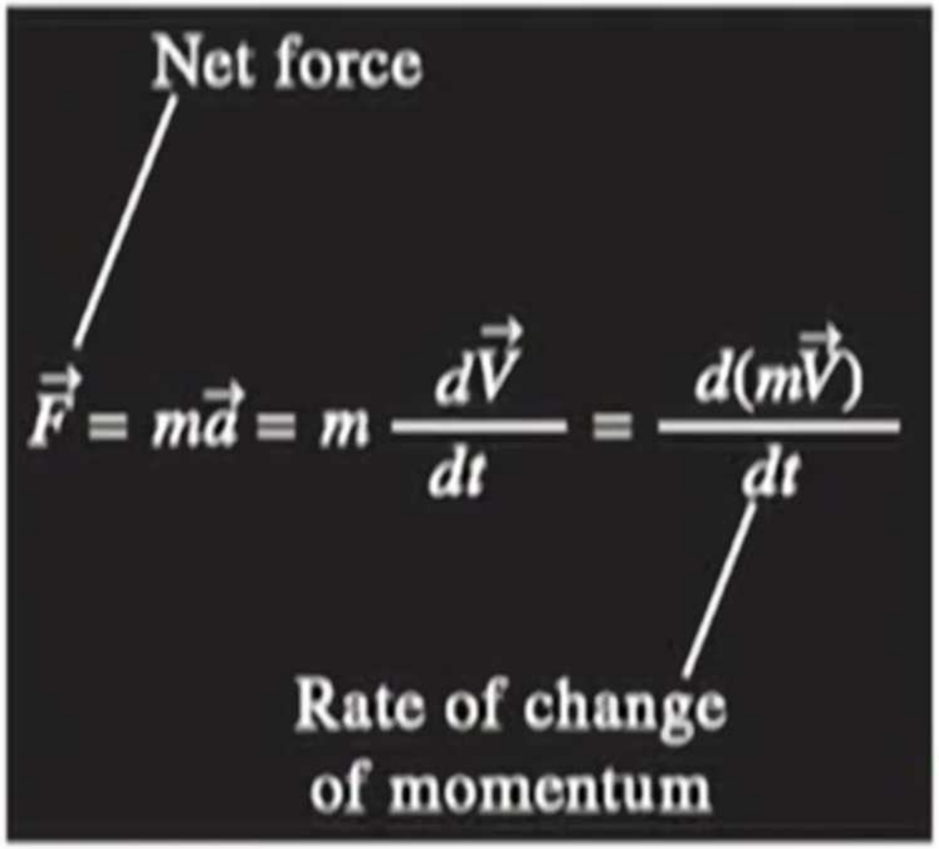


Conservation of momentum principle: The momentum of a system remains constant only when the net force acting on it is zero.



Advanced Fluid Dynamics

Newton's Law's of Motion



- Linear momentum is the product of mass and velocity, and its direction is the direction of velocity.
- Newton's second law is also expressed as the rate of change of the momentum of a body is equal to the net force acting on it.

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Advanced Fluid Dynamics

Newton's Law's of Motion

The counterpart of Newton's second law for rotating rigid bodies is expressed as $\vec{M} = I\vec{\alpha}$, where $\vec{M}$ is the net moment or torque applied on the body, I is the moment of inertia of the body about the axis of rotation, and $\vec{\alpha}$ is the angular acceleration. It can also be expressed in terms of the rate of change of angular momentum $d\vec{H}/dt$ as

Angular momentum equation:

$$\vec{M} = I\vec{\alpha} = I\frac{d\vec{\omega}}{dt} = \frac{d(I\vec{\omega})}{dt} = \frac{d\vec{H}}{dt} \qquad (6\text{–}2)$$

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Advanced Fluid Dynamics

Newton's Law's of Motion

Angular momentum about x-axis:

$$M_x = I_x \frac{d\omega_x}{dt} = \frac{dH_x}{dt}$$

- **The conservation of angular momentum Principle: The total angular momentum of a rotating body remains constant when the net torque acting on it is zero, and thus the angular momentum of such systems is conserved.**

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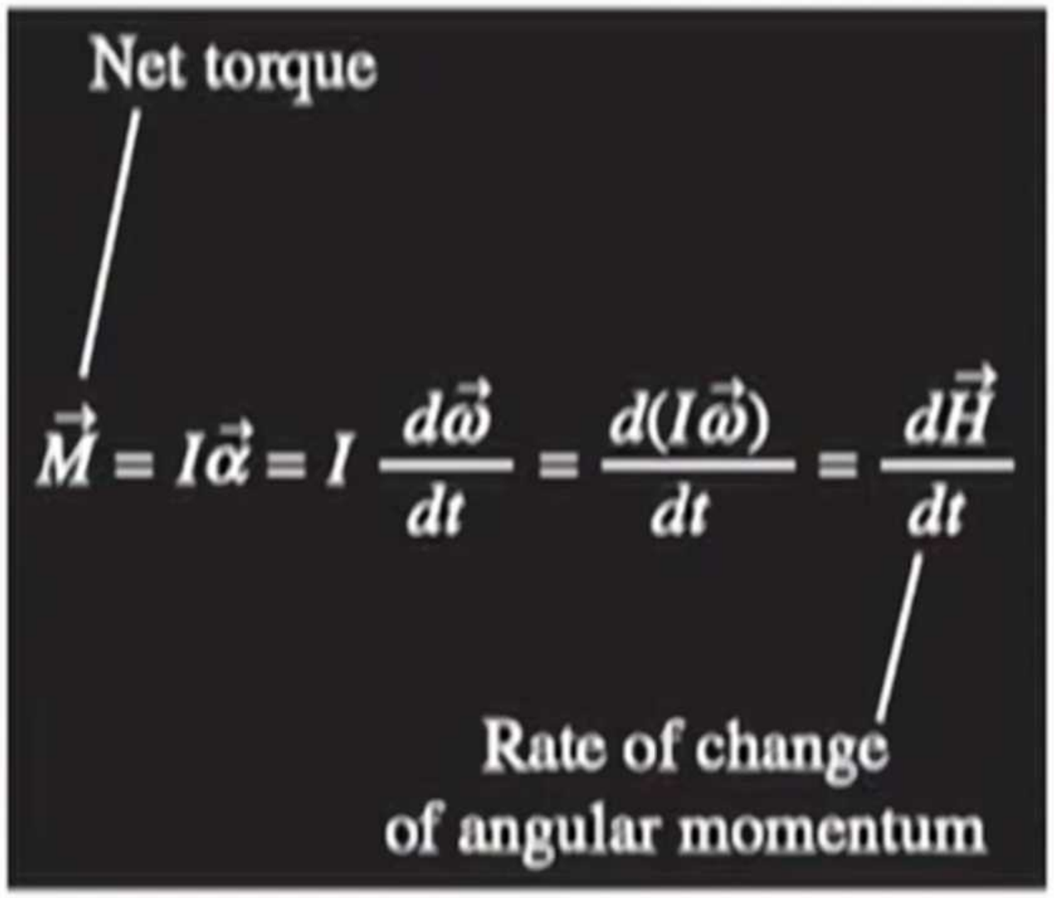
Advanced Fluid Dynamics

Newton's Law's of Motion



- **The rate of change of the angular momentum of a body is equal to the net torque acting on it.**

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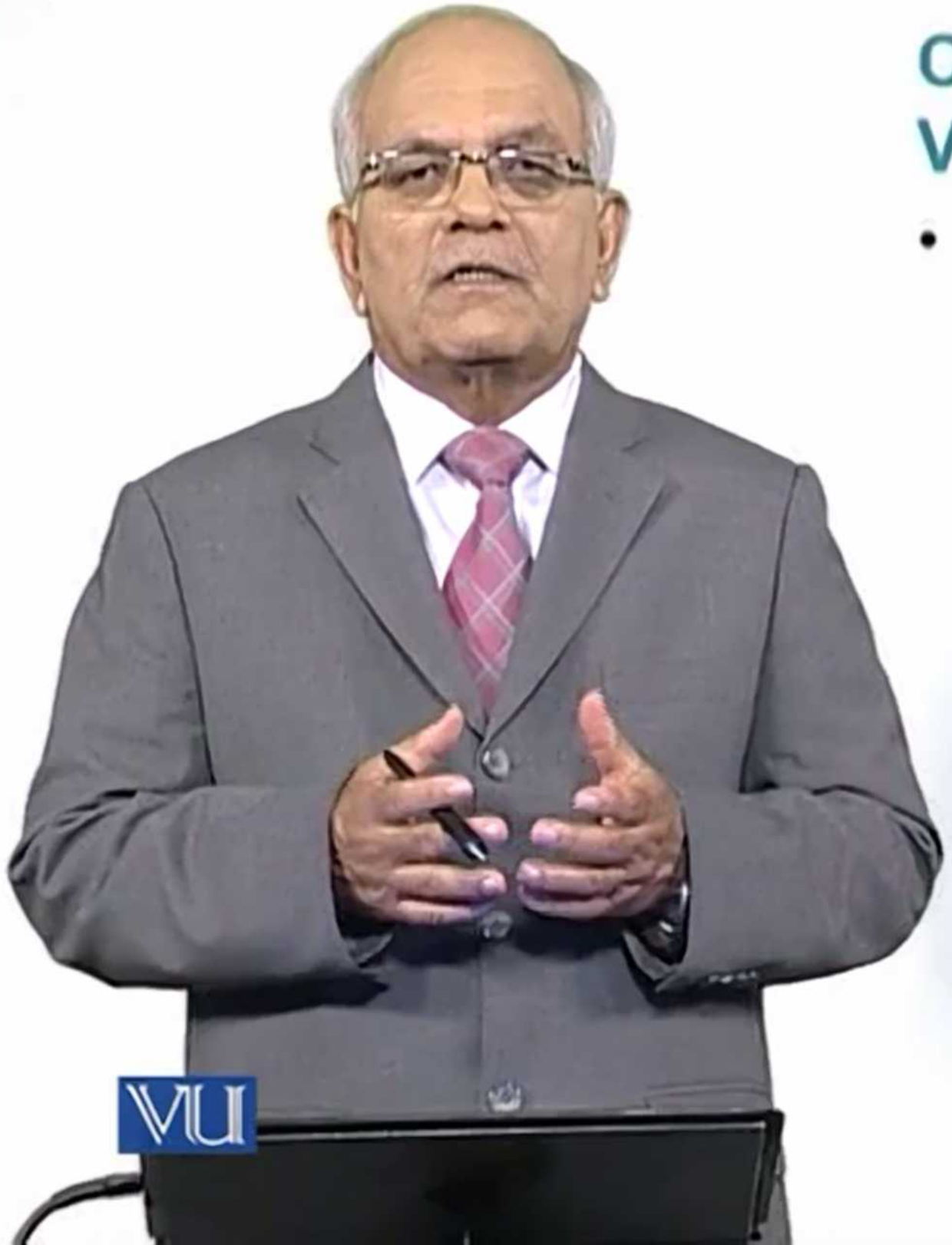

Choosing A Control Volume

- A control volume can be selected as any arbitrary region in space through which fluid flows, and its bounding control surface can be fixed, moving, and even deforming during flow.

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Choosing A Control Volume

- **Many flow systems involve stationary hardware firmly fixed to a stationary surface, and such systems are best analyzed using fixed control volumes.**

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Choosing A Control Volume

- **When analyzing flow systems that are moving or deforming, it is usually more convenient to allow the control volume to *move* or *deform*.**

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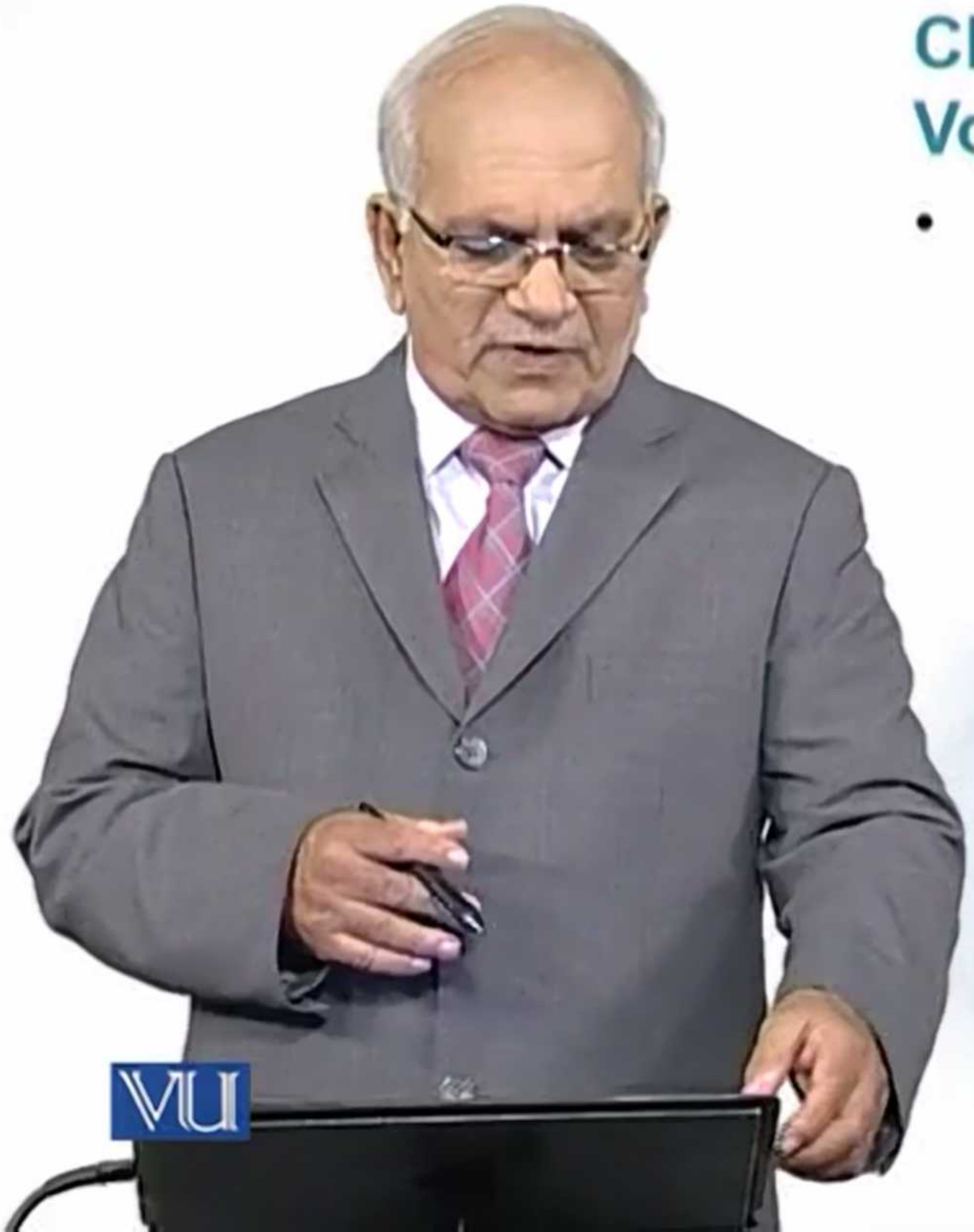

Choosing A Control Volume

- In deforming control volume, part of the control surface moves relative to other parts.

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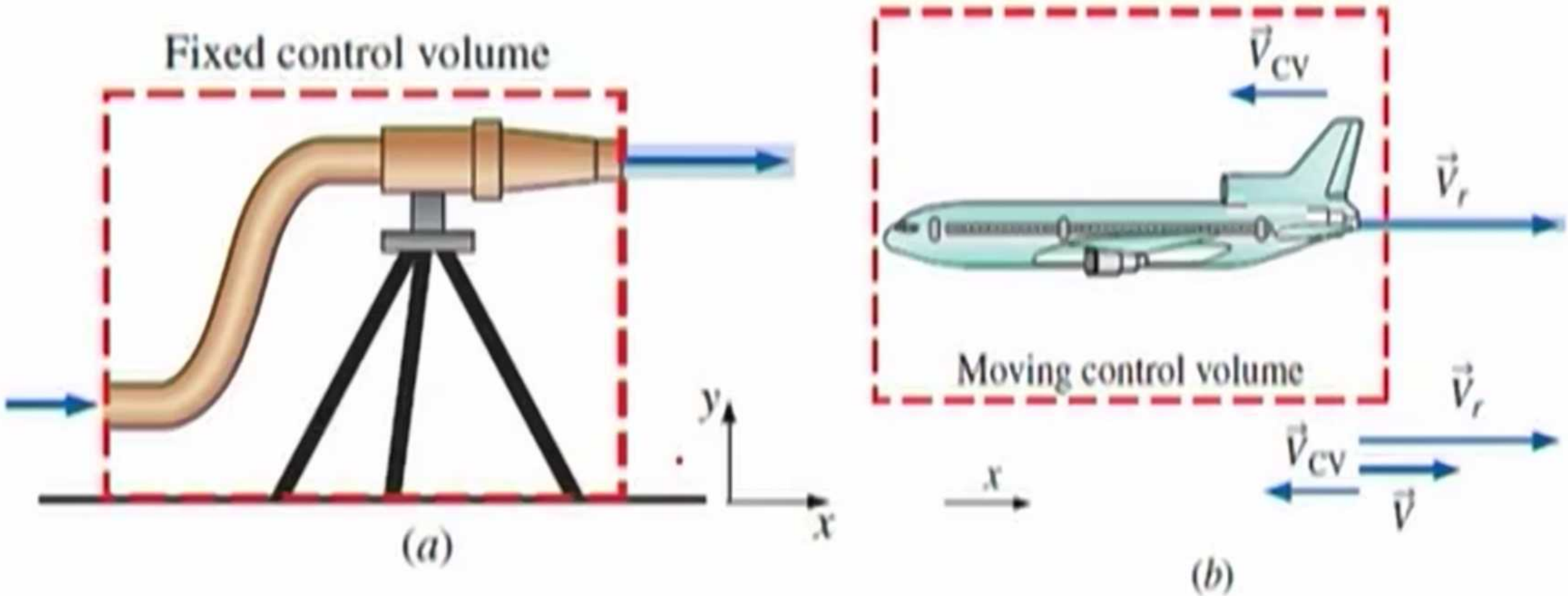
Choosing A Control Volume



Examples of (*a*) fixed, (*b*) moving,and

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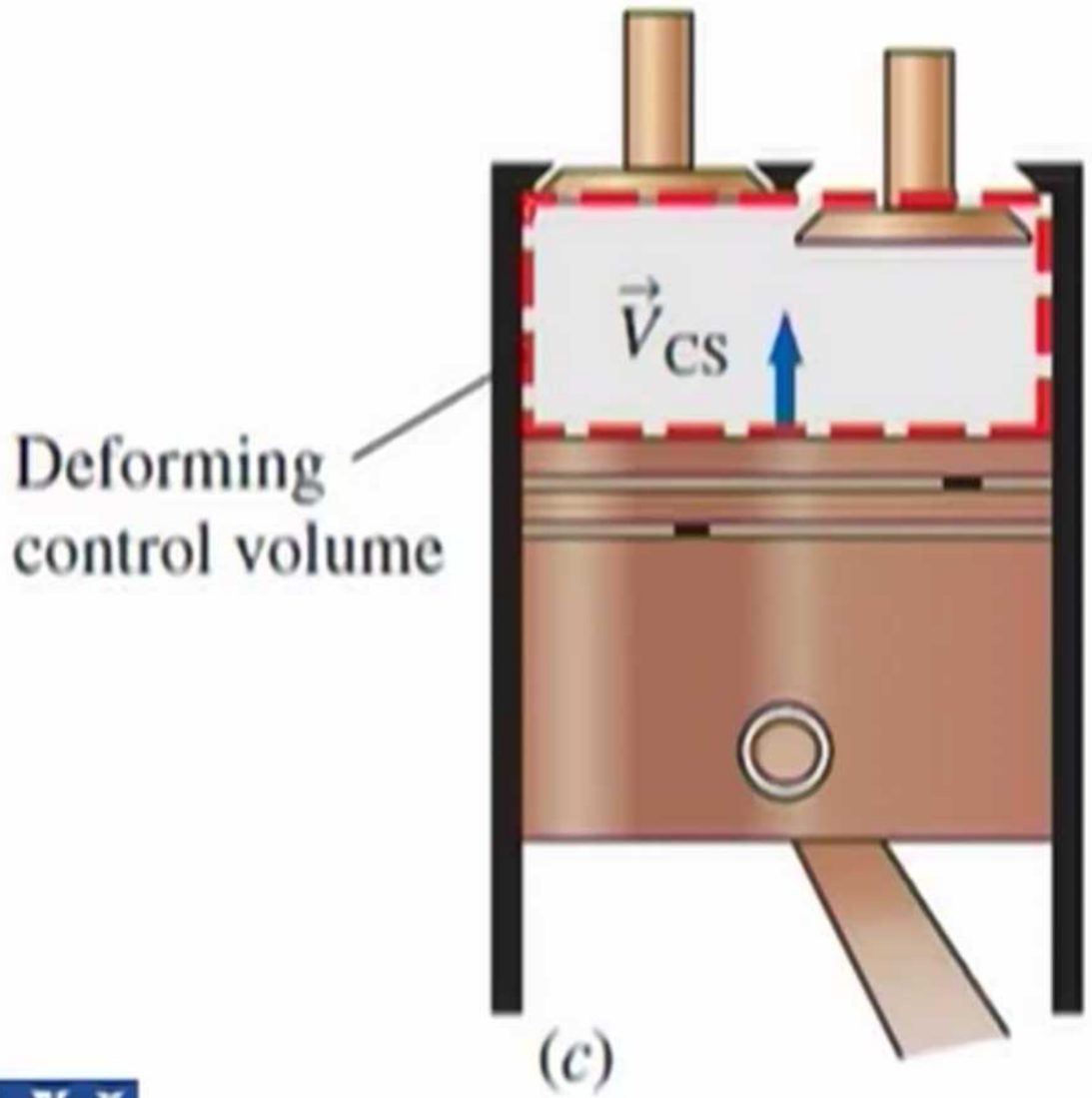
Newton's Law's of Motion



- Examples of

(*c*) deforming control volumes.

LECTURE 86

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Advanced Fluid Dynamics

Forces Acting On A Control Volume

- The forces acting on a control volume consist of
- Body forces that act throughout the entire body of the control volume (such as gravity, electric, and magnetic forces) and
- Surface forces that act on the control surface (such as

Advanced Fluid Dynamics

Forces Acting On A Control Volume

pressure and viscous forces and reaction forces at points of contact).

Only external forces are considered in the analysis.

Total force acting on control volume:

$$\sum \vec{F} = \sum \vec{F}_{\text{body}} + \sum \vec{F}_{\text{surface}}$$

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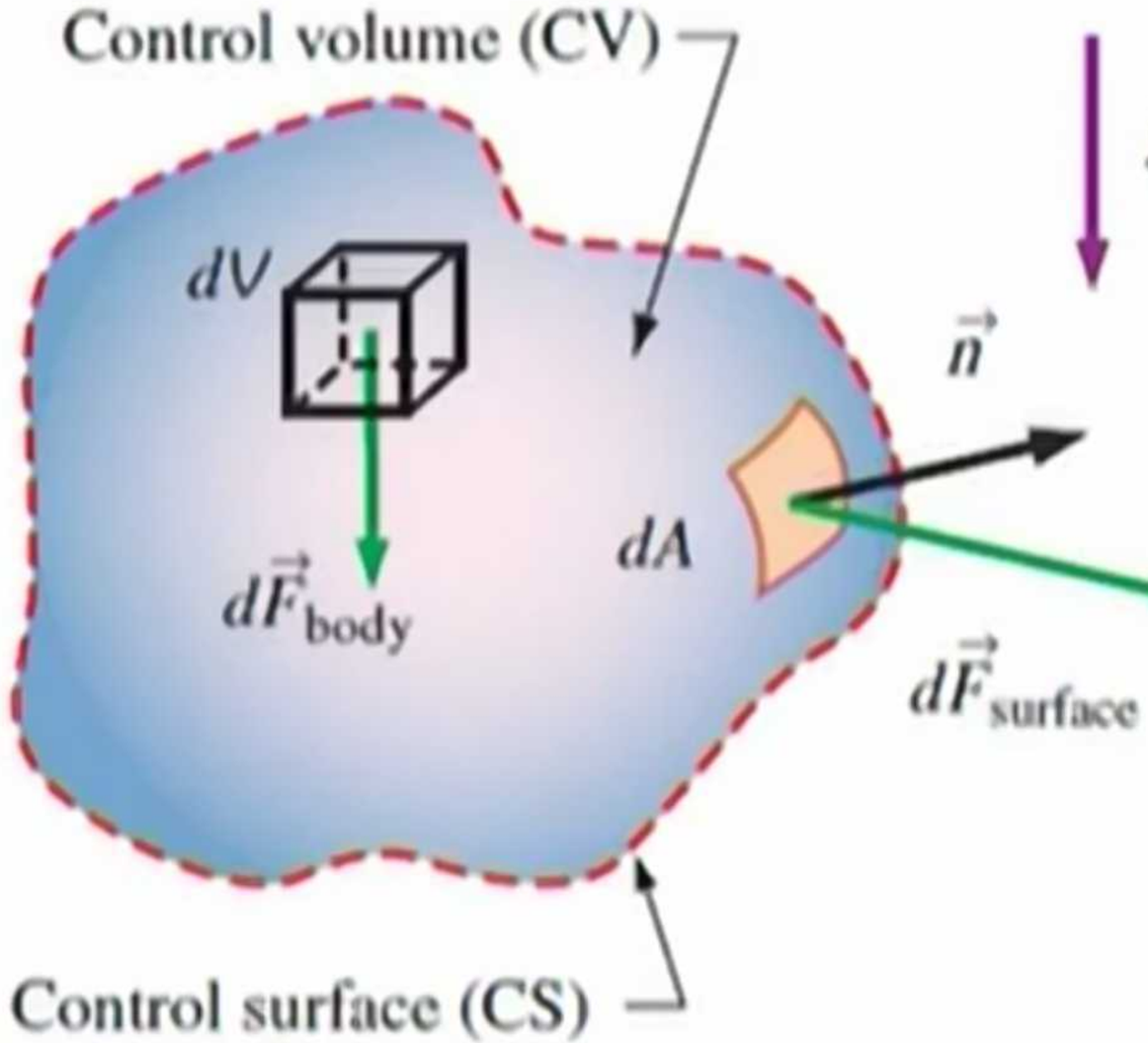
Forces Acting On A Control Volume



- **The total force acting on a control volume is composed of body forces and surface forces; body force is shown on a differential volume element, and surface force is shown on a differential surface element.**

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Forces Acting On A Control Volume

The most common body force is that of gravity, which exerts a downward force on every differential element of the control volume.

Gravitational force acting on a fluid element: $d\vec{F}_{\text{gravity}} = \rho\vec{g}\,dV$

Gravitational vector in Cartesian coordinates: $\vec{g} = -g\vec{k}$

Total body force acting on control volume: $\sum \vec{F}_{\text{body}} = \int_{CV} \rho\vec{g}\,dV = m_{CV}\vec{g}$

Total force: $\underbrace{\sum \vec{F}}_{\text{total force}} = \underbrace{\sum \vec{F}_{\text{gravity}}}_{\text{body force}} + \underbrace{\sum \vec{F}_{\text{pressure}} + \sum \vec{F}_{\text{viscous}} + \sum \vec{F}_{\text{other}}}_{\text{surface forces}}$

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LECTURE 87

MTH642 FLUID MECHANICS

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Forces Acting On A Control Volume

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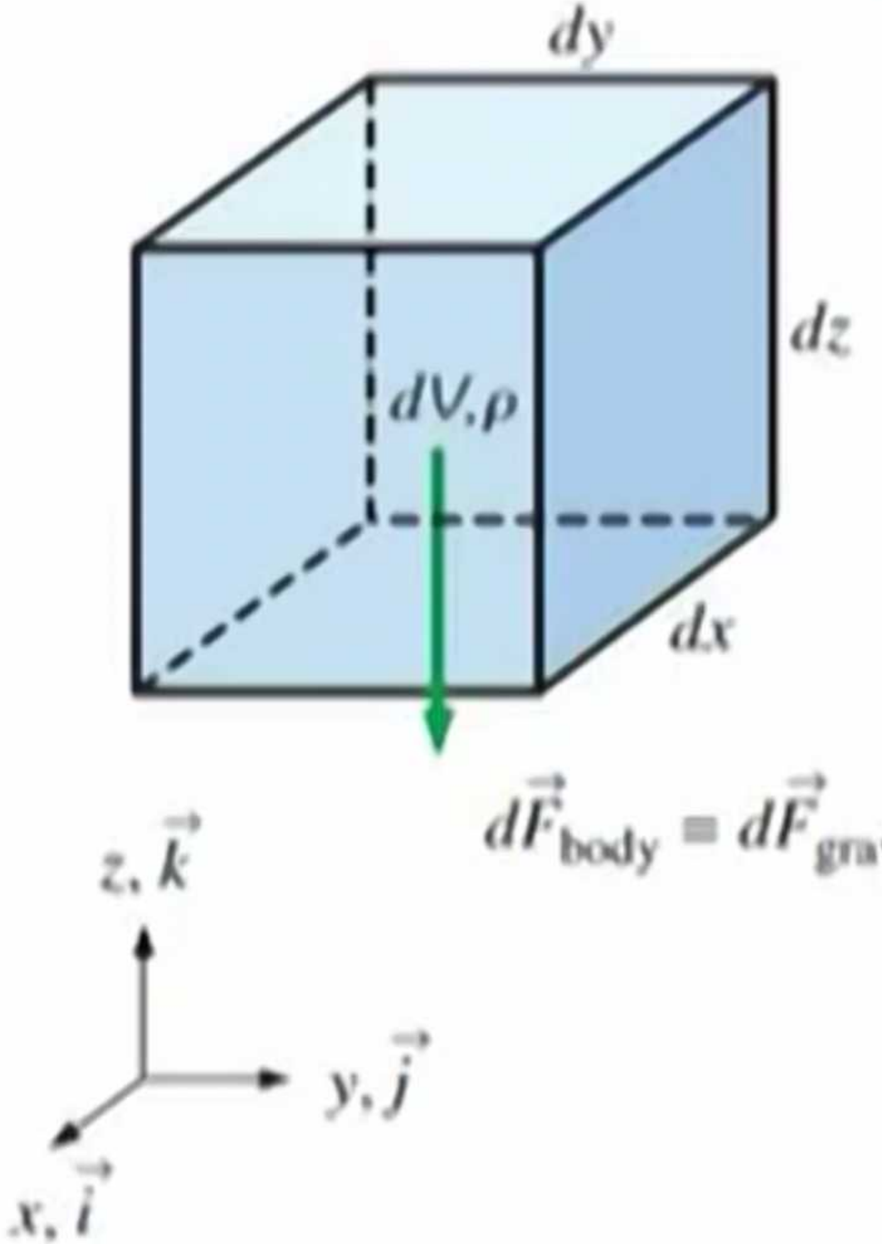
Forces Acting On A Control Volume



- **Surface forces are not as simple to analyze since they consist of both *normal* and *tangential* components.**
- **Normal stresses are composed of pressure (which always acts inwardly normal) and viscous stresses.**

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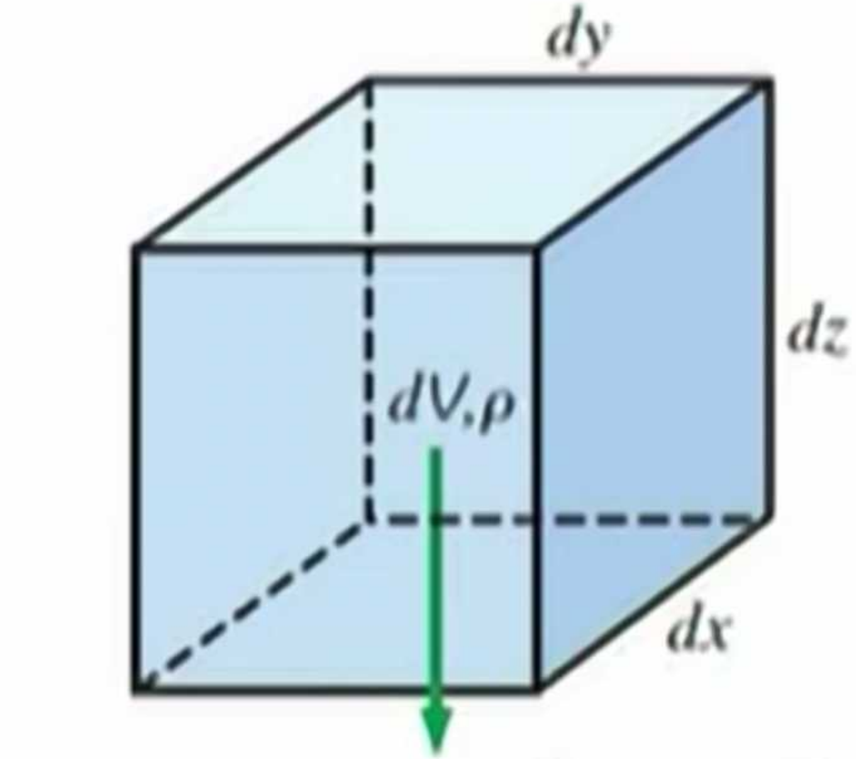
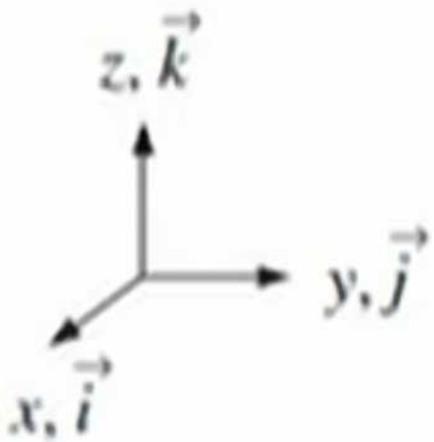
Forces Acting On A Control Volume



$$d\vec{F}_{\text{body}} = d\vec{F}_{\text{gravity}} = \rho\vec{g}\,dV$$



- **Shear stresses are composed entirely of viscous stresses.**
- **The gravitational force acting on a differential volume element of fluid is equal to its weight; the axes have been rotated so that the gravity vector acts *downward* in the negative *z*-direction.**

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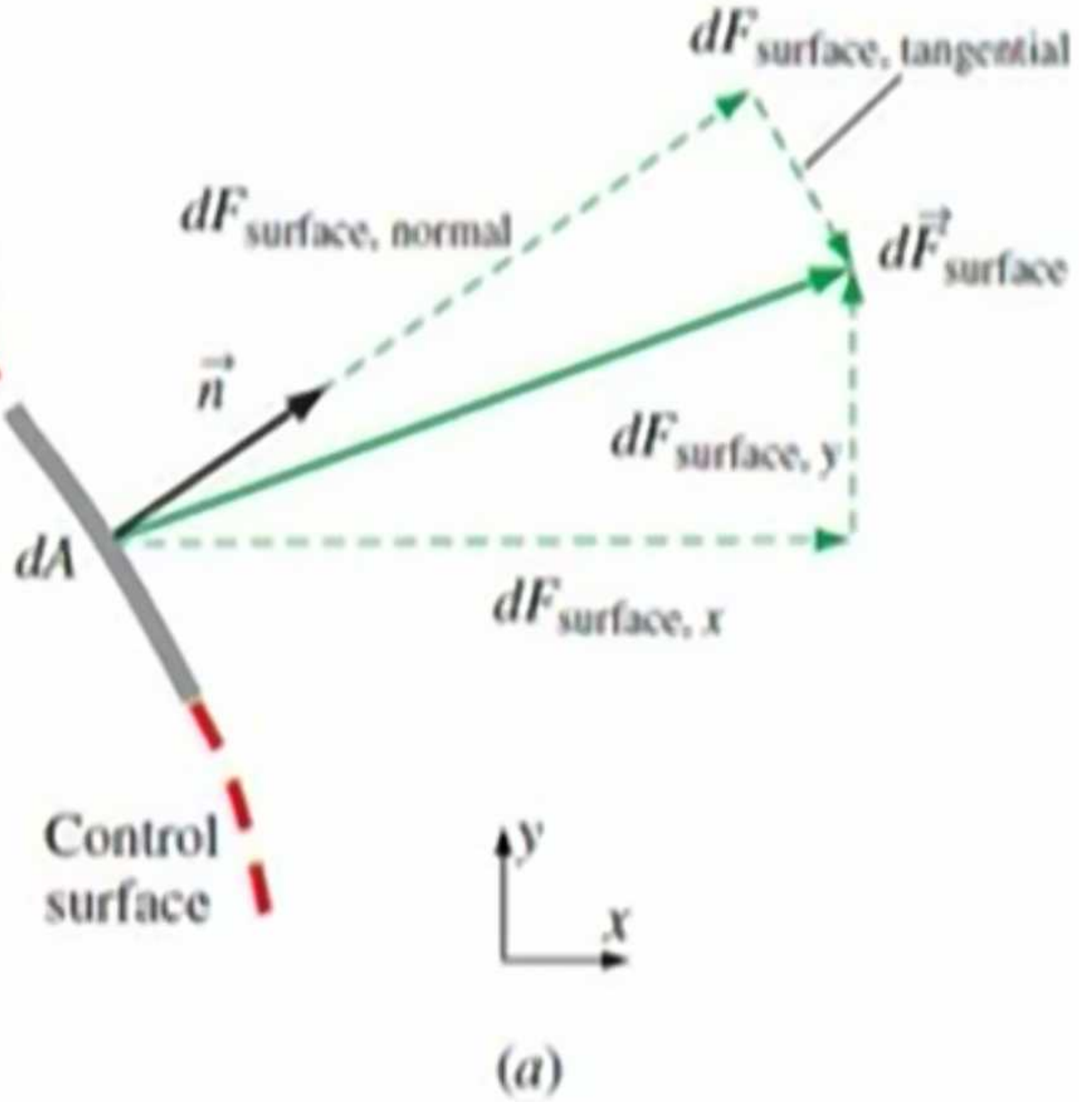
Forces Acting On A Control Volume



Surface force acting on a differential surface

$$d\vec{F}_{surface} = \sigma_{ij} \cdot \vec{n}\, dA$$

Total surface force acting on control surface:.

$$\sum \vec{F}_{surface} = \int_{CS} \sigma_{ij} \cdot \vec{n}\, dA$$

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Forces Acting On A Control Volume

$$\sum \vec{F} = \sum \vec{F}_{\text{body}} + \sum \vec{F}_{\text{surface}} = \int_{CV} \rho \vec{g}\, dV + \int_{CS} \sigma_{ij} \cdot \vec{n}\, dA$$

Total force:

$$\underbrace{\sum \vec{F}}_{\text{total force}} = \underbrace{\sum \vec{F}_{\text{gravity}}}_{\text{body force}} + \underbrace{\sum \vec{F}_{\text{pressure}} + \sum \vec{F}_{\text{viscous}} + \sum \vec{F}_{\text{other}}}_{\text{surface forces}}$$

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LECTURE 88

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Forces Acting On A Control Volume

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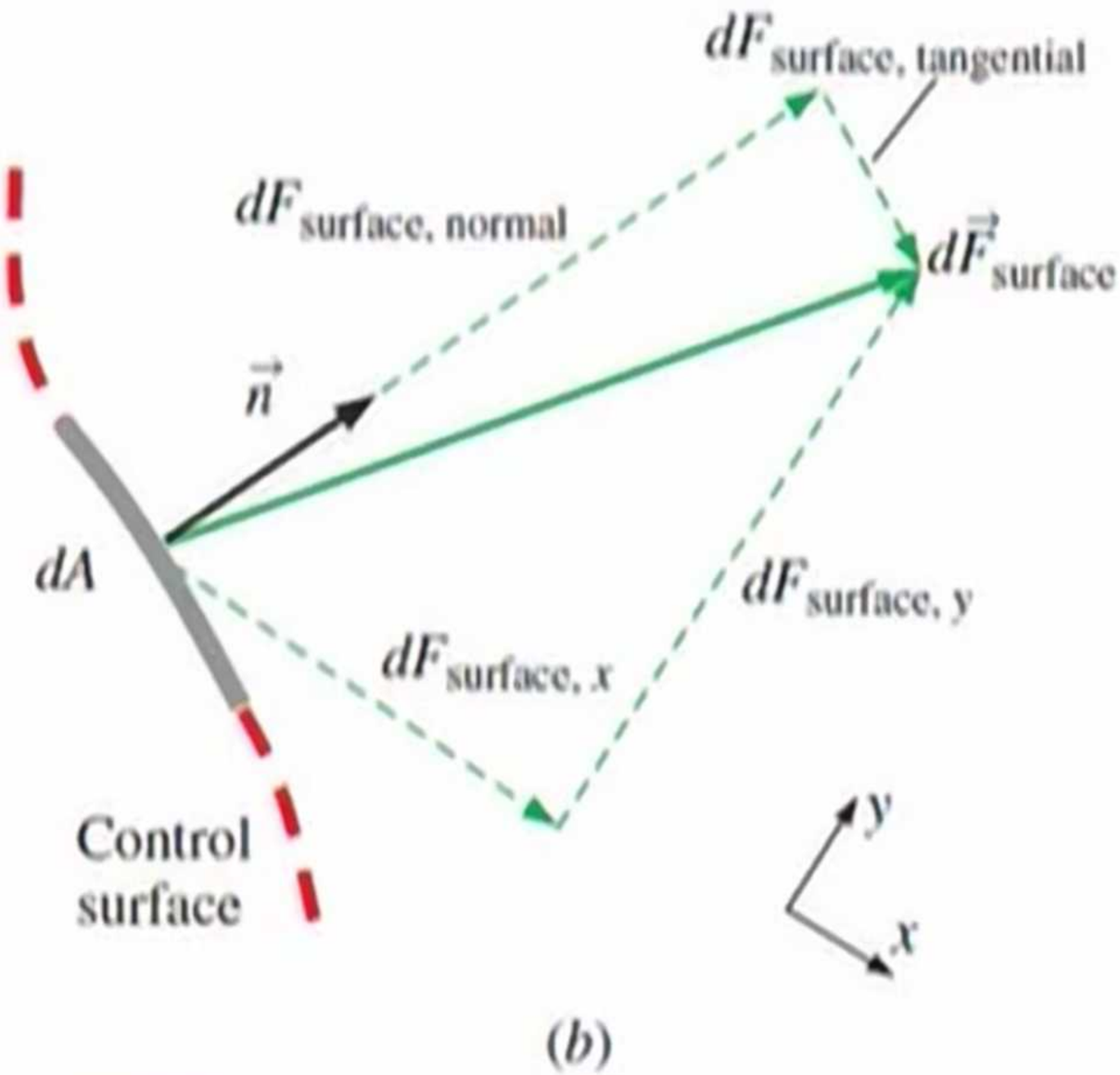
Forces Acting On A Control Volume



- **When coordinate axes are rotated (*a*) to (*b*), the components of the surface force change, even though the force itself remains the same; only two dimensions are shown here.**

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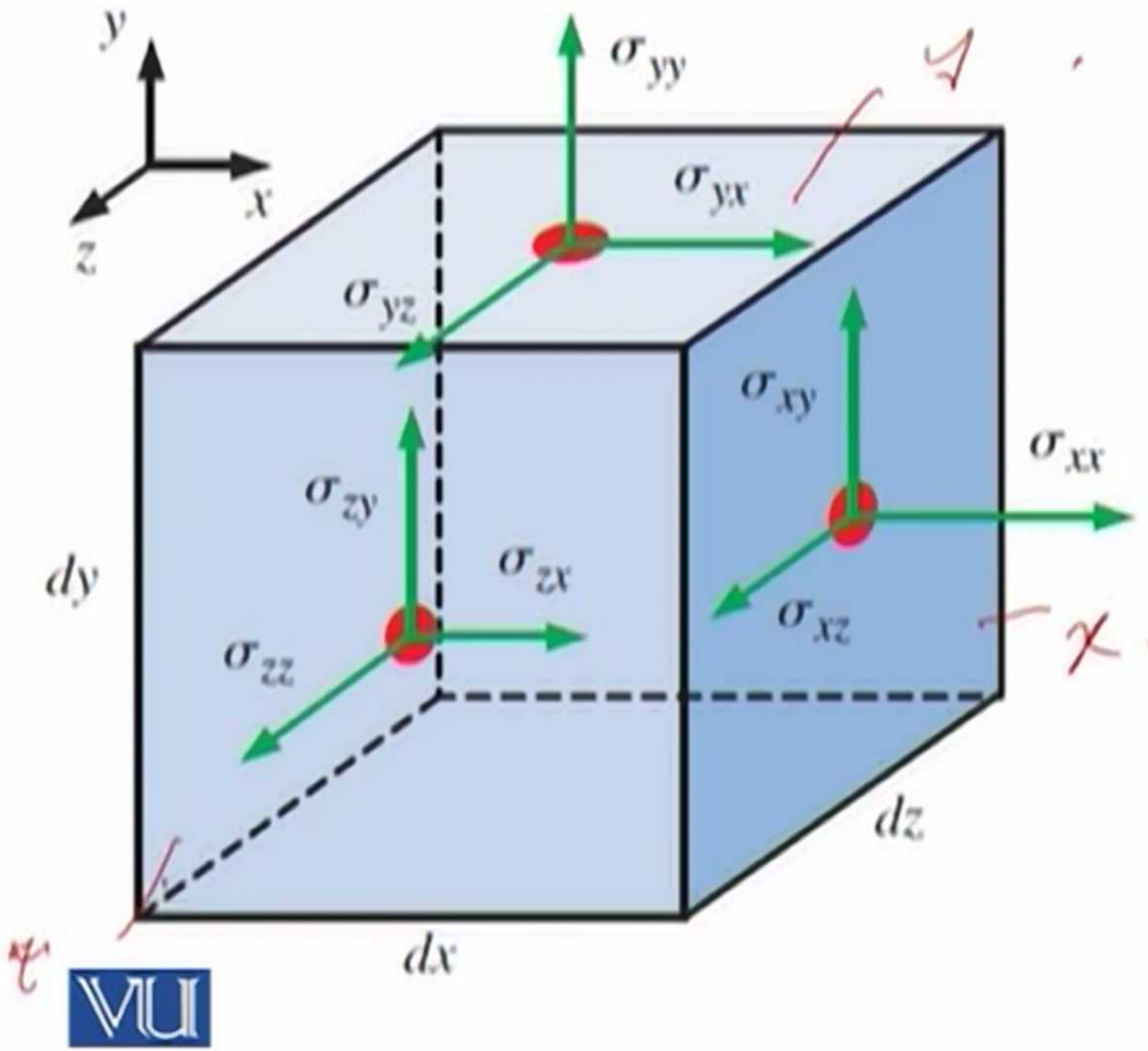
Forces Acting On A Control Volume



- **Components of the stress tensor in Cartesian coordinates on the right, top, and front faces.**

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Forces Acting On A Control Volume

- **A common simplification in the application of Newton's laws of motion is to subtract the *atmospheric pressure* and work with gage pressures.**

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Forces Acting On A Control Volume

- **This is because atmospheric pressure acts in all directions, and its effect cancels out in every direction.**

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Forces Acting On A Control Volume

- This means we can also ignore the pressure forces at outlet sections where the fluid is discharged to the atmosphere since the discharge pressure in such cases is very near atmospheric pressure at subsonic velocities.

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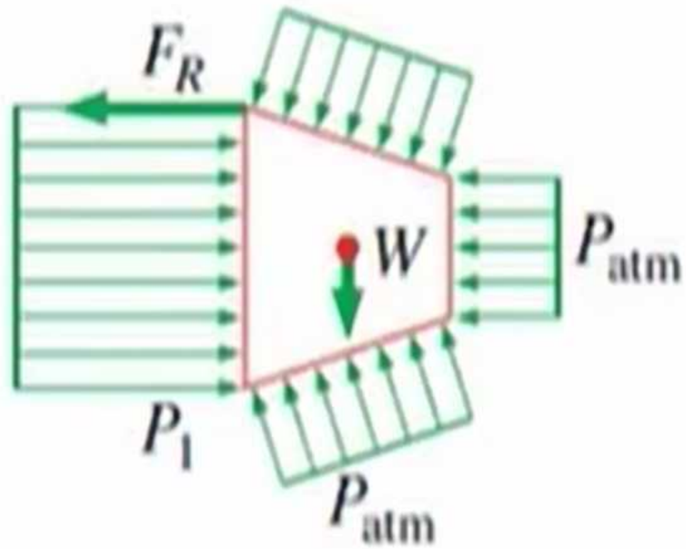
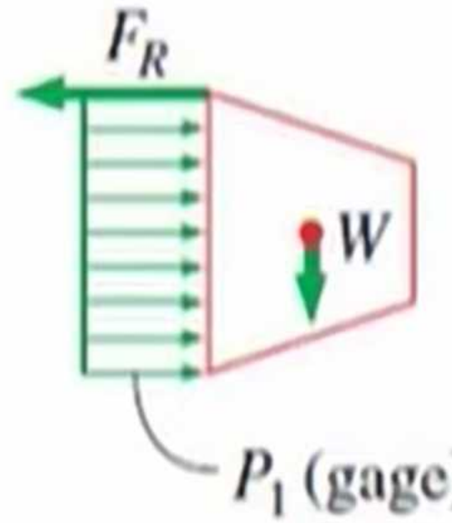
Forces Acting On A Control Volume

- **Atmospheric pressure acts in all directions, and thus it can be ignored when performing force balances since its effect cancels out in every direction.**



With atmospheric pressure considered



With atmospheric pressure cancelled out

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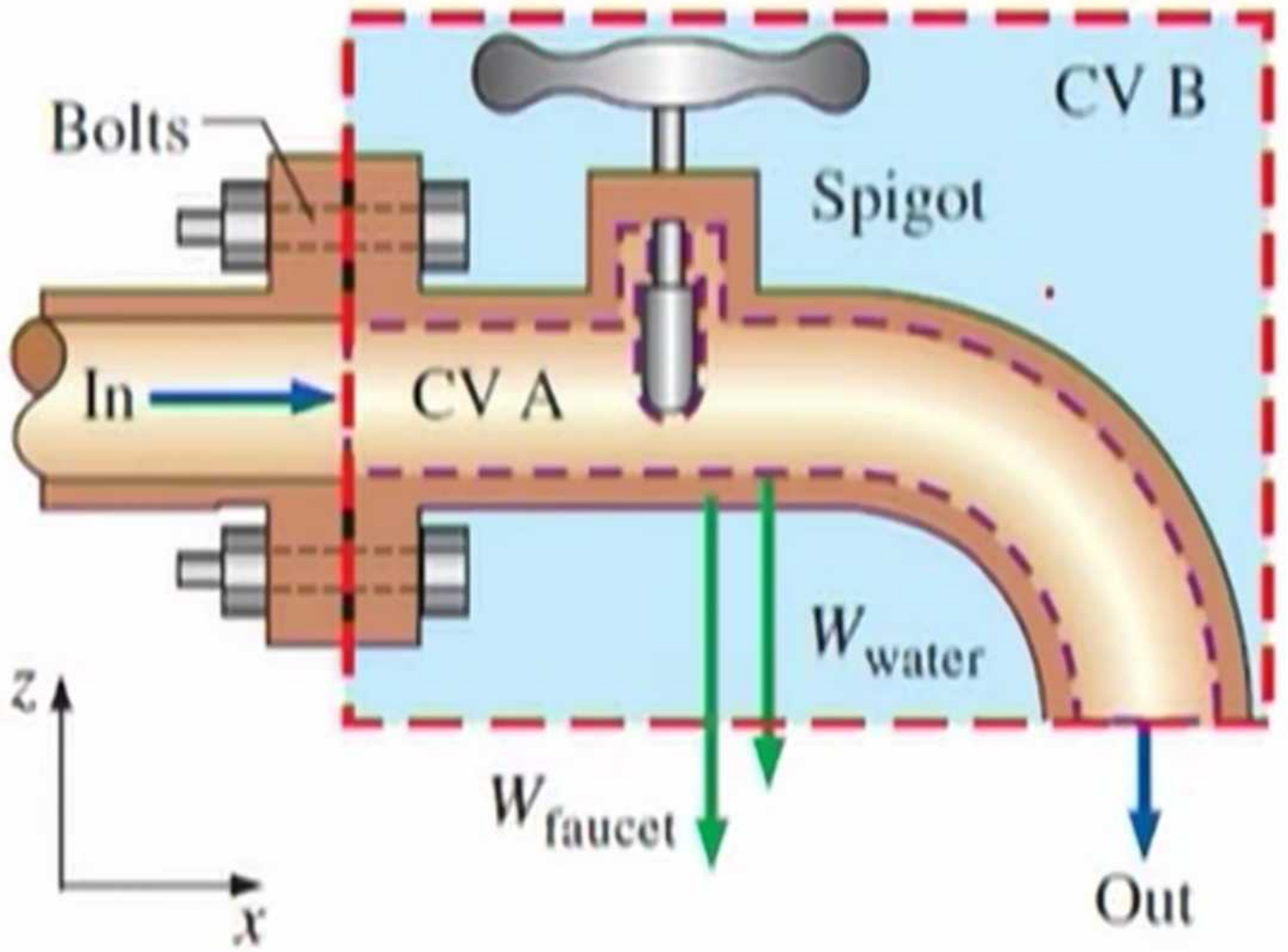
Forces Acting On A Control Volume

- **Cross section through a faucet assembly, illustrating the importance of choosing a control volume wisely; CV B is much easier to work with than CV A.**

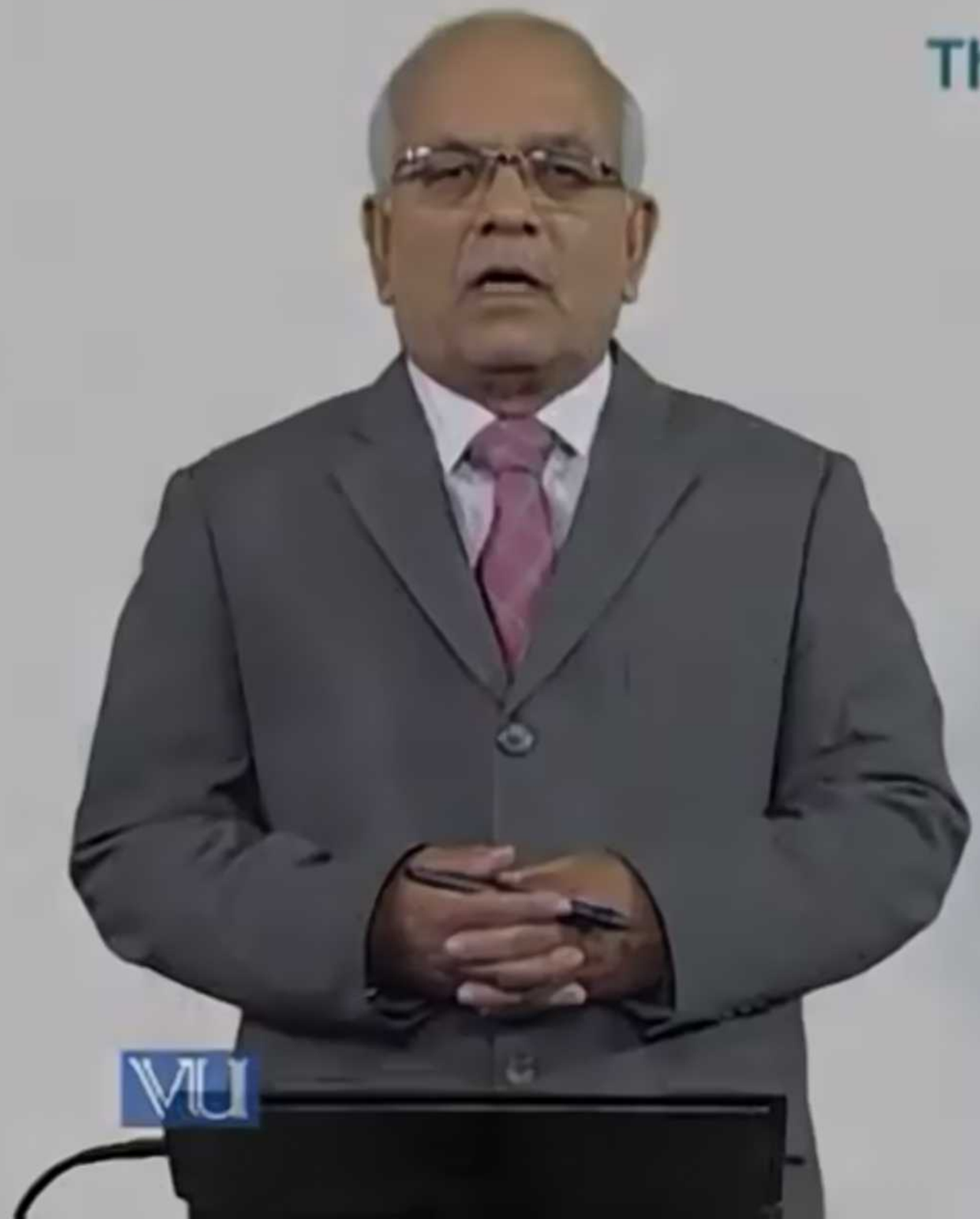


LECTURE 89

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

The Linear Momentum Equation



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The Linear Momentum Equation

Newton's second law for a system of mass m subjected to net force $\Sigma\vec{F}$ is expressed as

$$\sum \vec{F} = m\vec{a} = m\frac{d\vec{V}}{dt} = \frac{d}{dt}(m\vec{V}) \qquad \textbf{(6–13)}$$

where $m\vec{V}$ is the **linear momentum** of the system. Noting that both the density and velocity may change from point to point within the system, Newton's second law can be expressed more generally as

$$\sum \vec{F} = \frac{d}{dt}\int_{sys} \rho\vec{V}\, dV \qquad \textbf{(6–14)}$$

where $\rho\vec{V}\, dV$ is the momentum of a differential element dV, which has mass

Advanced Fluid Dynamics

The Linear Momentum Equation

$$\delta m = \rho dV$$

- **Newton's second law can be stated as**
- ***The sum of all external forces acting on a system is equal to the time rate of change of linear momentum of the system.***
- **This statement is valid for a coordinate system that is at rest or moves with a constant velocity, called an *inertial coordinate system* or *inertial reference frame*.**

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The Linear Momentum Equation

$$\frac{d(m\vec{V})_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\vec{V}\, d\forall + \int_{CS} \rho\vec{V}\,(\vec{V}_r \cdot \vec{n})\, dA$$

General:

$$\sum \vec{F} = \frac{d}{dt}\int_{CV} \rho\vec{V}\, d\forall + \int_{CS} \rho\vec{V}(\vec{V}_r \cdot \vec{n})\, dA$$

The sum of all external forces acting on a CV	=	The time rate of change of the linear momentum of the contents of the CV

+

The net flow rate of linear momentum out of the control surface by mass flow

Fixed CV:

$$\sum \vec{F} = \frac{d}{dt}\int_{CV} \rho\vec{V}\, d\forall + \int_{CS} \rho\vec{V}(\vec{V} \cdot \vec{n})\, dA$$

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The Linear Momentum Equation

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho b\, dV + \int_{CS} \rho b(\vec{V}_r \cdot \vec{n})\, dA$$

$B = m\vec{V}$ $\quad$ $b = \vec{V}$ $\quad$ $b = \vec{V}$

$$\frac{d(m\vec{V})_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\vec{V}\, dV + \int_{CS} \rho\vec{V}(\vec{V}_r \cdot \vec{n})\, dA$$

The linear momentum equation is obtained by replacing B in the Reynolds transport theorem by the momentum $m\vec{V}$, and b by the momentum per unit mass $\vec{V}$.

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The Linear Momentum Equation

The linear momentum equation is obtained by replacing B in the Reynolds transport theorem by the momentum $m\vec{V}$, and b by the momentum per unit mass $\vec{V}$.

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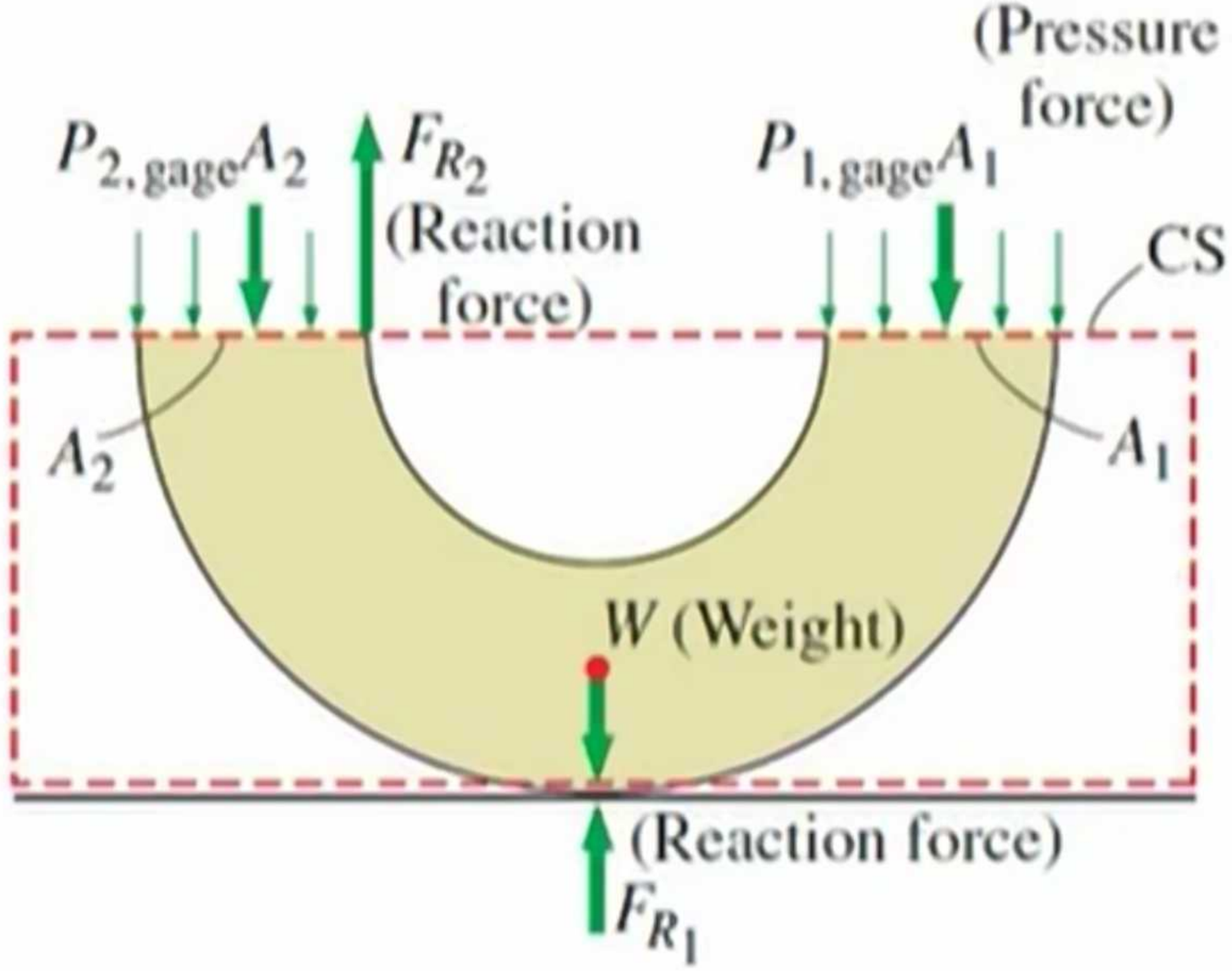
The Linear Momentum Equation



An 180° elbow supported by the ground

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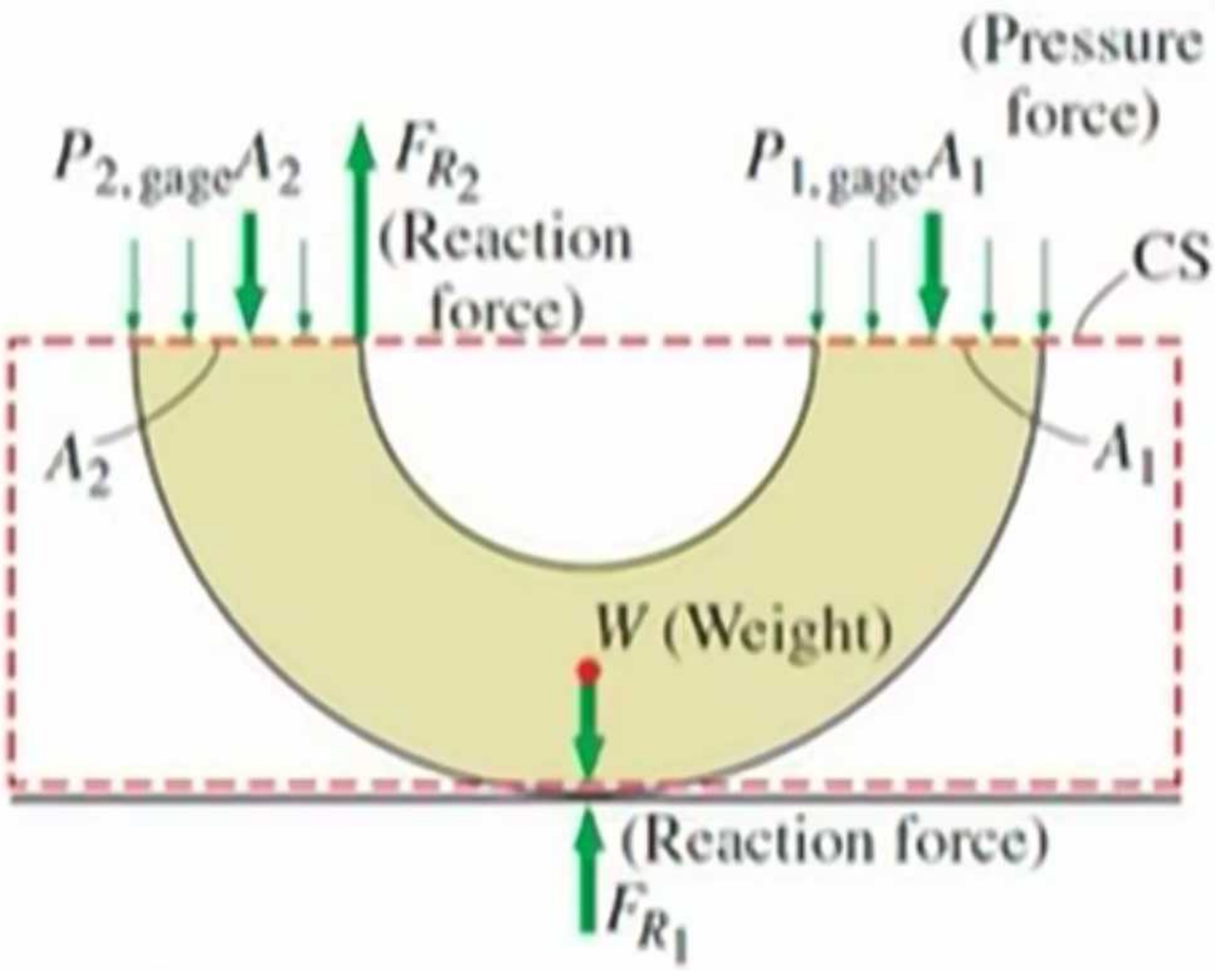
The Linear Momentum Equation



An 180° elbow supported by the ground

- **The momentum equation is commonly used to calculate the forces (usually on support systems or connectors) induced by the flow.**

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The Linear Momentum Equation

In most flow systems, the sum of forces $\Sigma\vec{F}$ consists of weights, pressure forces, and reaction forces. Gage pressures are used here since atmospheric pressure cancels out on all sides of the control surface.

$$\sum \vec{F} = \frac{d}{dt}\int_{CV} \rho\vec{V}\, d\mathcal{V} + \int_{CS} \rho\vec{V}(\vec{V}\cdot\vec{n})\, dA$$

Advanced Fluid Dynamics

The Linear Momentum Equation

Special Cases

Steady flow

$$\sum \vec{F} = \int_{CS} \rho \vec{V} \, (\vec{V}_r \cdot \vec{n}) dA$$

$$\dot{m} = \int_{A_c} \rho(\vec{V} \cdot \vec{n}) dA_c = \rho V_{avg} A_c$$

- **Mass flow rate across an inlet or outlet**

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The Linear Momentum Equation

$$\int_{A_c} \rho \vec{V}(\vec{V} \cdot \vec{n}) dA_c = \rho V_{\text{avg}} A_c \vec{V}_{\text{avg}} = \dot{m} \vec{V}_{\text{avg}}$$

- **Momentum flow rate across a uniform inlet or outlet:**

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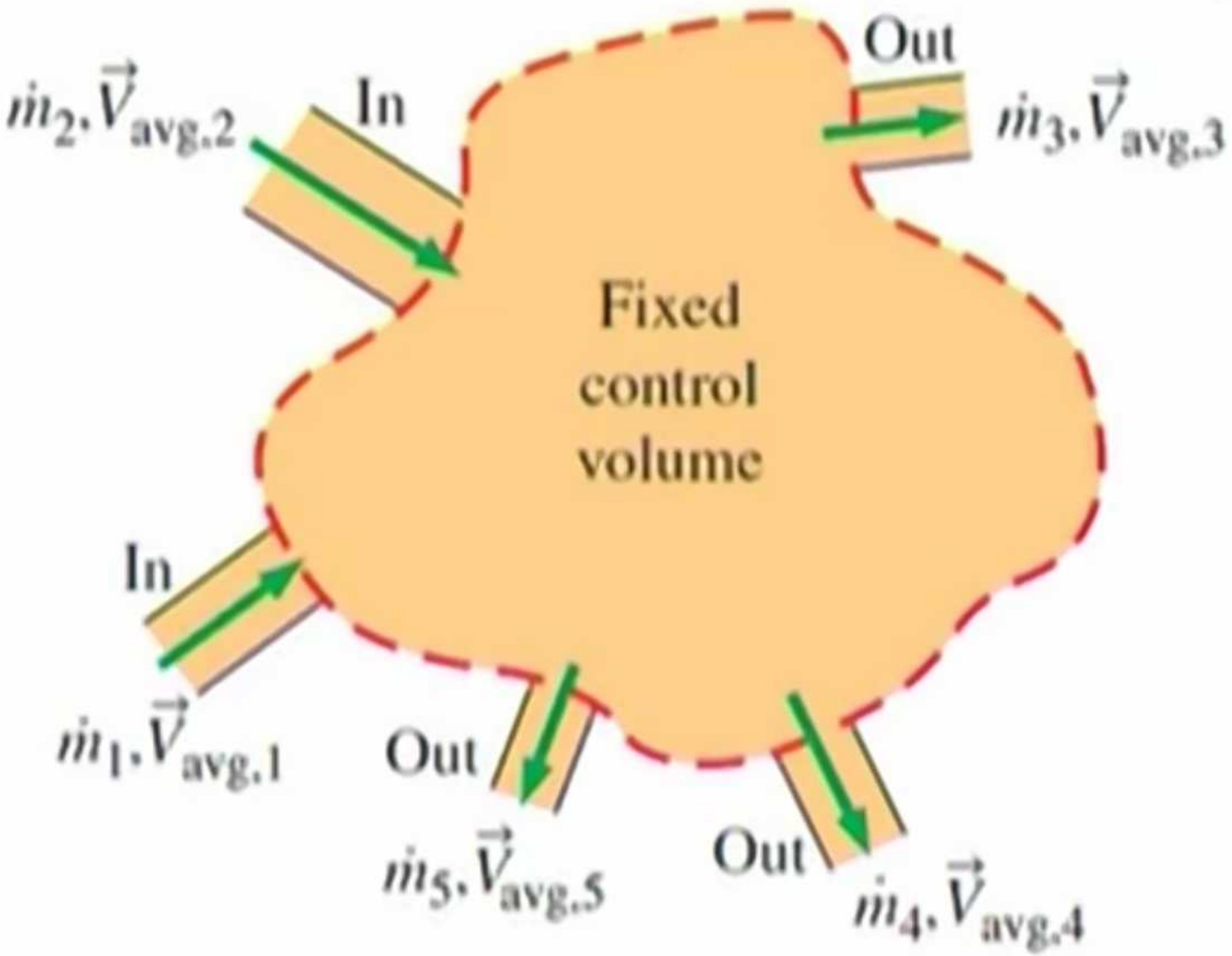
The Linear Momentum Equation



- **In a typical problem, the control volume may contain many inlets and outlets; at each inlet or outlet we define the mass flow rate and the average velocity.**

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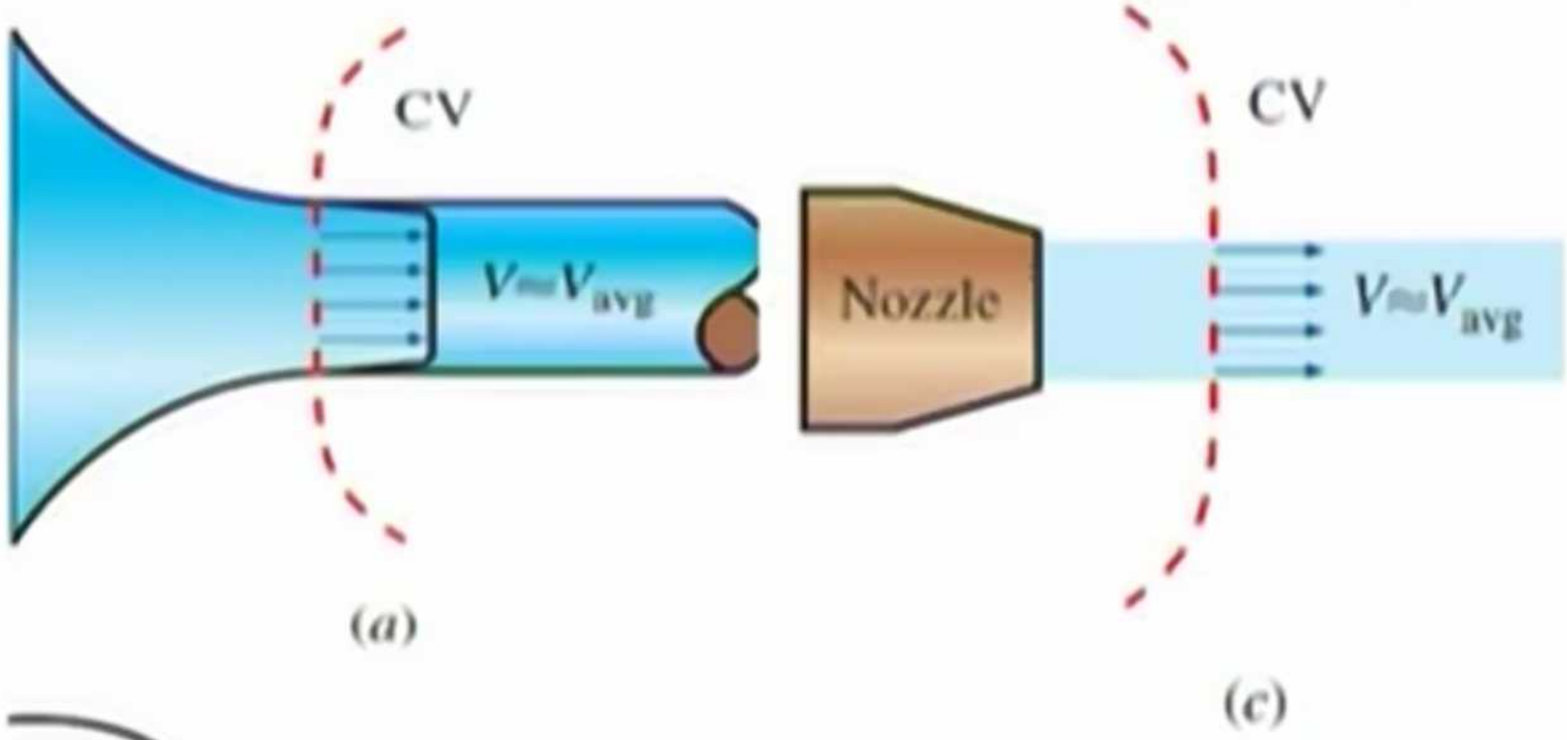
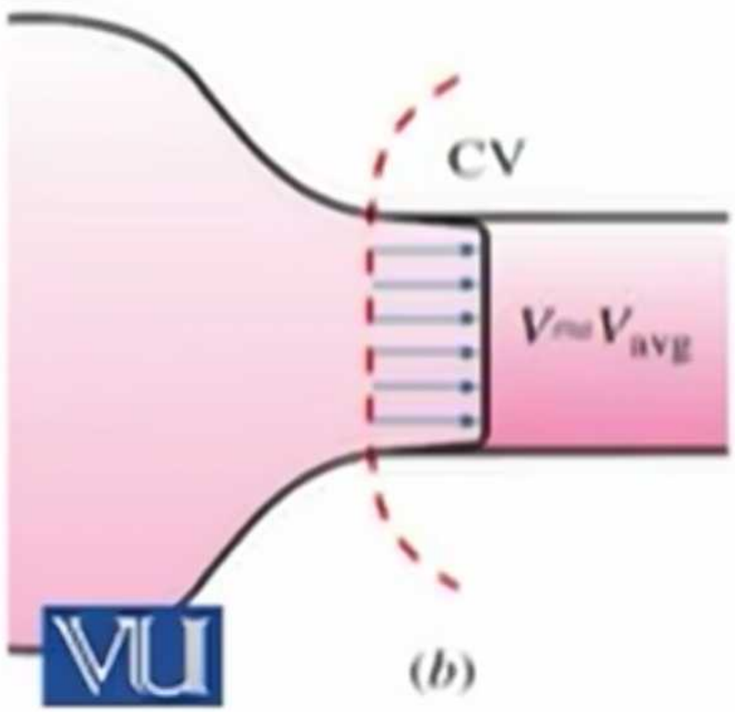
The Linear Momentum Equation





Examples of inlets or outlets in which the uniform flow approximation is reasonable: (*a*) the well-rounded entrance to a pipe,

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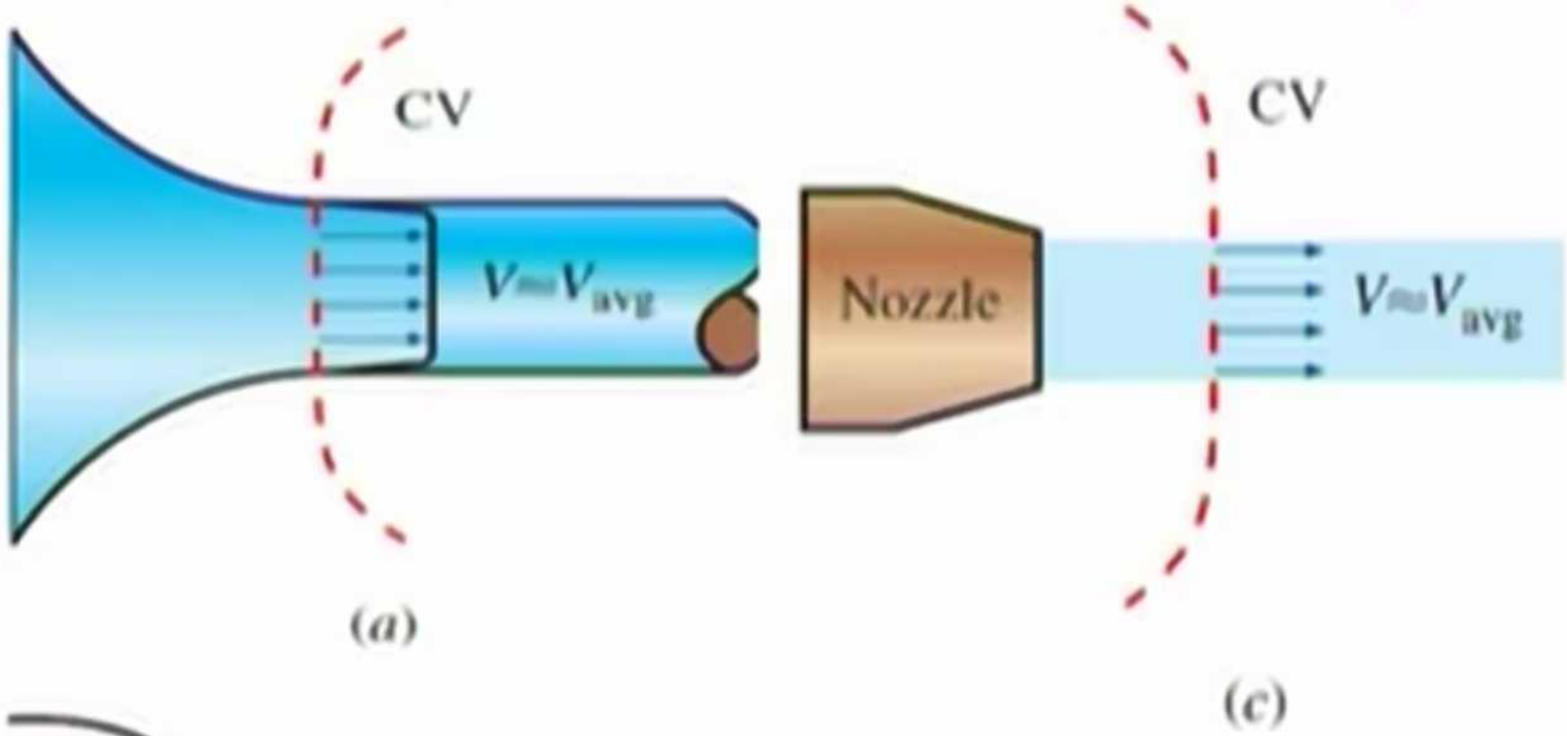
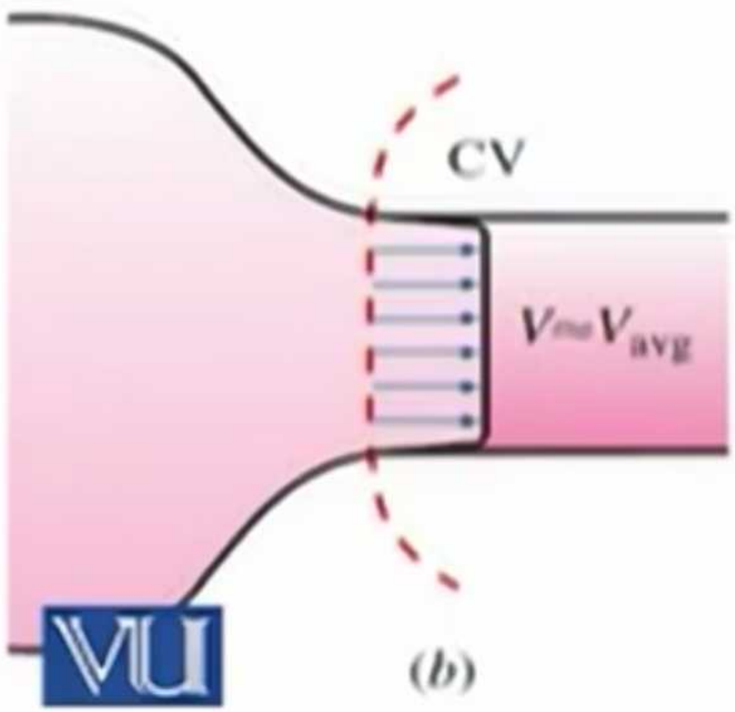
The Linear Momentum Equation





(*b*) the entrance to a wind tunnel test section, and
(*c*) a slice through a free water jet in air.

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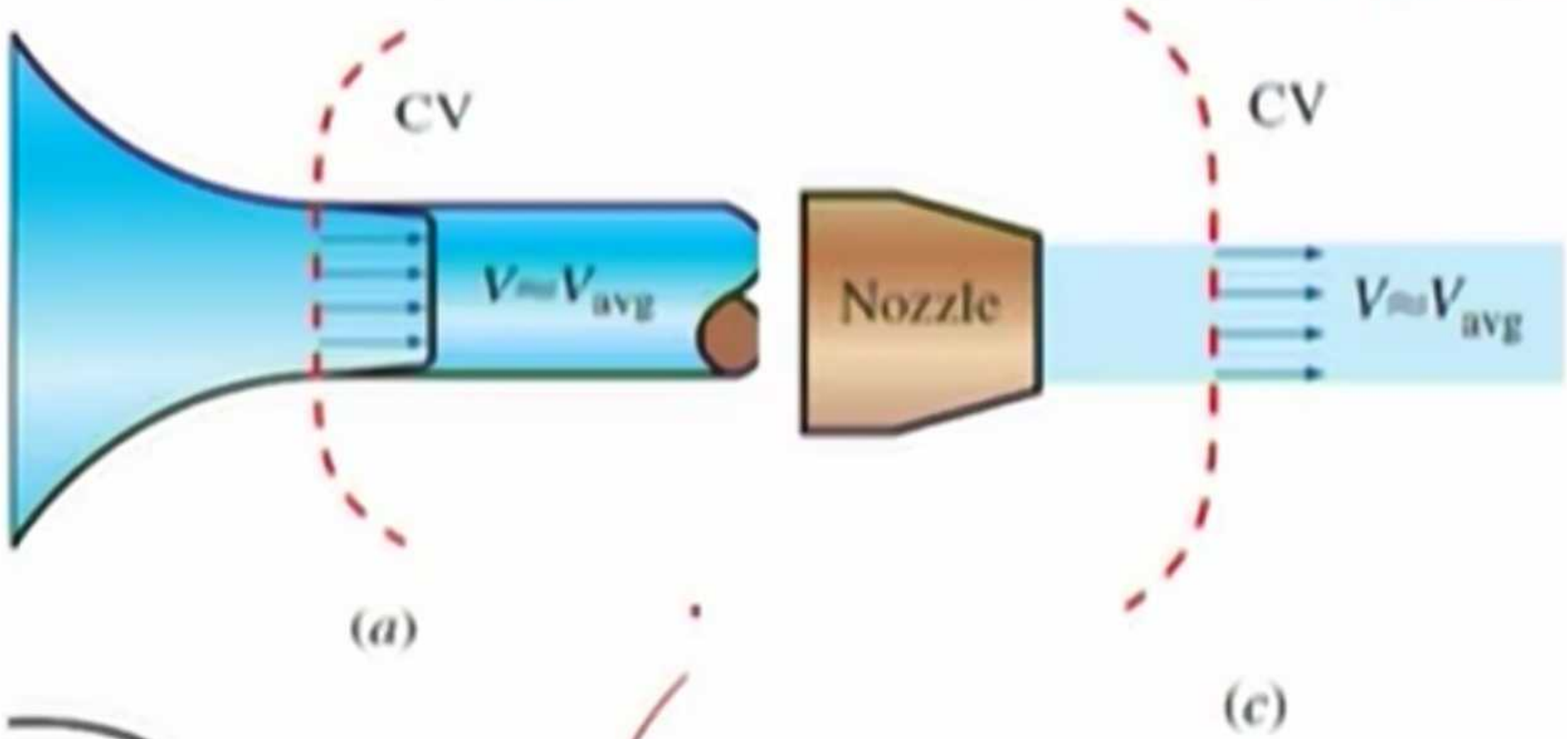
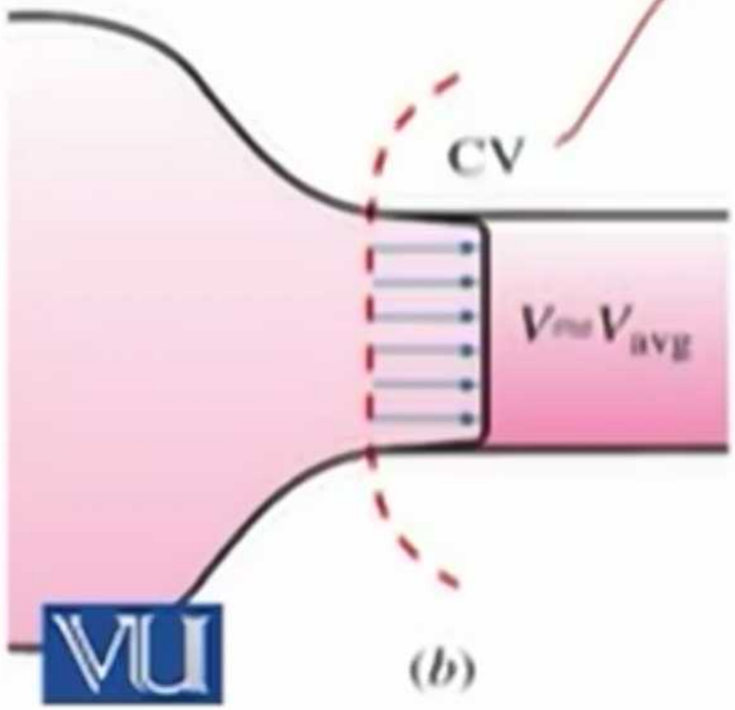
The Linear Momentum Equation





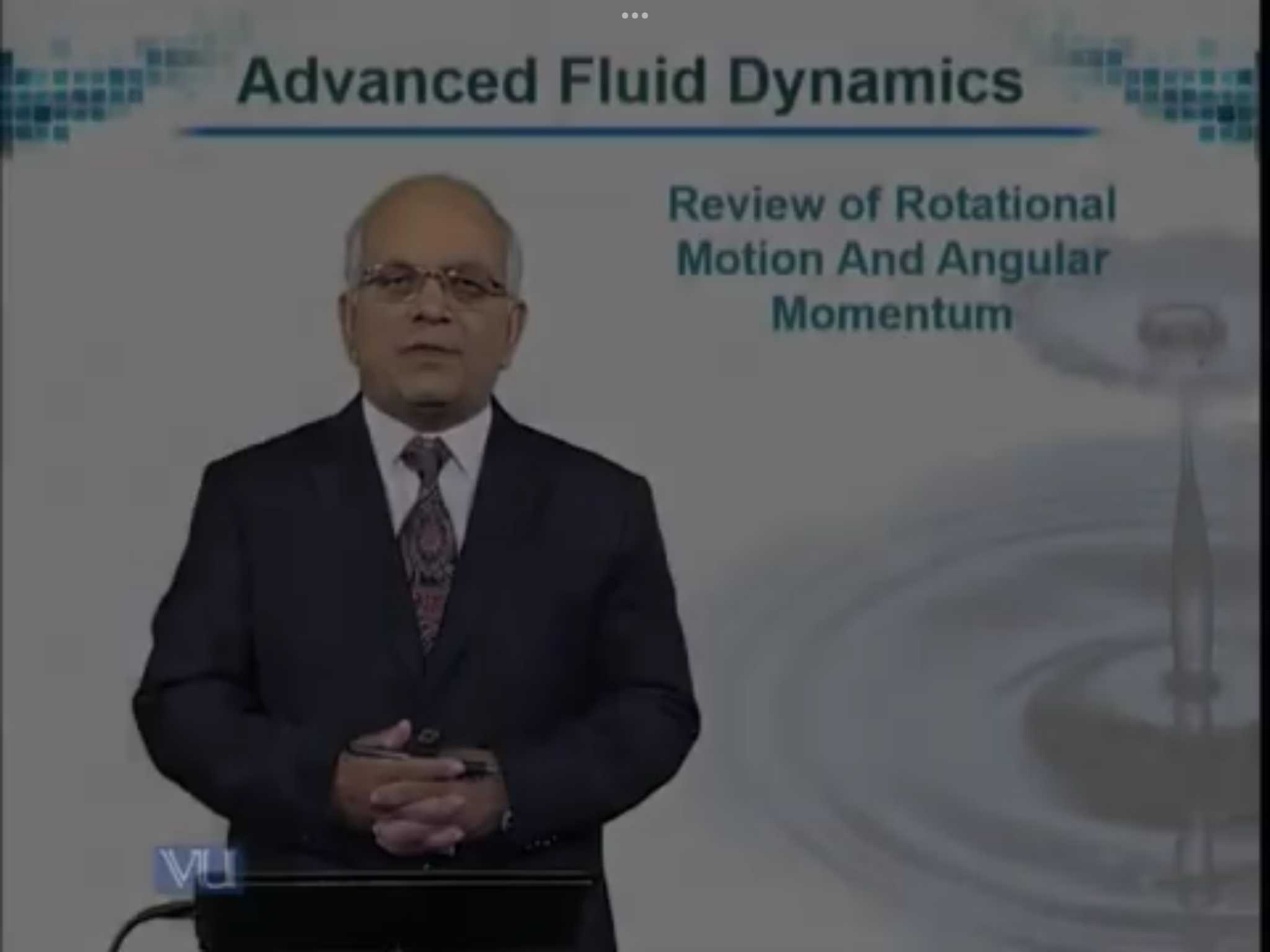
(*b*) the entrance to a wind tunnel test section, and
(*c*) a slice through a free water jet in air.

LECTURE 90

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Review of Rotational Motion And Angular Momentum



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Review of Rotational Motion And Angular Momentum

$$\omega = \frac{d\theta}{dt} = \frac{d(l/r)}{dt} = \frac{1}{r}\frac{dl}{dt} = \frac{V}{r}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = \frac{1}{r}\frac{dV}{dt} = \frac{a_t}{r}$$

$$V = r\omega \quad \text{and} \quad a_t = r\alpha$$

Rotational motion: A motion during which all points in the body move in circles about the axis of rotation.

- **Rotational motion is described with angular quantities such as the angular distance θ, angular velocity ω, and angular acceleration α.**

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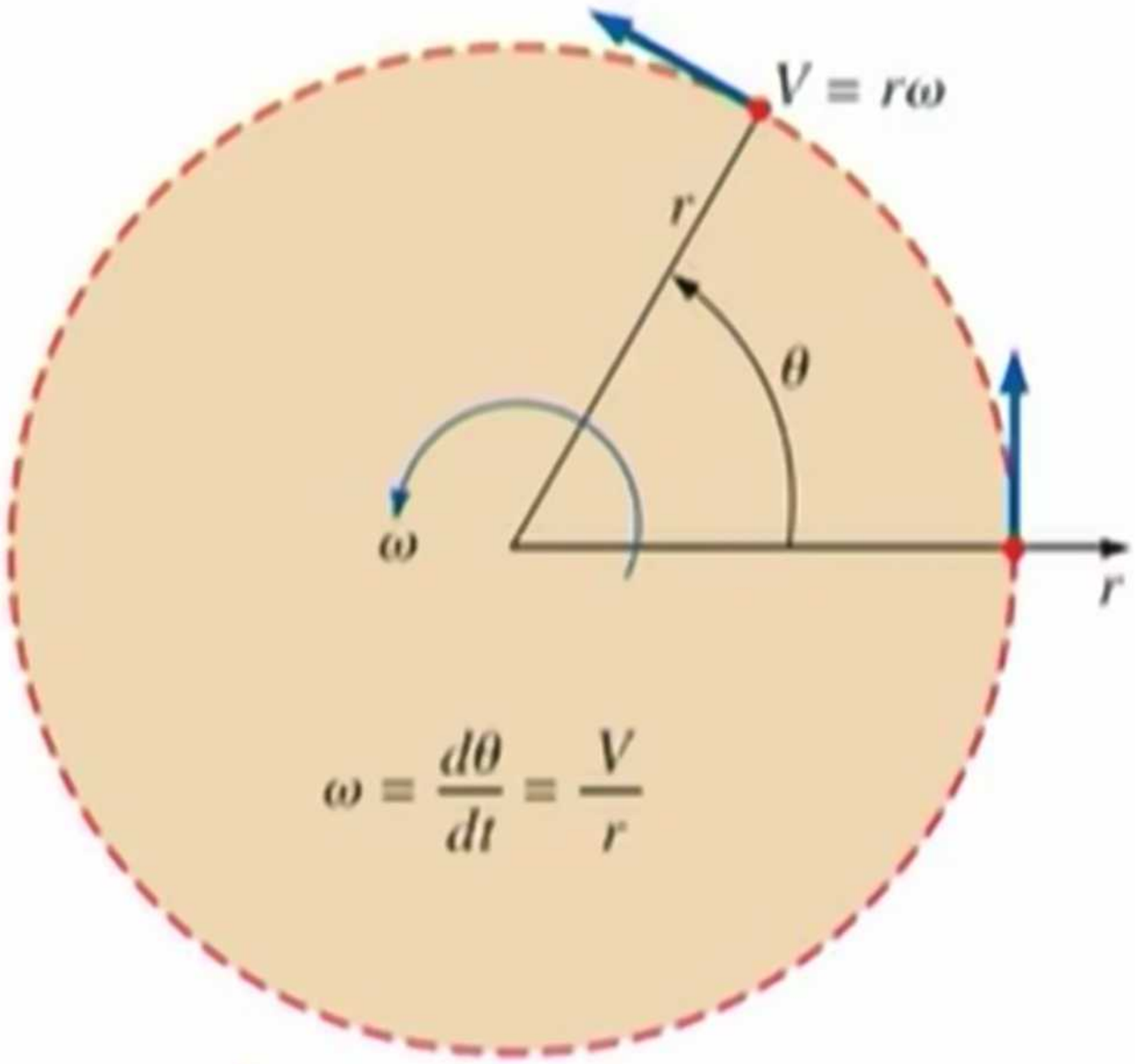
Review of Rotational Motion And Angular Momentum



The relations between angular distance θ, angular velocity ω, and linear velocity V.

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Review of Rotational Motion And Angular Momentum

- Newton's second law requires that there must be a force acting in the tangential direction to cause angular acceleration

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Review of Rotational Motion And Angular Momentum

- **The strength of the rotating effect, called the *moment* or *torque*, is proportional to the magnitude of the force and its distance from the axis of rotation.**

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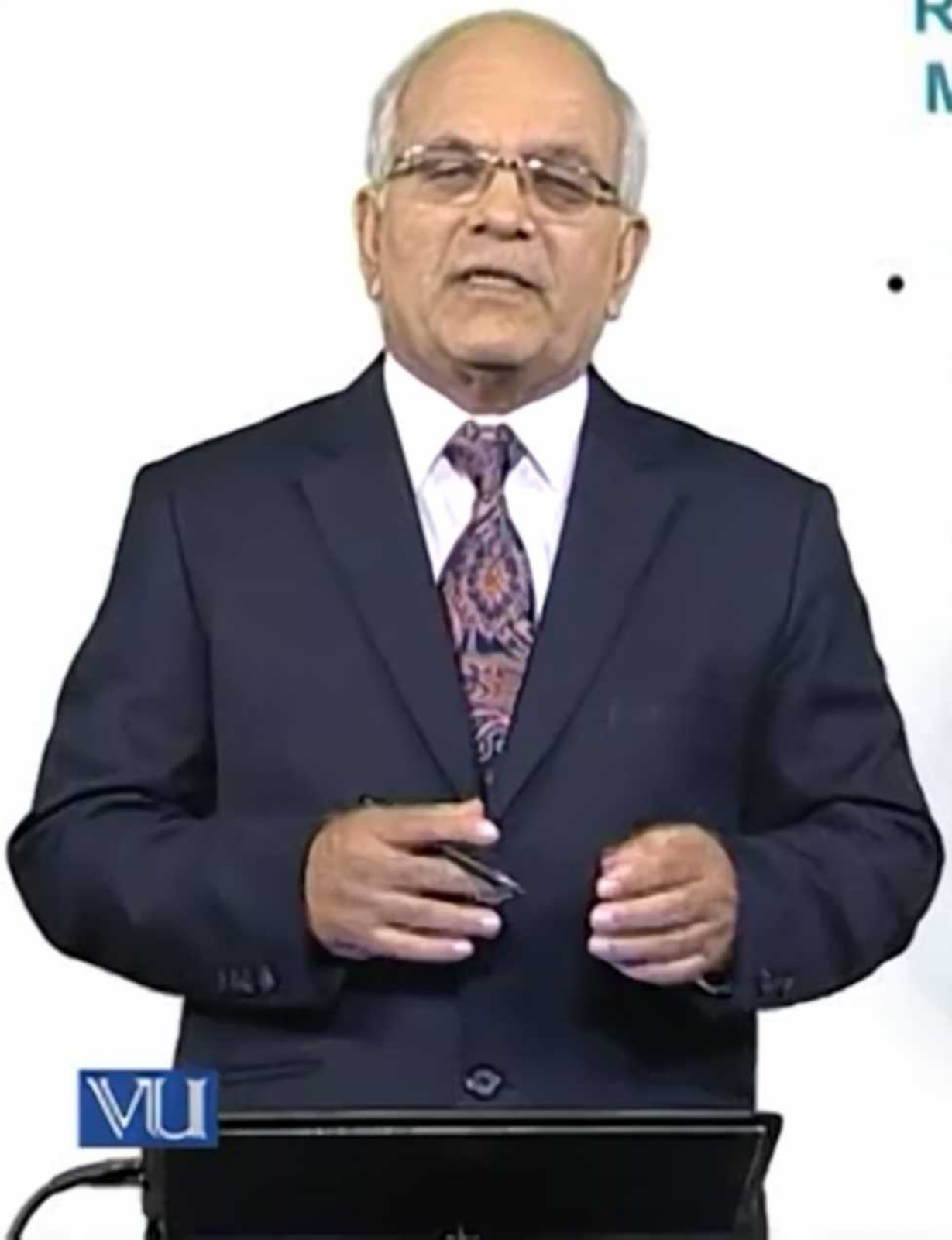

Review of Rotational Motion And Angular Momentum

- **The perpendicular distance from the axis of rotation to the line of action of the force is called the *moment arm*, and the torque *M* acting on a point mass *m* at a normal distance *r* from the axis of rotation is expressed as**

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Review of Rotational Motion And Angular Momentum

$$M = rF_t = rma_t = mr^2\alpha$$ Torque

$$M = \int_{\text{mass}} r^2\alpha\,\delta m = \left[\int_{\text{mass}} r^2\,\delta m\right]\alpha = I\alpha$$

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I **is the *moment of inertia* of the body about the axis of rotation, which is a measure of the inertia of a body against rotation.**

Mass, m	Moment of inertia, I
Linear acceleration, $\vec{a}$	Angular acceleration, $\vec{\alpha}$
Linear velocity, $\vec{V}$	Angular velocity, $\vec{\omega}$
Linear momentum	Angular momentum
$m\vec{V}$	$I\vec{\omega}$
Force, $\vec{F}$	Torque, $\vec{M}$
$\vec{F} = m\vec{a}$	$\vec{M} = I\vec{\alpha}$
Moment of force, $\vec{M}$	Moment of momentum, $\vec{H}$
$\vec{M} = \vec{r} \times \vec{F}$	$\vec{H} = \vec{r} \times m\vec{V}$

Analogy between corresponding linear and angular quantities.

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Unlike mass, the rotational inertia of a body also depends on the distribution of the mass of the body with respect to the axis of rotation.

Mass, m	Moment of inertia, I
Linear acceleration, $\vec{a}$	Angular acceleration, $\vec{\alpha}$
Linear velocity, $\vec{V}$	Angular velocity, $\vec{\omega}$
Linear momentum	Angular momentum
$m\vec{V}$	$I\vec{\omega}$
Force, $\vec{F}$	Torque, $\vec{M}$
$\vec{F} = m\vec{a}$	$\vec{M} = I\vec{\alpha}$
Moment of force, $\vec{M}$	Moment of momentum, $\vec{H}$
$\vec{M} = \vec{r} \times \vec{F}$	$\vec{H} = \vec{r} \times m\vec{V}$

Analogy between corresponding linear and angular quantities.

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Review of Rotational Motion And Angular Momentum

Angular momentum

$$H = \int_{\text{mass}} r^2 \omega \, \delta m = \left[\int_{\text{mass}} r^2 \, \delta m \right] \omega = I\omega \qquad \vec{H} = I\vec{\omega}$$

$$\vec{M} = I\vec{\alpha} = I \frac{d\vec{\omega}}{dt} = \frac{d(I\vec{\omega})}{dt} = \frac{d\vec{H}}{dt}$$

Angular momentum equation

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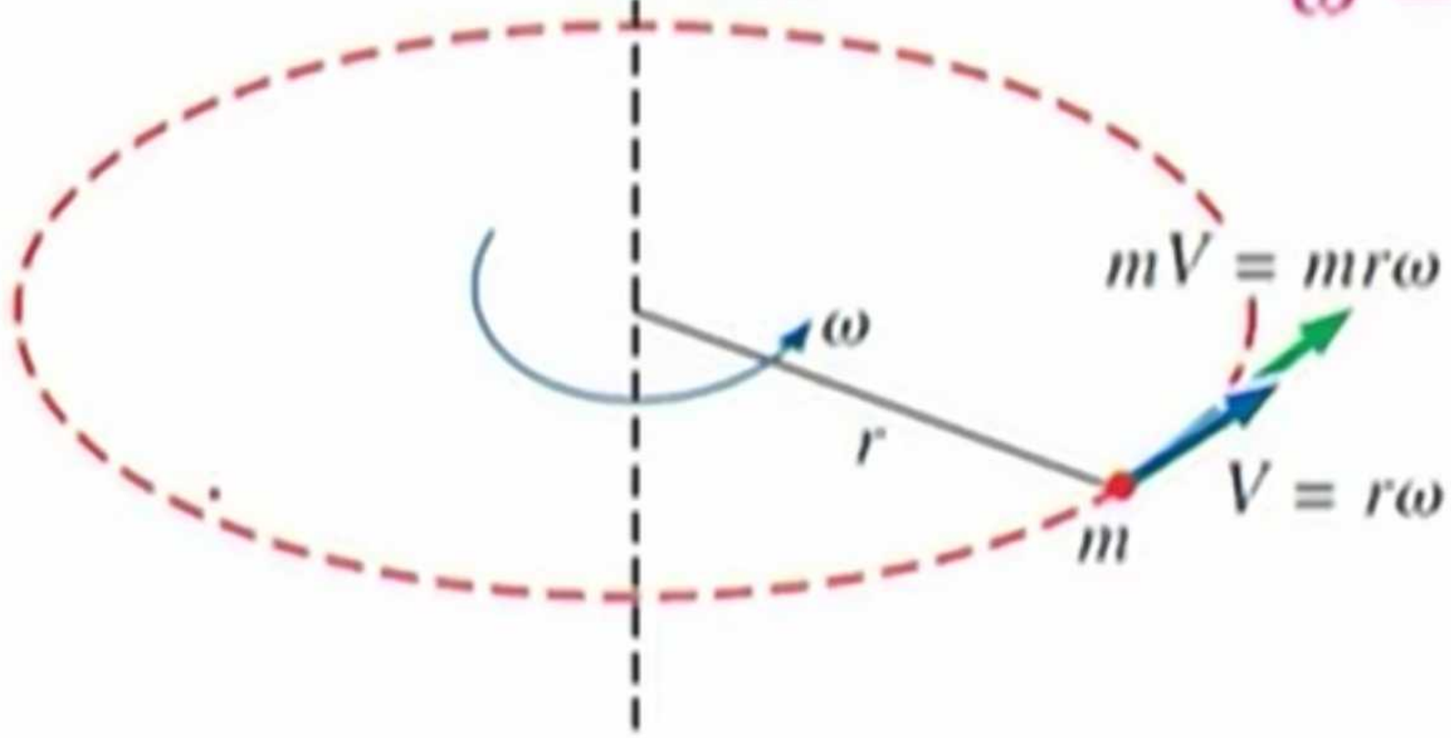
Review of Rotational Motion And Angular Momentum

$$H = rmV$$
$$= rm(r\omega)$$
$$= r^2 m\omega$$
$$= I\omega$$

$$\omega = \frac{2\pi \dot{n}}{60} \quad \text{(rad/s)}$$



Angular momentum of point mass *m* rotating at angular velocity ω at distance *r* from the axis of rotation.

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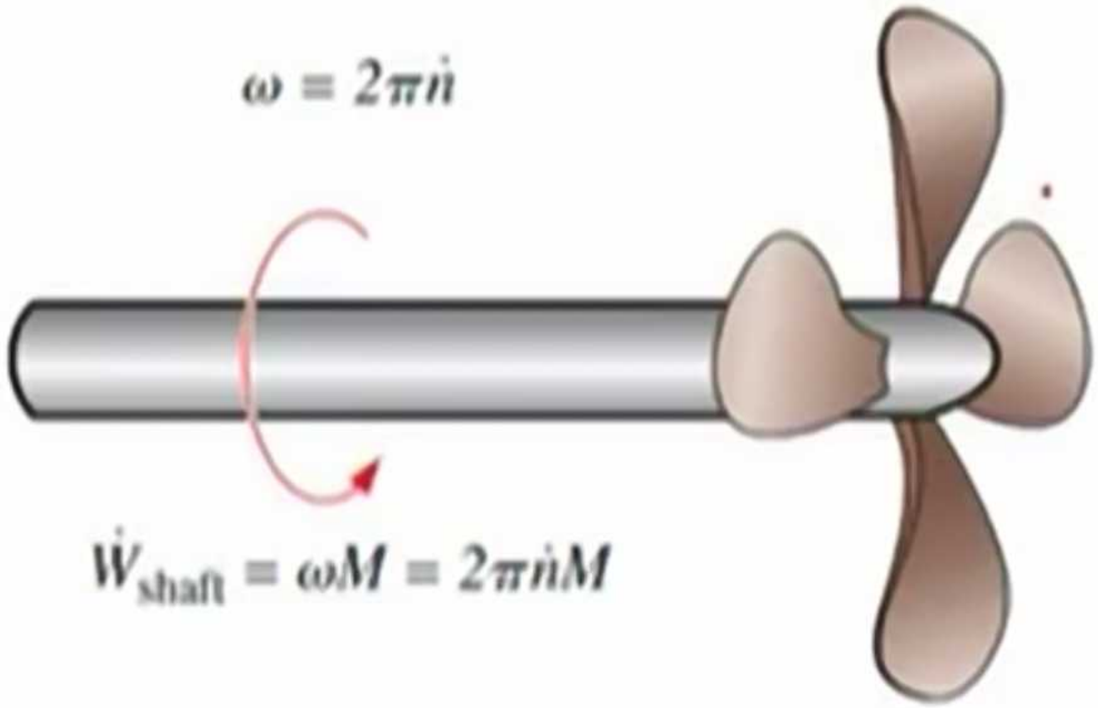
Review of Rotational Motion And Angular Momentum



Angular velocity versus rpm

$$\omega = \frac{2\pi\dot{n}}{60} \quad \text{(rad/s)}$$

Angular momentum of point mass *m* rotating at angular velocity ω at distance *r* from the axis of rotation.

The relations between angular velocity, rpm, and the power transmitted through a shaft.

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Review of Rotational Motion And Angular Momentum

$$\dot{W}_{\text{shaft}} = FV = Fr\omega = M\omega$$

$$\dot{W}_{\text{shaft}} = \omega M = 2\pi \dot{n} M \qquad (\text{W})$$ **Shaft power**

$$\text{KE}_r = \frac{1}{2} I \omega^2$$ **Rotational kinetic energy**

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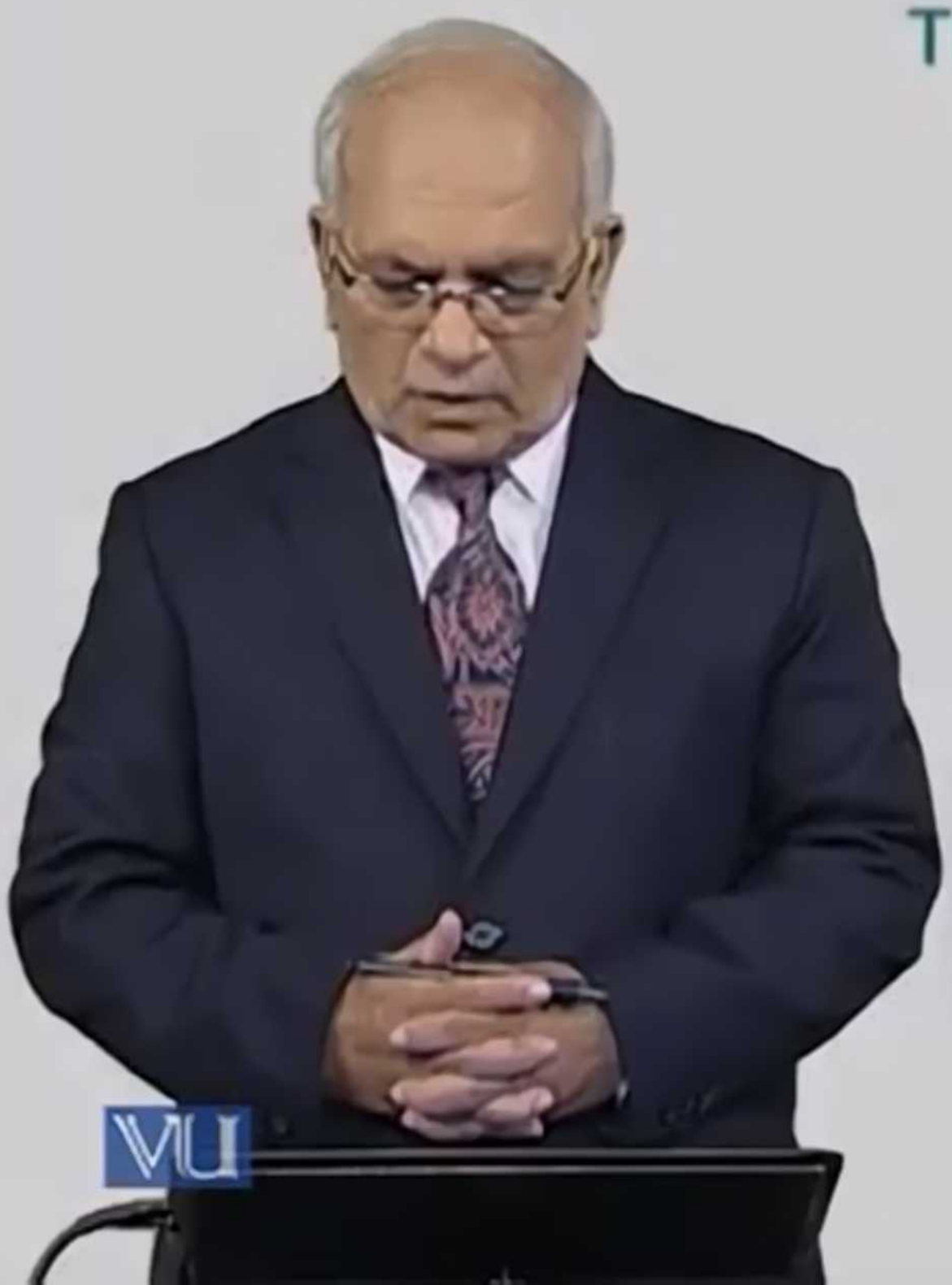

The Angular Momentum Equation

- Many engineering problems involve the moment of the linear momentum of flow streams, and the rotational effects caused by them.

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The Angular Momentum Equation

- **Such problems are best analyzed by the *angular momentum equation*, also called the *moment of momentum equation*.**

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The Angular Momentum Equation

- **An important class of fluid devices, called *turbomachines*, which include centrifugal pumps, turbines, and fans, is analyzed by the angular momentum equation.**

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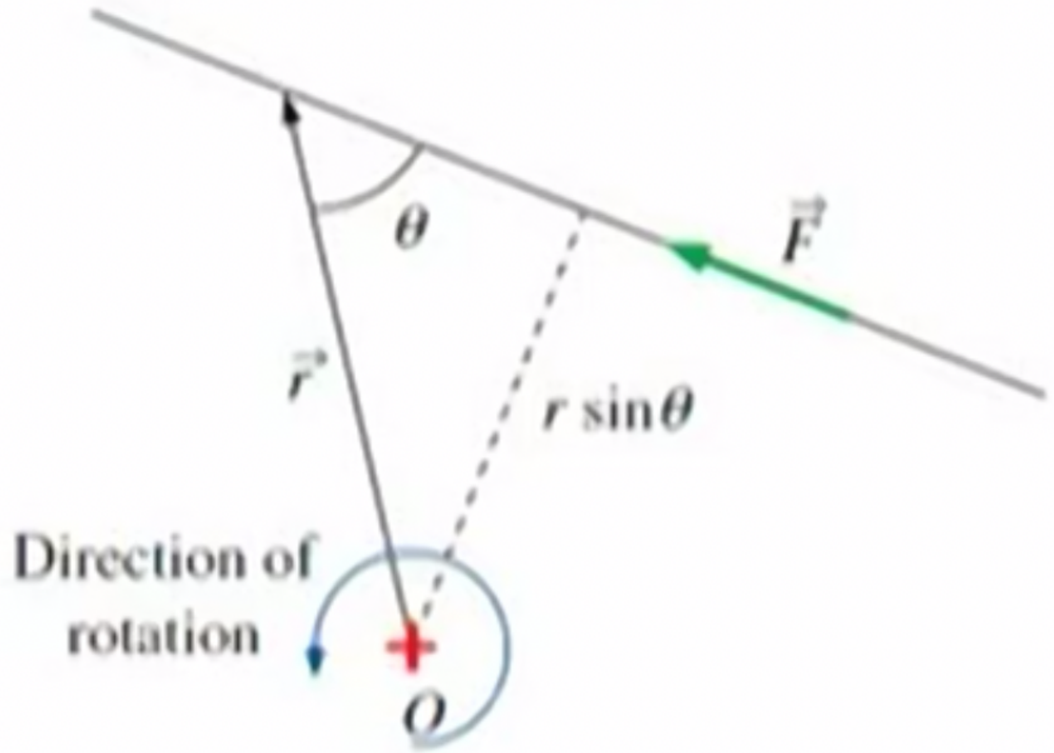
The Angular Momentum Equation



A force whose line of action passes through point *O* produces zero moment about point *O*.

$$\vec{M} = \vec{r} \times \vec{F}$$

$$M = Fr\sin\theta$$

The moment of a force $\vec{F}$ about a point O is the vector product of the position vector $\vec{r}$ and $\vec{F}$.

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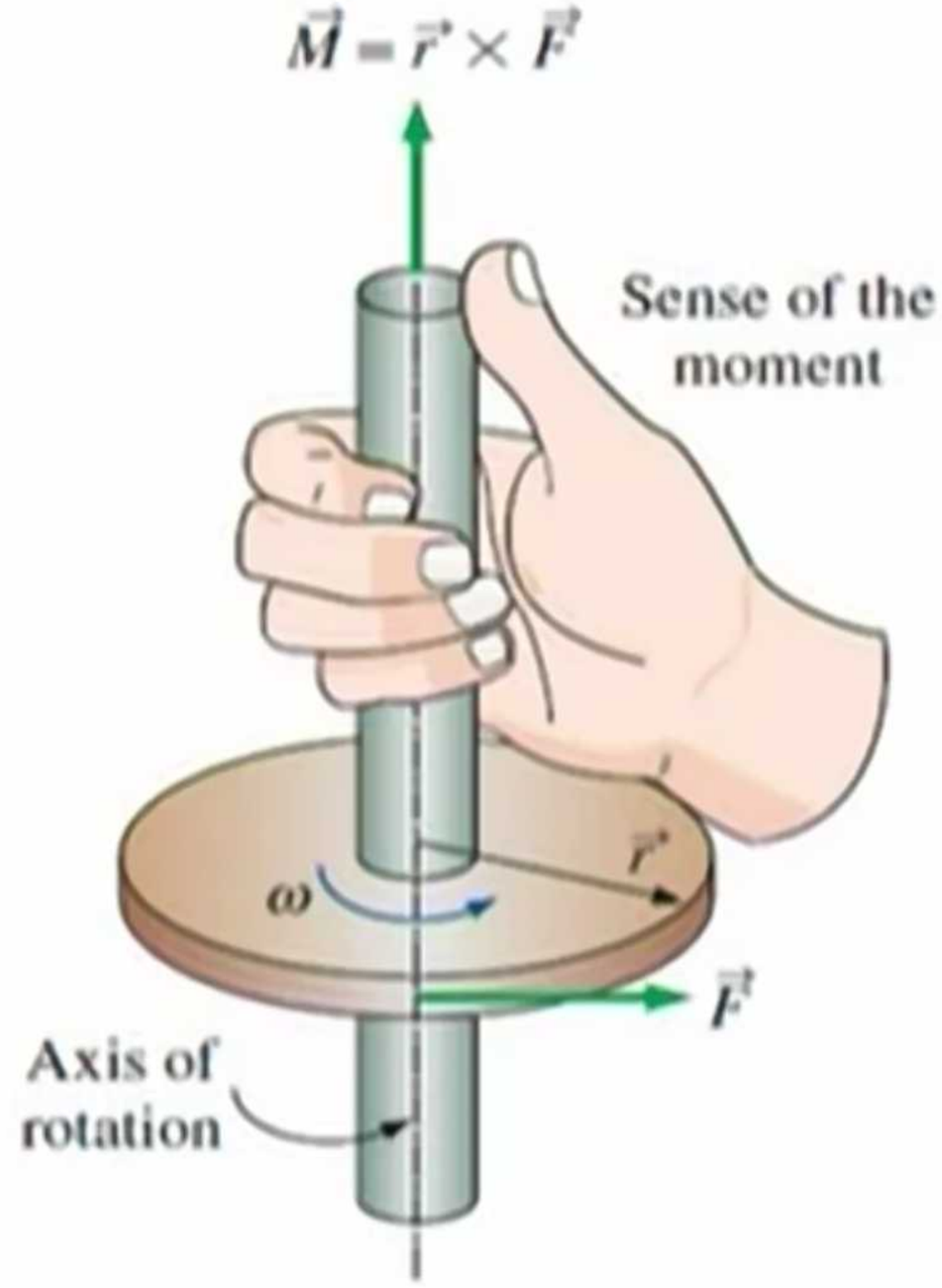
The Angular Momentum Equation

The determination of the direction of the moment by the right-hand rule.



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The Angular Momentum Equation

Moment of momentum

$$\vec{H} = \vec{r} \times m\vec{V}$$

Moment of momentum (system)

$$\vec{H}_{sys} = \int_{sys} (\vec{r} \times \vec{V})\rho \, d\forall$$

$$\frac{d\vec{H}_{sys}}{dt} = \frac{d}{dt}\int_{sys} (\vec{r} \times \vec{V})\rho \, d\forall$$

Rate of change of moment of momentum

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The Angular Momentum Equation

Rate of change of moment of momentum

$$\sum \vec{M} = \frac{d\vec{H}_{sys}}{dt} \qquad \sum \vec{M} = \sum (\vec{r} \times \vec{F})$$

$$\frac{d\vec{H}_{sys}}{dt} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho (\vec{V}_r \cdot \vec{n}) \, dA$$

The angular momentum equation is obtained by replacing B in the Reynolds transport theorem by the angular momentum $\vec{H}$, and b by the angular momentum per unit mass $\vec{r} \times \vec{V}$.

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The Angular Momentum Equation

General:

$$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V}_r \cdot \vec{n})dA$$

$$\begin{pmatrix}\text{The sum of all}\\ \text{external moments}\\ \text{acting on a CV}\end{pmatrix} = \begin{pmatrix}\text{The time rate of change}\\ \text{of the angular momentum}\\ \text{of the contents of the CV}\end{pmatrix} + \begin{pmatrix}\text{The net flow rate of}\\ \text{angular momentum}\\ \text{out of the control}\\ \text{surface by mass flow}\end{pmatrix}$$

Fixed CV:

$$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V} \cdot \vec{n})dA$$

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The Angular Momentum Equation

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho b\, dV + \int_{CS} \rho b (V_r \cdot \vec{n})\, dA$$

$B = \vec{H}$ $\quad b = \vec{r} \times \vec{V}$ $\quad b = \vec{r} \times \vec{V}$

$$\frac{d\vec{H}_{sys}}{dt} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho\, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V}_r \cdot \vec{n})\, dA$$

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The Angular Momentum Equation

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The Angular Momentum Equation

Special Cases

A rotating lawn sprinkler is a good example of application of the angular momentum equation.

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The Angular Momentum Equation

Steady flow:

$$\sum \vec{M} = \int_{CS} (\vec{r} \times \vec{V}) \rho (\vec{V}_r \cdot \vec{n}) \, dA$$

An approximate form of the angular momentum equation in terms of average properties at inlets and outlets:

$$\sum \vec{M} = \frac{d}{dt} \int_{CV} (\vec{r} \times \vec{V}) \rho \, dV + \sum_{out} \vec{r} \times \dot{m} \vec{V} - \sum_{in} \vec{r} \times \dot{m} \vec{V}$$

Steady flow:

$$\sum \vec{M} = \sum_{out} \vec{r} \times \dot{m} \vec{V} - \sum_{in} \vec{r} \times \dot{m} \vec{V}$$

$$\sum M = \sum_{out} r \dot{m} V - \sum_{in} r \dot{m} V$$

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The Angular Momentum Equation

$$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \sum_{out} \vec{r} \times \dot{m}\vec{V} - \sum_{in} \vec{r} \times \dot{m}\vec{V}$$

Steady flow:

$$\sum \vec{M} = \sum_{out} \vec{r} \times \dot{m}\vec{V} - \sum_{in} \vec{r} \times \dot{m}\vec{V}$$

The net torque acting on the control volume during steady flow is equal to the difference between the outgoing and incoming angular momentum flow rates.

$$\sum M = \sum_{out} r\dot{m}V - \sum_{in} r\dot{m}V$$

scalar form of angular momentum equation

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The Angular Momentum Equation

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The Angular Momentum Equation

Steady flow:

$$\sum \vec{M} = \int_{CS} (\vec{r} \times \vec{V}) \rho (\vec{V}_r \cdot \vec{n}) \, dA$$

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$$\sum M = \sum_{out} r \dot{m} V - \sum_{in} r \dot{m} V$$

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The Angular Momentum Equation

$$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \sum_{out} \vec{r} \times \dot{m}\vec{V} - \sum_{in} \vec{r} \times \dot{m}\vec{V}$$

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The net torque acting on the control volume during steady flow is equal to the difference between the outgoing and incoming angular momentum flow rates.

$$\sum M = \sum_{out} r\dot{m}V - \sum_{in} r\dot{m}V$$

scalar form of angular momentum equation

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Flow with No External Moments

$$\textit{No external moments:} \qquad 0 = \frac{d\vec{H}_{CV}}{dt} + \sum_{out} (\vec{r} \times \dot{m}\vec{V}) - \sum_{in} (\vec{r} \times \dot{m}\vec{V})$$

In the absence of external moments, the rate of change of the angular momentum of a control volume is equal to the difference between the incoming and outgoing angular momentum fluxes.

When the moment of inertia *I* of the control volume remains constant, the irst term on the right side of the above equation becomes simply moment of inertia times angular acceleration. Therefore, the control volume in this case can be treated as a solid body, with a net torque of

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Flow with No External Moments

$$\vec{M}_{\text{body}} = I_{\text{body}}\vec{\alpha} = \sum_{\text{in}}(\vec{r}\times\dot{m}\vec{V}) - \sum_{\text{out}}(\vec{r}\times\dot{m}\vec{V})$$

- **This approach can be used to determine the angular acceleration of space vehicles and aircraft when a rocket is fired in a direction different than the direction of motion.**

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Radial-Flow Devices

- **Radial-flow devices: Many rotary-flow devices such as centrifugal pumps and fans involve flow in the radial direction normal to the axis of rotation.**
- **Axial-flow devices are easily analyzed using the linear momentum equation.**
- **Radial-flow devices involve large changes in angular momentum of the fluid and are best analyzed with the help of the angular momentum equation.**

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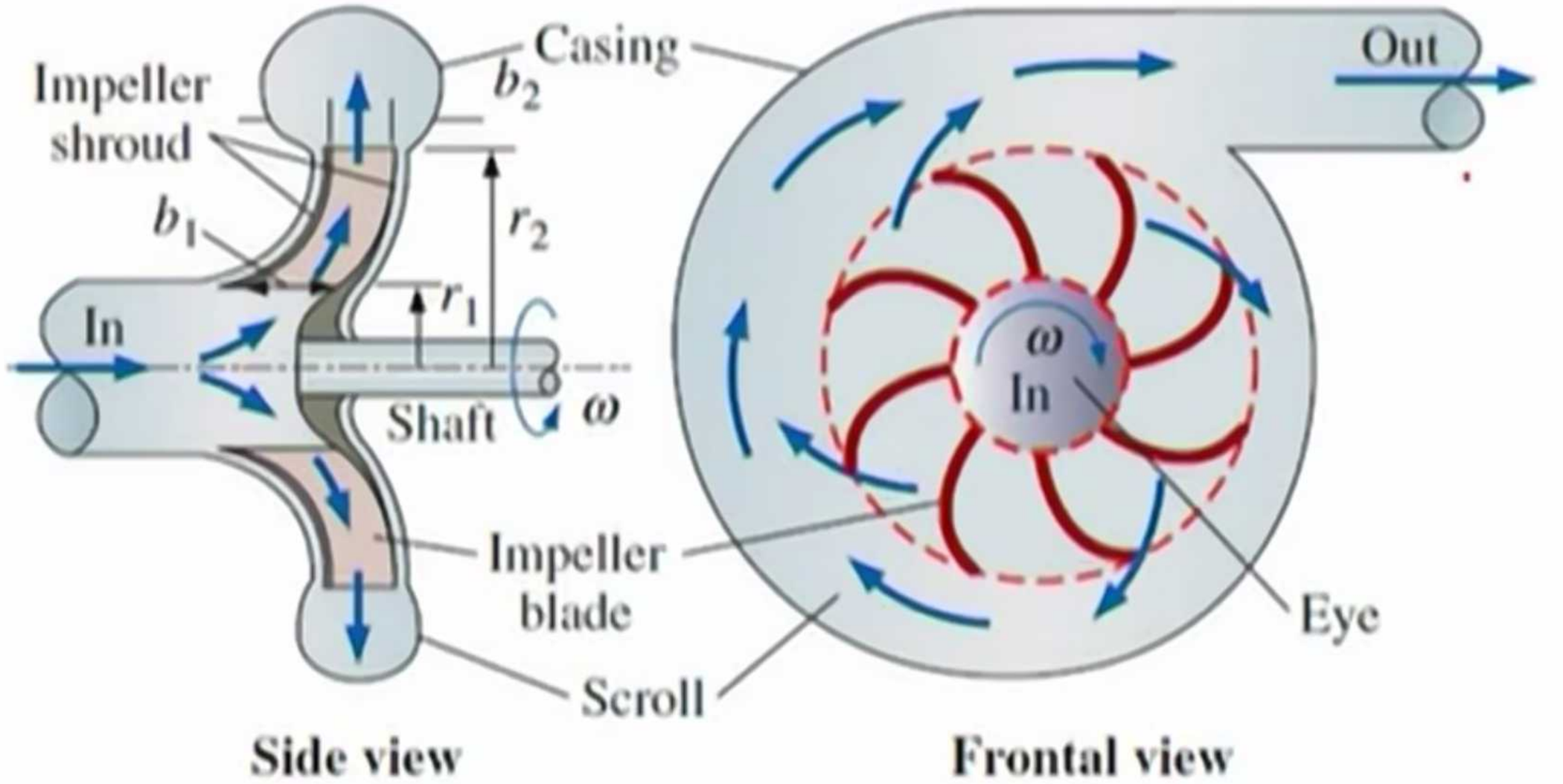
Radial-Flow Devices



Side and frontal views of a typical centrifugal pump.

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Radial-Flow Devices

The conservation of mass equation for steady incompressible flow

$$\dot{V}_1 = \dot{V}_2 = \dot{V} \quad \rightarrow \quad (2\pi r_1 b_1)V_{1,n} = (2\pi r_2 b_2)V_{2,n}$$

$$V_{1,n} = \frac{\dot{V}}{2\pi r_1 b_1} \quad \text{and} \quad V_{2,n} = \frac{\dot{V}}{2\pi r_2 b_2}$$

$$\sum M = \sum_{\text{out}} rmV - \sum_{\text{in}} rmV$$ **angular momentum equation**

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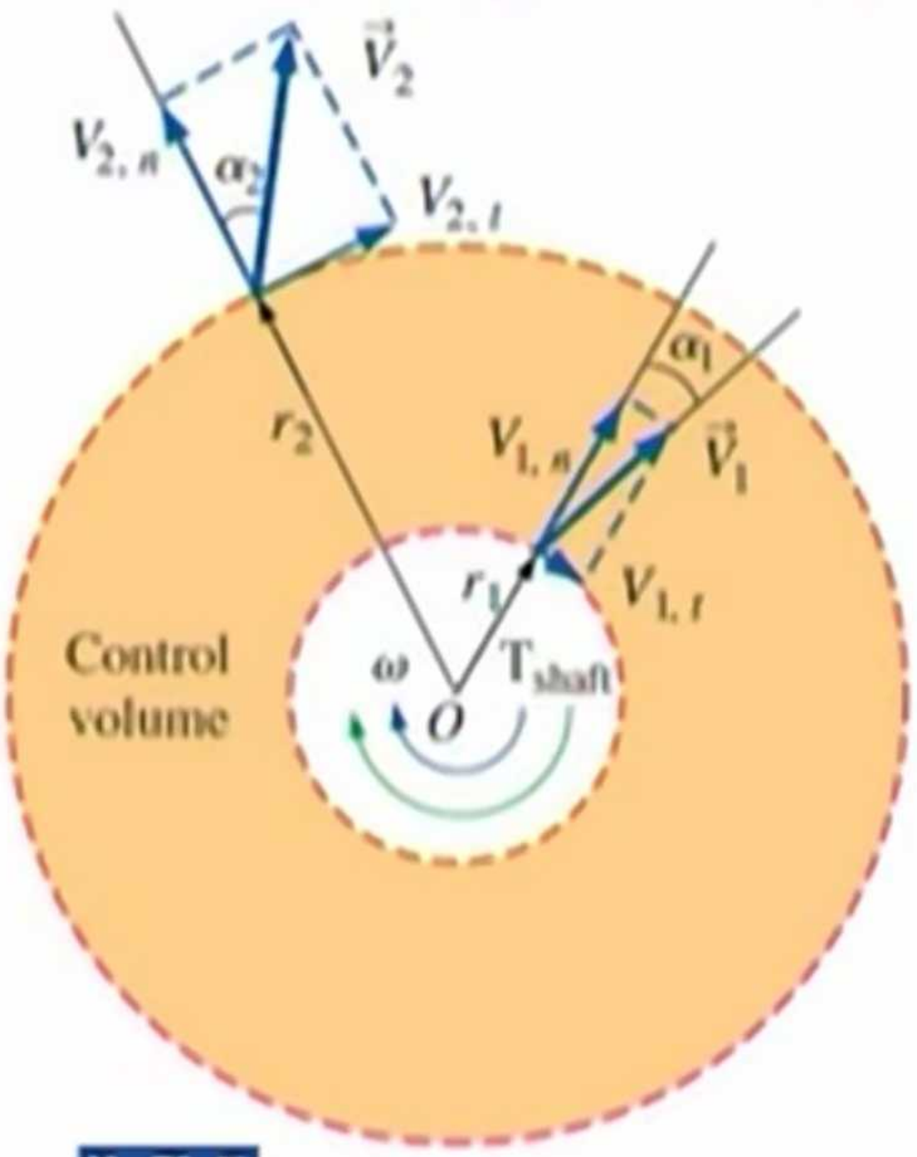
Radial-Flow Devices

$$\dot{T}_{shaft} = \dot{m}(r_2 V_{2,t} - r_1 V_{1,t})$$ **Euler's turbine formula**



$$\dot{W}_{shaft} = \dot{m}(r_2 V_2 \sin\alpha_2 - r_1 V_1 \sin\alpha_1)$$

$$\dot{T}_{shaft,\,ideal} = \dot{m}\omega(r_2^2 - r_1^2)$$

$$\dot{W}_{shaft} = \omega \dot{T}_{shaft} = 2\pi \dot{n} \dot{T}_{shaft}.$$

An annular control volume that encloses the impeller section of a centrifugal pump.

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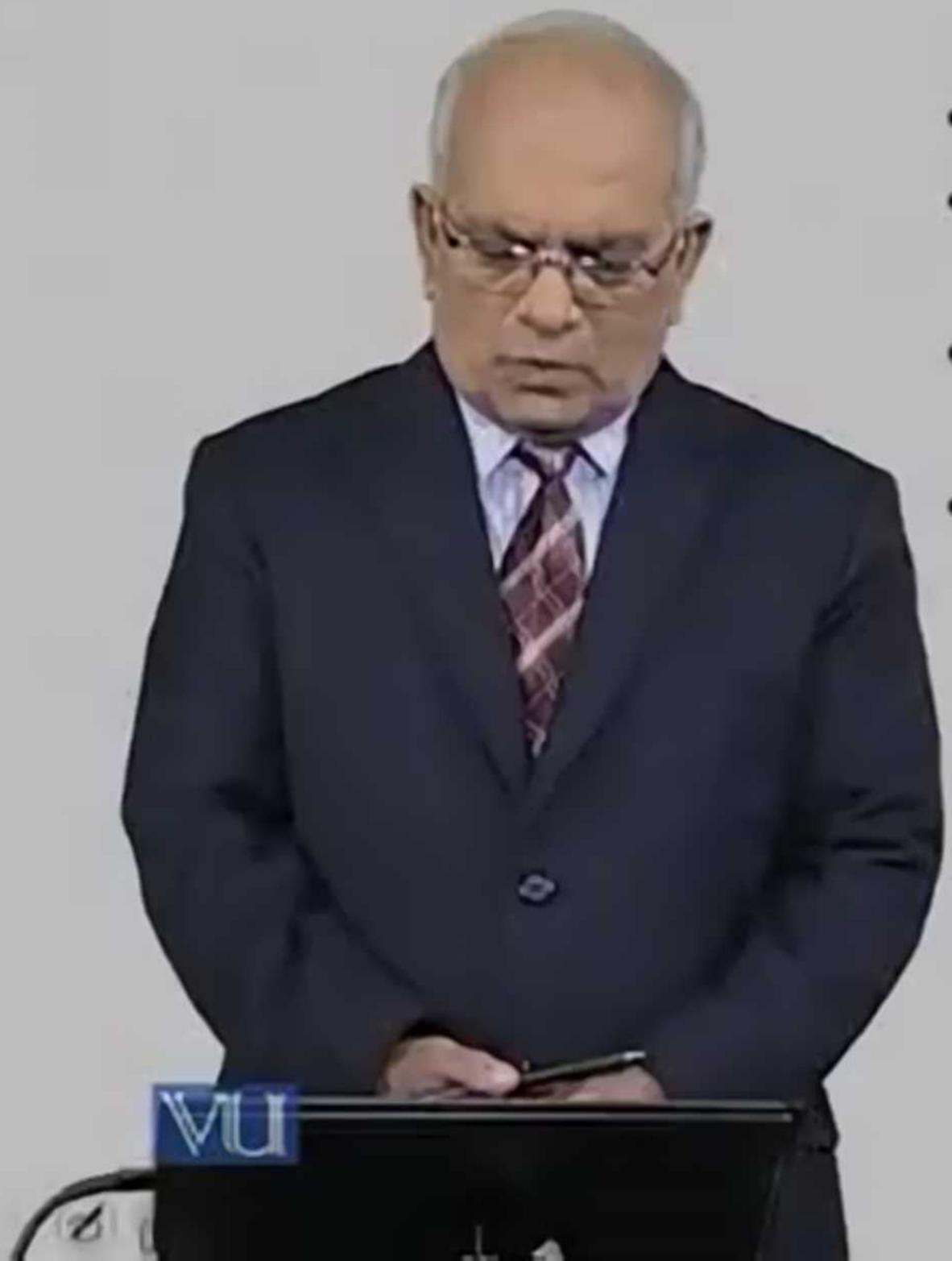

Summary

- Newton's Laws
- Choosing a Control Volume
- Forces Acting on a Control Volume
- The Linear Momentum Equation
  - Special Cases
  - Momentum-Flux Correction Factor, β
  - Steady Flow
  - Flow with No External Forces

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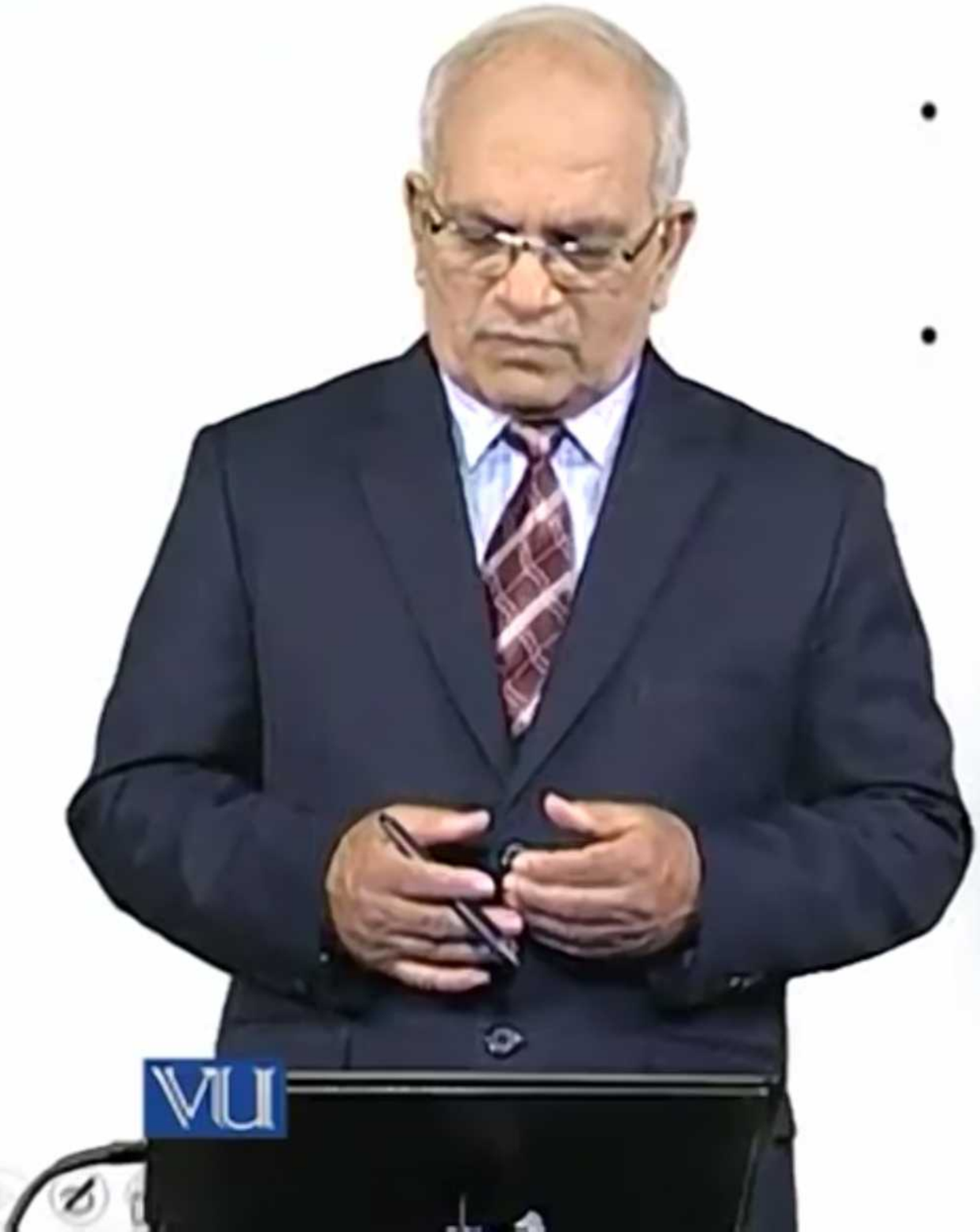

Summary

- Review of Rotational Motion and Angular Momentum
- The Angular Momentum Equation

- Special Cases
- Flow with No External Moments
- Radial-Flow Devices

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Dimensional Analysis

Background

- **Although many practical engineering problems involving fluid mechanics can be solved by equations and analytical procedures, yet a large number of problems rely on experimental data for their solution**
- **In fact, very few problems involving real fluids can be solved by analytical analysis alone.**

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Dimensional Analysis Background

In general, solution is obtained through the use of a combination of analysis and experimental data

An obvious goal of any experiment is to make the results as widely applicable as possible

To achieve this goal, the concept of similitude is often used so that

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Dimensional Analysis

Background

measurements made on one system (Laboratory) can be used to describe the behavior of other systems (Outside of laboratory)

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Dimensional Analysis

Background

- The laboratory systems are usually thought of as models and are used to study the phenomenon of interest under carefully controlled conditions

- From these model studies, empirical formulations can be developed, or specific predictions of one or more characteristics of some other similar system can be made.

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Dimensional Analysis

Background

- However, to do this, it is necessary to establish the
- relationship between the laboratory model and the "other" system

- *In present topic, we will learn how to achieve this in a systematic manner*

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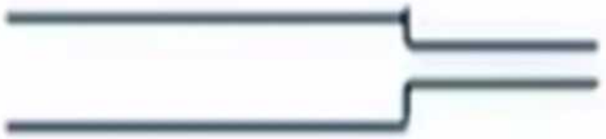

Sudden contraction in pipes

Wind Turbine

Dam/spillway

Dimensional Analysis to predict the physical parameters that will significantly influence phenomenon under study

Similitude and Model analysis to investigate the complexity of phenomenon in details

Application of knowledge on actual/prototype model

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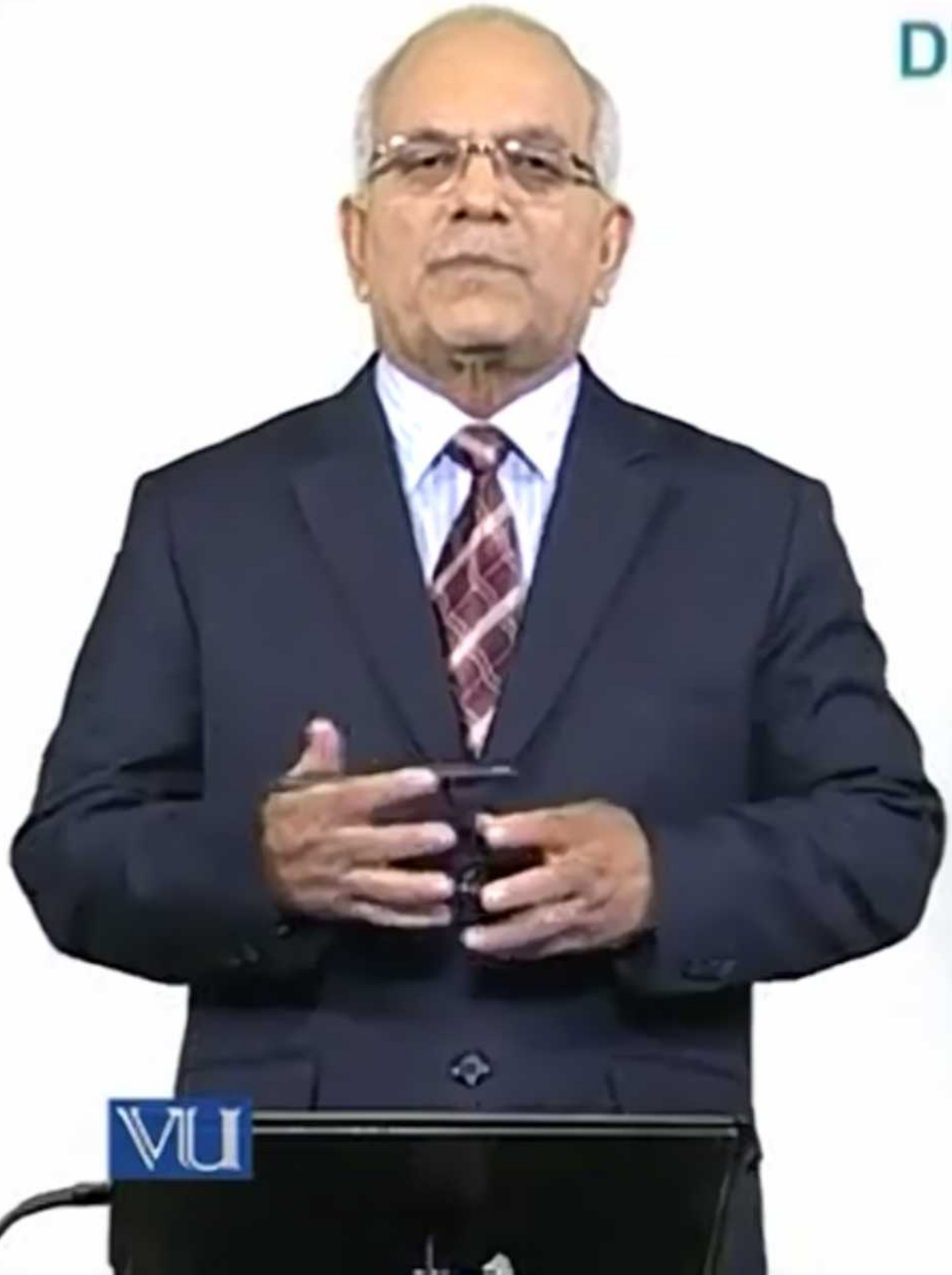
Dimensional Analysis



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Dimensional Analysis

At the University of Iowa. A 1:46.6 scale model of an Arleigh Burke class U.S. Navy fleet destroyer being tested in the 100-m long towing tank The model is 3.048 m long. In tests like this, the Froude number is the most important nondimensional parameter.

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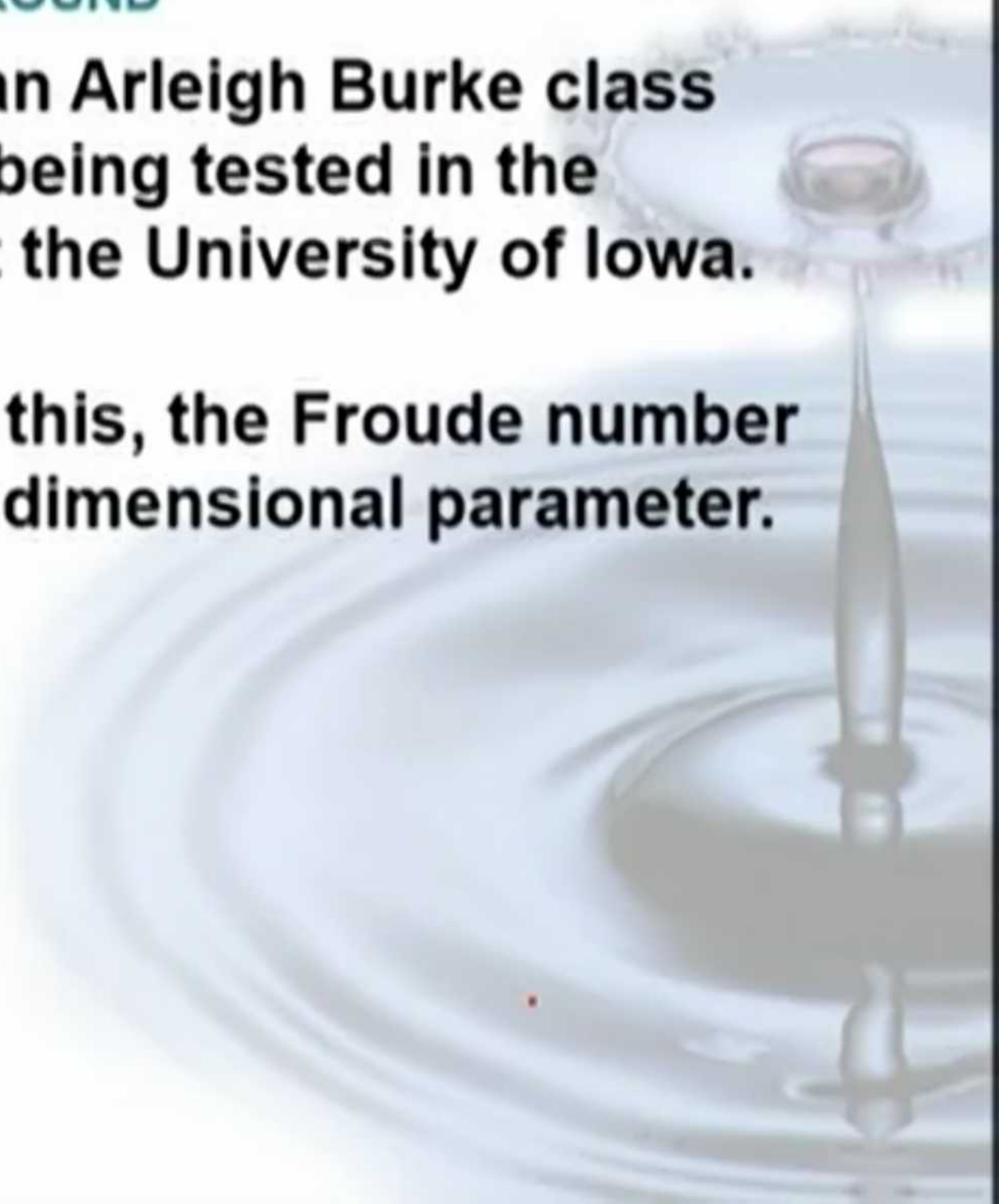
Dimensional Analysis
BACKGROUND

A 1 : 46.6 scale model of an Arleigh Burke class U.S. Navy fleet destroyer being tested in the 100-m long towing tank at the University of Iowa. The model is 3.048 m long. In tests like this, the Froude number is the most important nondimensional parameter.



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Dimensional Analysis

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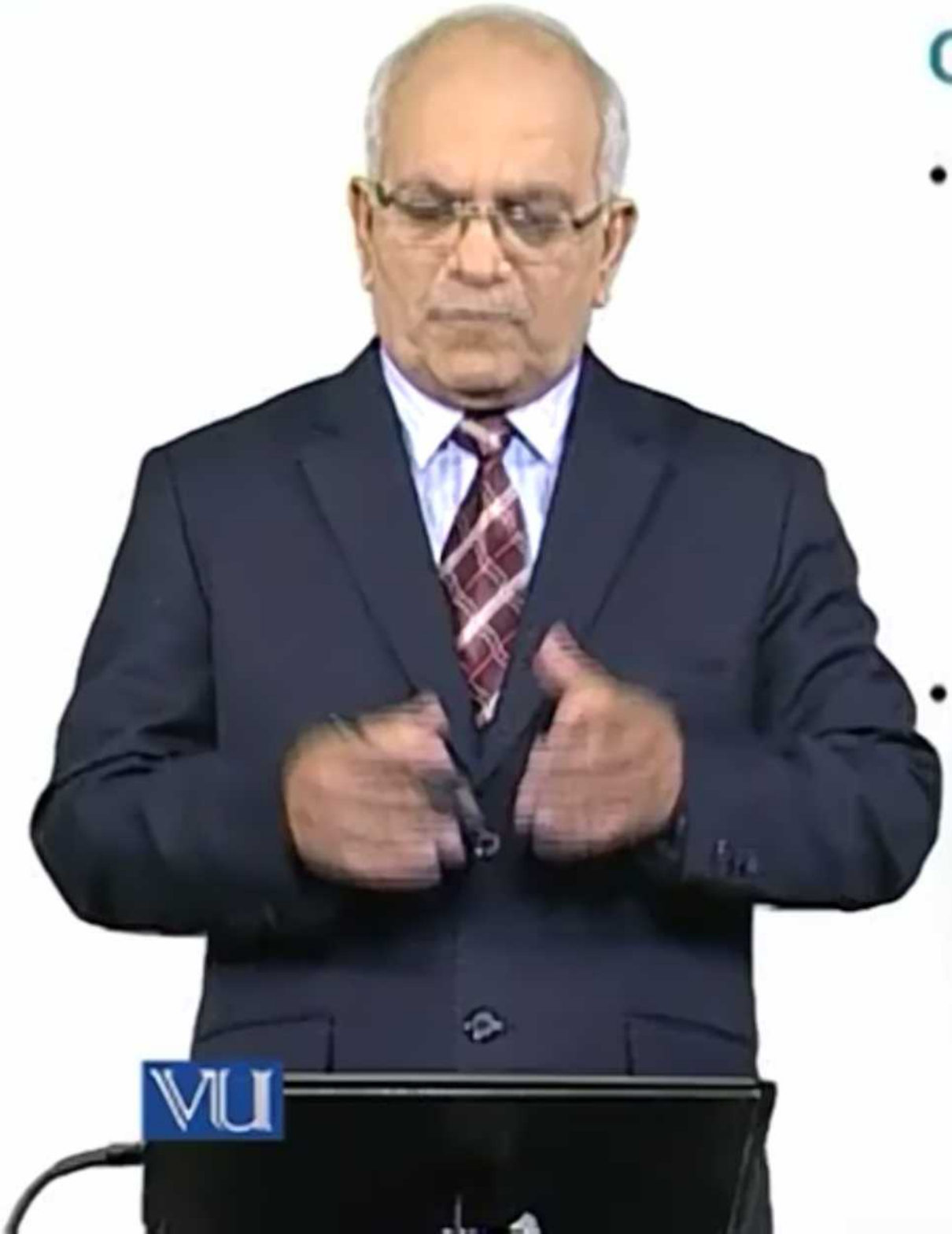

Objectives

- Develop a better understanding of dimensions, units, and dimensional homogeneity of equations
- Understand the numerous benefits of dimensional analysis

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DIMENSIONS AND UNITS

- **Dimension: A measure of a physical quantity (without numerical values).**
- **Unit: A way to assign a *number* to that dimension.**
- **There are seven primary dimensions (also called fundamental or basic dimensions):**

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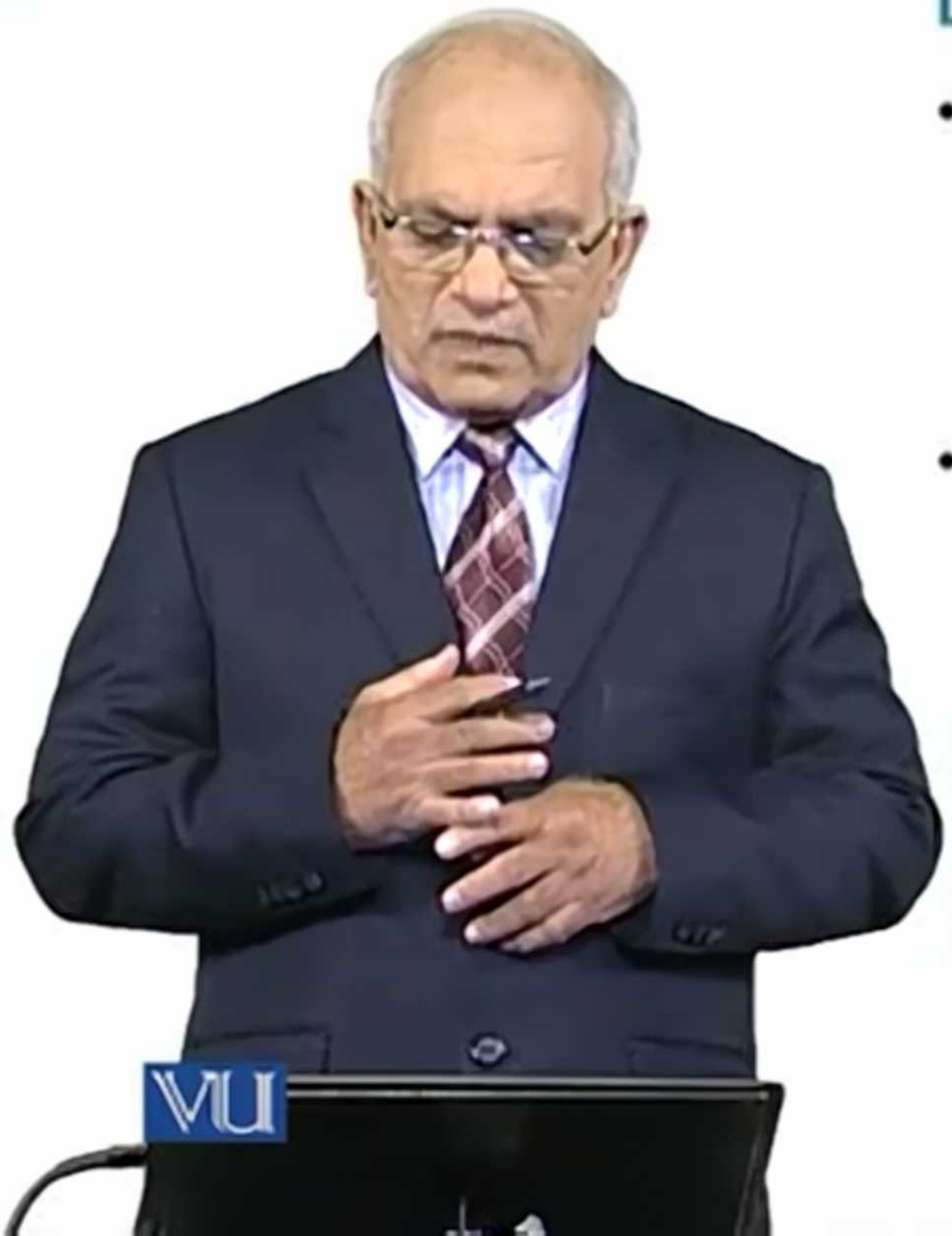

DIMENSIONS AND UNITS

- mass, length, time, temperature, electric current, amount of light, and amount of matter.
- All nonprimary dimensions can be formed by some combination of the seven primary dimensions.

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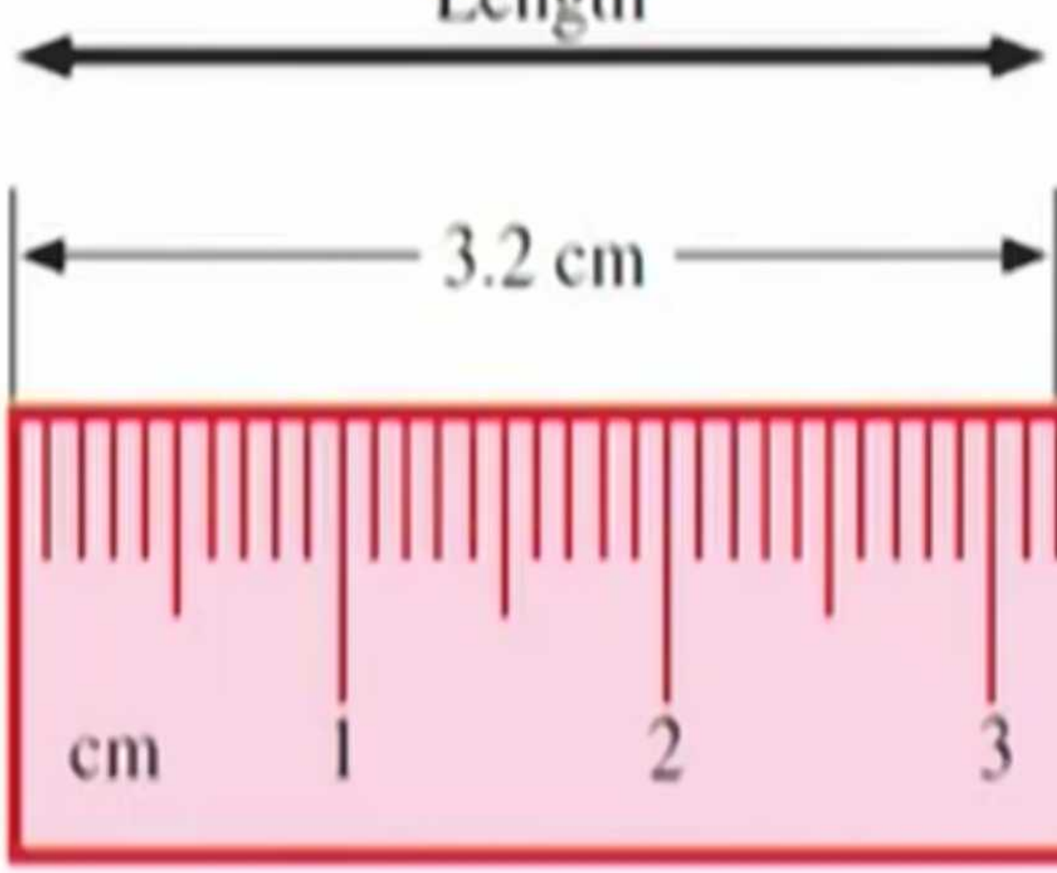
Dimensions And Units

$$\textit{Dimensions of force:} \quad \{\text{Force}\} = \left\{\text{Mass}\,\frac{\text{Length}}{\text{Time}^2}\right\} = \{\text{mL/t}^2\}$$



- **A *dimension* is a measure of a physical quantity without numerical values, while a *unit* is a way to assign a number to the dimension. For example, length is a dimension, but centimeter is a unit.**

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Dimensions And Units

Primary dimensions and their associated primary SI and English units

Dimension	Symbol*	SI Unit	English Unit
Mass	m	kg (kilogram)	lbm (pound-mass)
Length	L	m (meter)	ft (foot)
Time†	t	s (second)	s (second)
Temperature	T	K (kelvin)	R (rankine)
Electric current	I	A (ampere)	A (ampere)
Amount of light	C	cd (candela)	cd (candela)
Amount of matter	N	mol (mole)	mol (mole)

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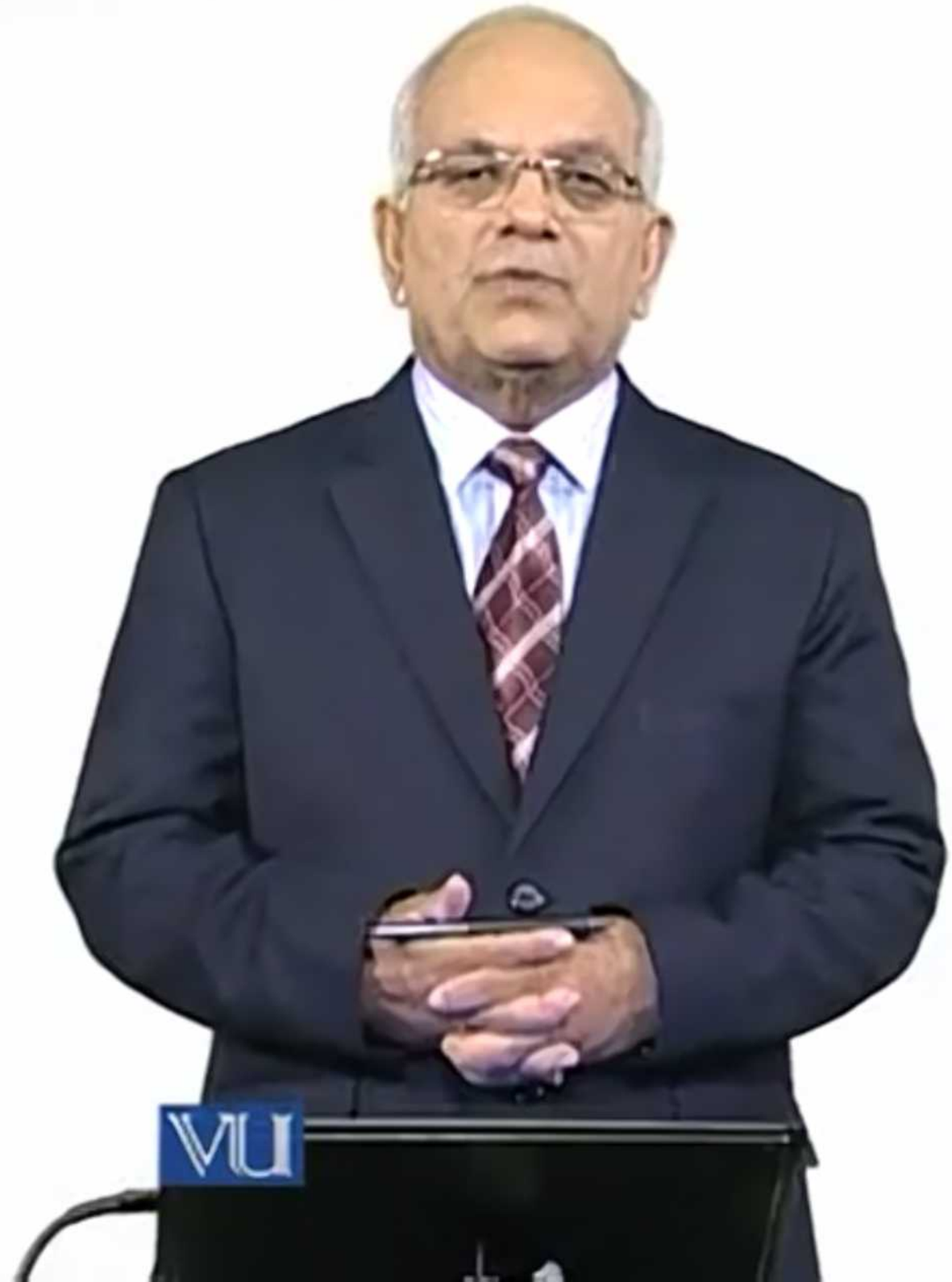
DIMENSIONAL HOMOGENEITY



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DIMENSIONAL HOMOGENEITY

EXAMPLE 7–2 Dimensional Homogeneity of the Bernoulli Equation

Probably the most well-known (and most misused) equation in fluid mechanics is the Bernoulli equation (Fig. 7–6), discussed in Chap. 5. One standard form of the Bernoulli equation for incompressible irrotational fluid flow is

Bernoulli equation:
$$P + \frac{1}{2}\rho V^2 + \rho g z = C \qquad (1)$$

(*a*) Verify that each additive term in the Bernoulli equation has the same dimensions. (*b*) What are the dimensions of the constant *C*?

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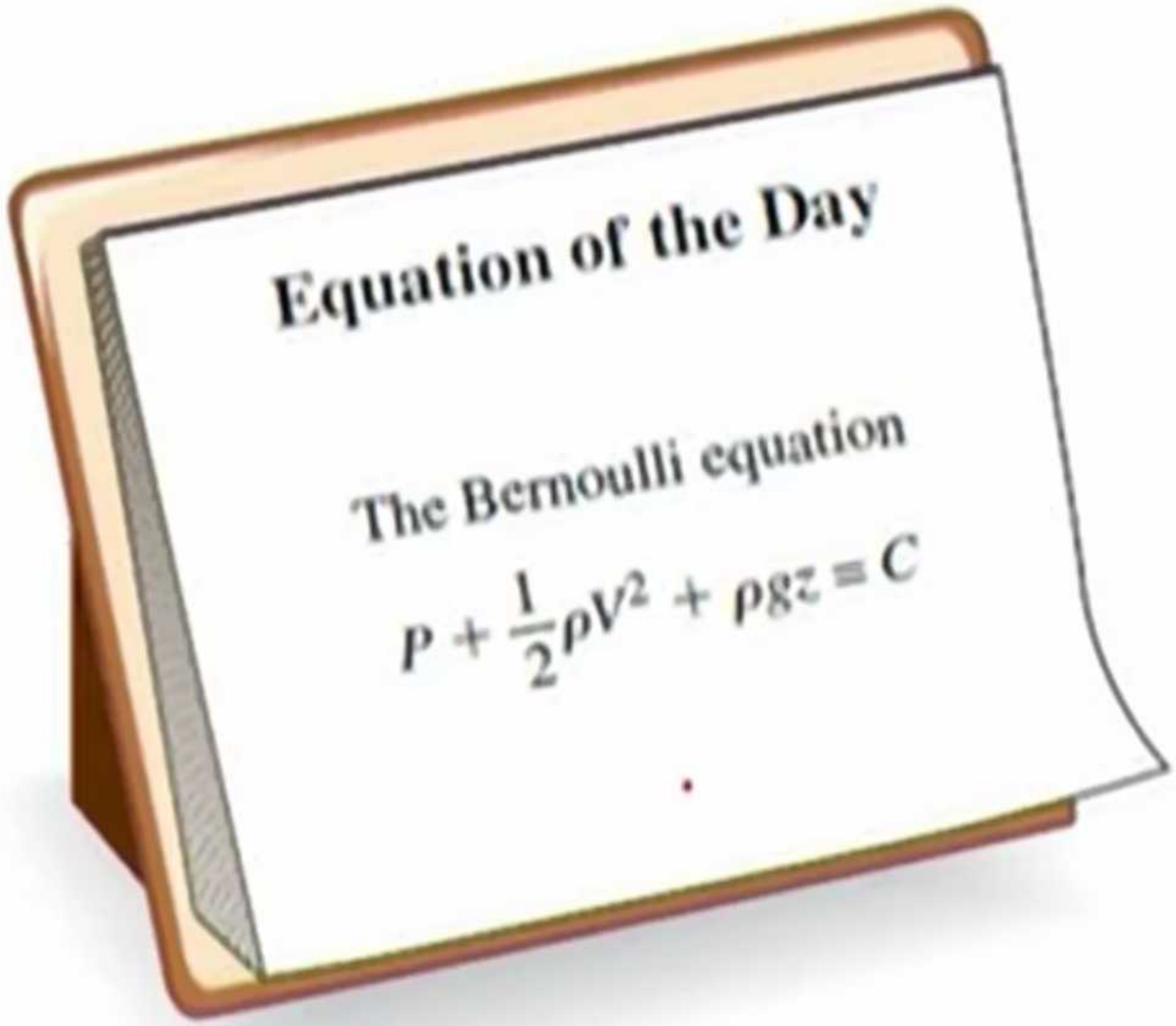
DIMENSIONAL HOMOGENEITY



The Bernoulli equation is a good example of a *dimensionally homogeneous* equation. All additive terms, including the constant, have the *same* dimensions, namely that of pressure. In terms of primary dimensions, each term has dimensions $\{m/(t^2L)\}$.

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DIMENSIONAL HOMOGENEITY

SOLUTION We are to verify that the primary dimensions of each additive term in Eq. 1 are the same, and we are to determine the dimensions of constant *C*.

Analysis (*a*) Each term is written in terms of primary dimensions,

$$\{P\} = \{\text{Pressure}\} = \left\{\frac{\text{Force}}{\text{Area}}\right\} = \left\{\text{Mass}\,\frac{\text{Length}}{\text{Time}^2}\,\frac{1}{\text{Length}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\left\{\frac{1}{2}\rho V^2\right\} = \left\{\frac{\text{Mass}}{\text{Volume}}\left(\frac{\text{Length}}{\text{Time}}\right)^2\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\{\rho g z\} = \left\{\frac{\text{Mass}}{\text{Volume}}\cdot\frac{\text{Length}}{\text{Time}^2}\,\text{Length}\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

Indeed, **all three additive terms have the same dimensions.**

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DIMENSIONAL HOMOGENEITY

(*b*) From the law of dimensional homogeneity, the constant must have the same dimensions as the other additive terms in the equation. Thus,

Primary dimensions of the Bernoulli constant: $\{C\} = \left\{\frac{m}{t^2 L}\right\}$

Discussion If the dimensions of any of the terms were different from the others, it would indicate that an error was made somewhere in the analysis.

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DIMENSIONAL HOMOGENEITY

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DIMENSIONAL HOMOGENEITY

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(*a*) Verify that each additive term in the Bernoulli equation has the same dimensions. (*b*) What are the dimensions of the constant *C*?

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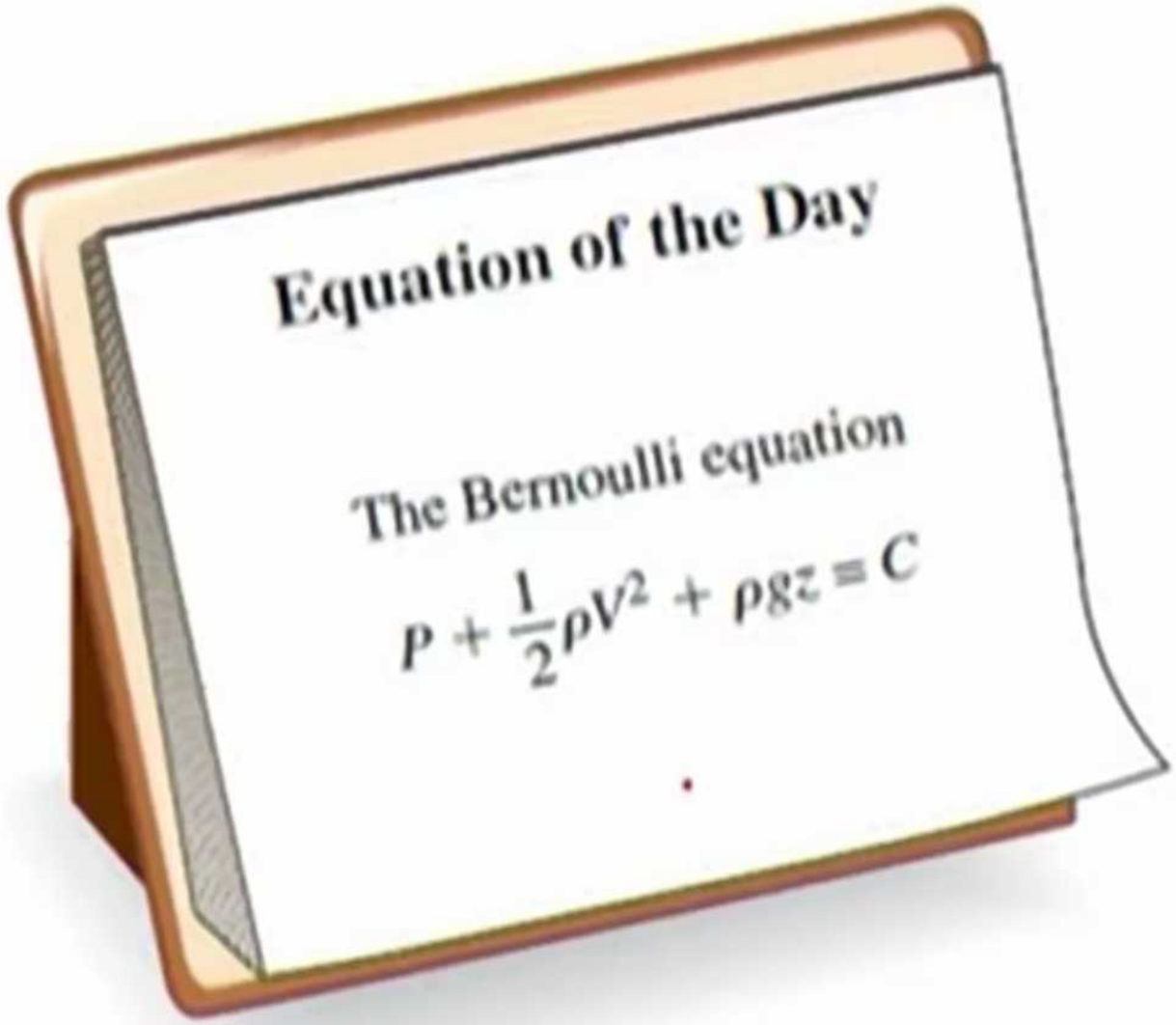
DIMENSIONAL HOMOGENEITY



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DIMENSIONAL HOMOGENEITY

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$$\{P\} = \{\text{Pressure}\} = \left\{\frac{\text{Force}}{\text{Area}}\right\} = \left\{\text{Mass}\,\frac{\text{Length}}{\text{Time}^2}\,\frac{1}{\text{Length}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\left\{\frac{1}{2}\rho V^2\right\} = \left\{\frac{\text{Mass}}{\text{Volume}}\left(\frac{\text{Length}}{\text{Time}}\right)^2\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\{\rho g z\} = \left\{\frac{\text{Mass} \cdot}{\text{Volume}}\,\frac{\text{Length}}{\text{Time}^2}\,\text{Length}\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

Indeed, **all three additive terms have the same dimensions.**

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DIMENSIONAL HOMOGENEITY

(*b*) From the law of dimensional homogeneity, the constant must have the same dimensions as the other additive terms in the equation. Thus,

Primary dimensions of the Bernoulli constant: $\{C\} = \left\{\frac{m}{t^2 L}\right\}$

Discussion If the dimensions of any of the terms were different from the others, it would indicate that an error was made somewhere in the analysis.

LECTURE 100

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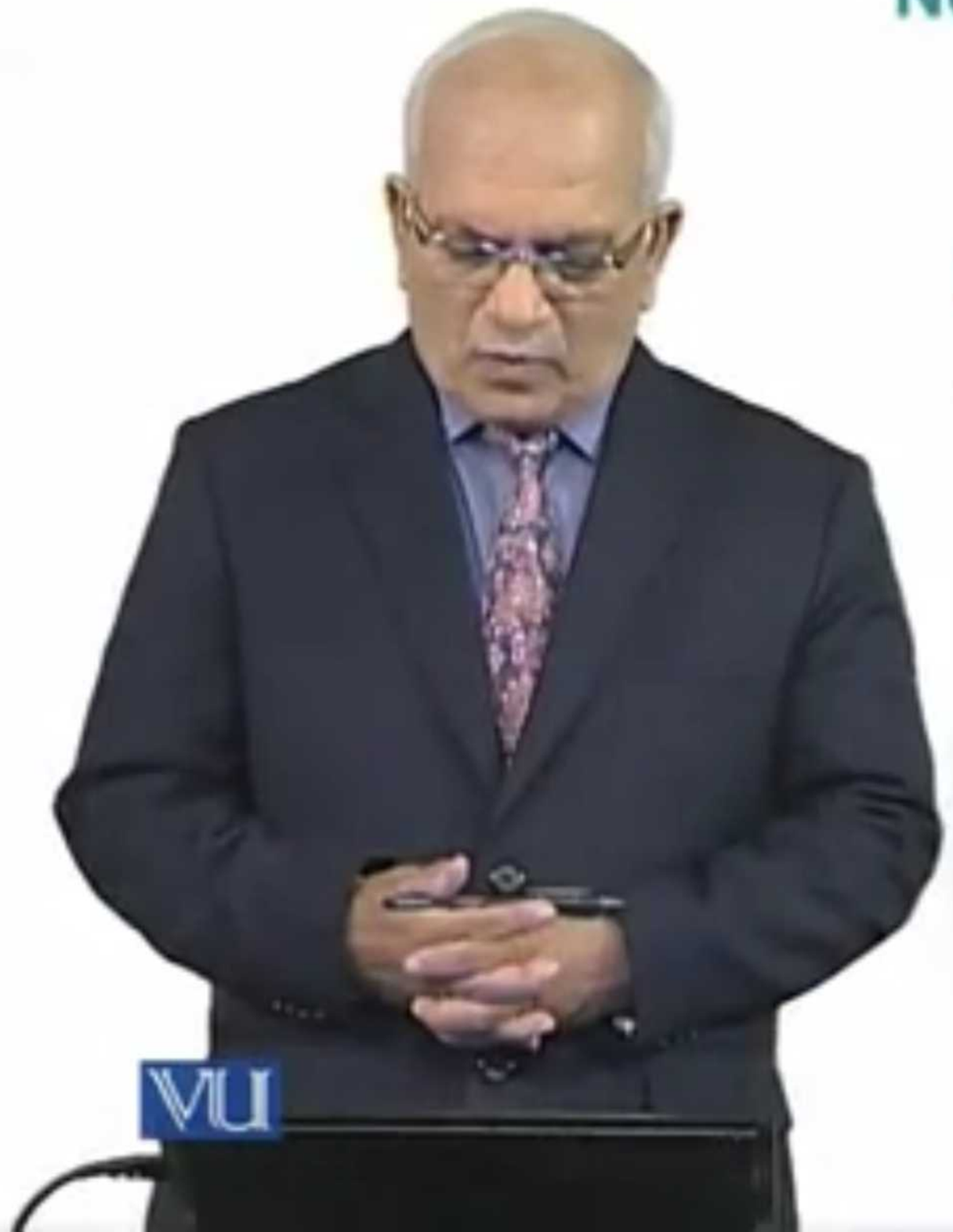

Non Dimensionalisation of Equations

Normalized Equatiion:

- **If the nondimensional terms in the equation are of order unity, the equation is called normalized.**
- **Each term in a nondimensional equation is dimensionless.**

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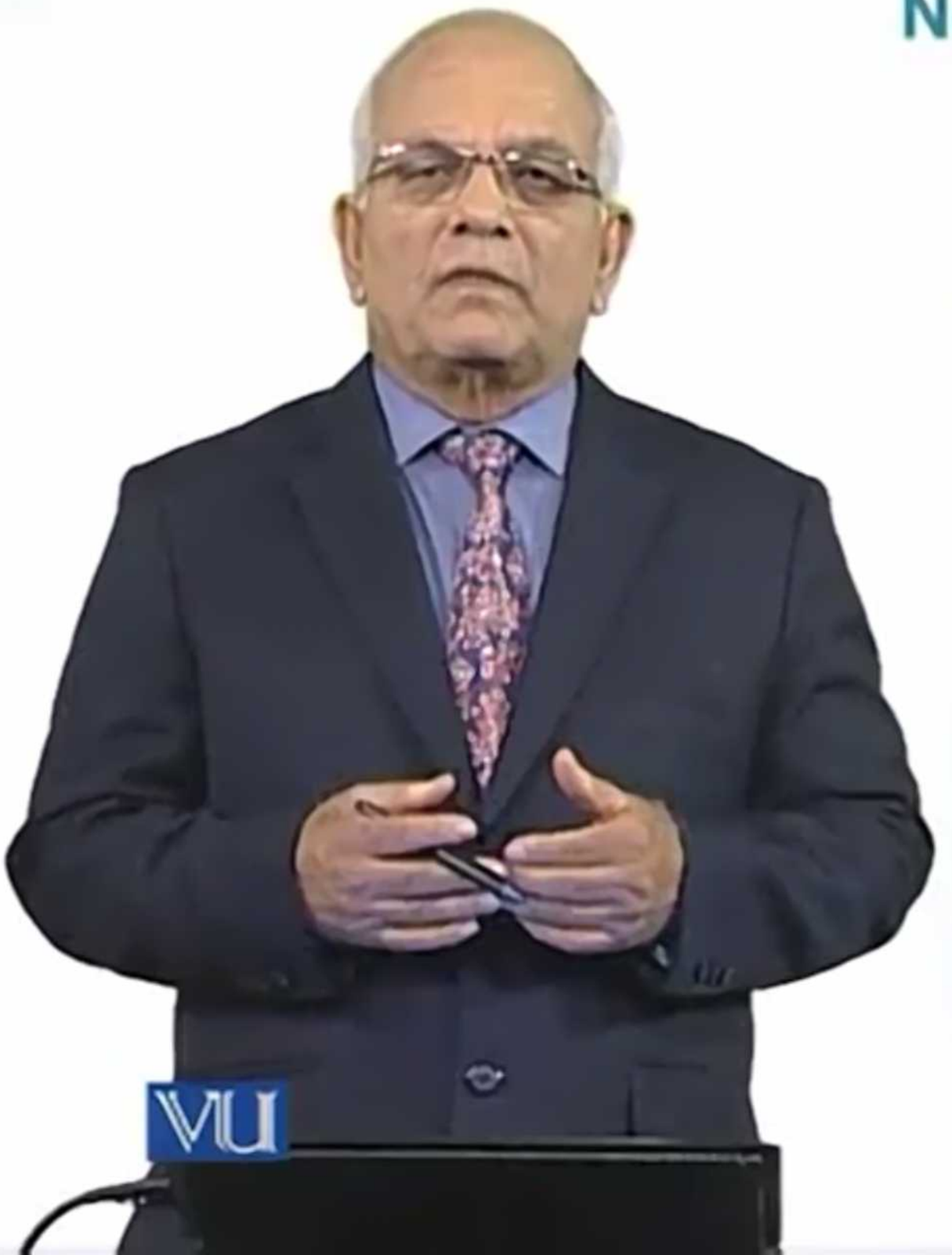

Non Dimensionalisation of Equations

Nondimensional parameters:

In the process of nondimensionalizing an equation of motion, nondimensional parameters often appear—most of which are named after a notable scientist or engineer (e.g., the Reynolds number and the Froude number).

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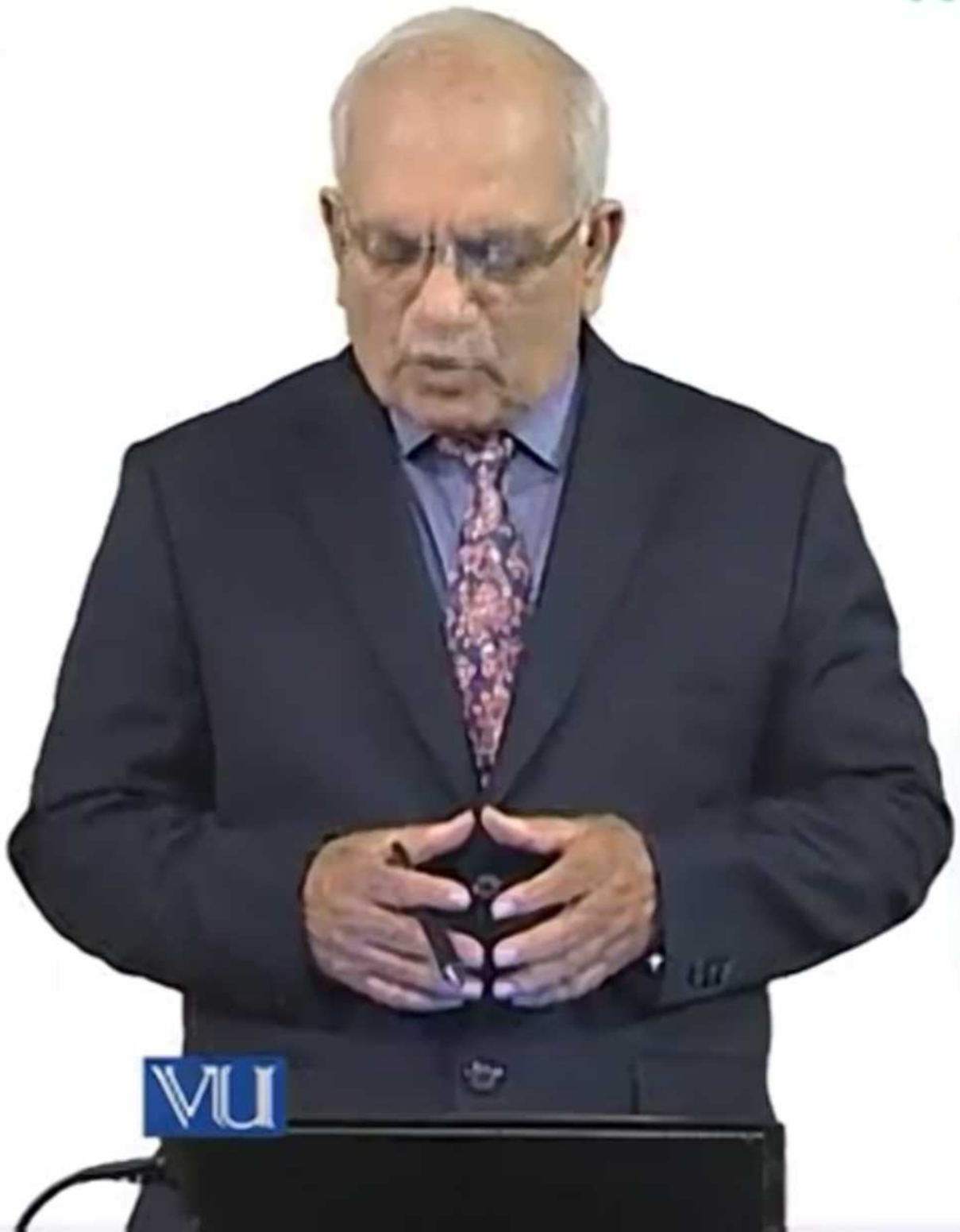

Non Dimensionalisation of Equations

Nondimensional parameters:

- **This process is referred to by some authors as inspectional analysis.**

LECTURE 101

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Non Dimensionalisation of Equations

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Non Dimensionalisation Of Equations

The nondimensionalized Bernoulli equation

$$\frac{P}{P_\infty} + \frac{\rho V^2}{2P_\infty} + \frac{\rho g z}{P_\infty} = \frac{C}{P_\infty}$$

$$\{1\} \quad \{1\} \quad \{1\} \quad \{1\}$$

A *nondimensionalized* form of the Bernoulli equation is formed by dividing each additive term by a pressure (here we use P_∞). Each resulting term is *dimensionless* (dimensions of {1}).

Advanced Fluid Dynamics

Non Dimensionalisation Of Equations

Equation of motion:

$$\frac{d^2z}{dt^2} = -g$$

Dimensional result:

$$z = z_0 + w_0 t - \frac{1}{2} g t^2$$

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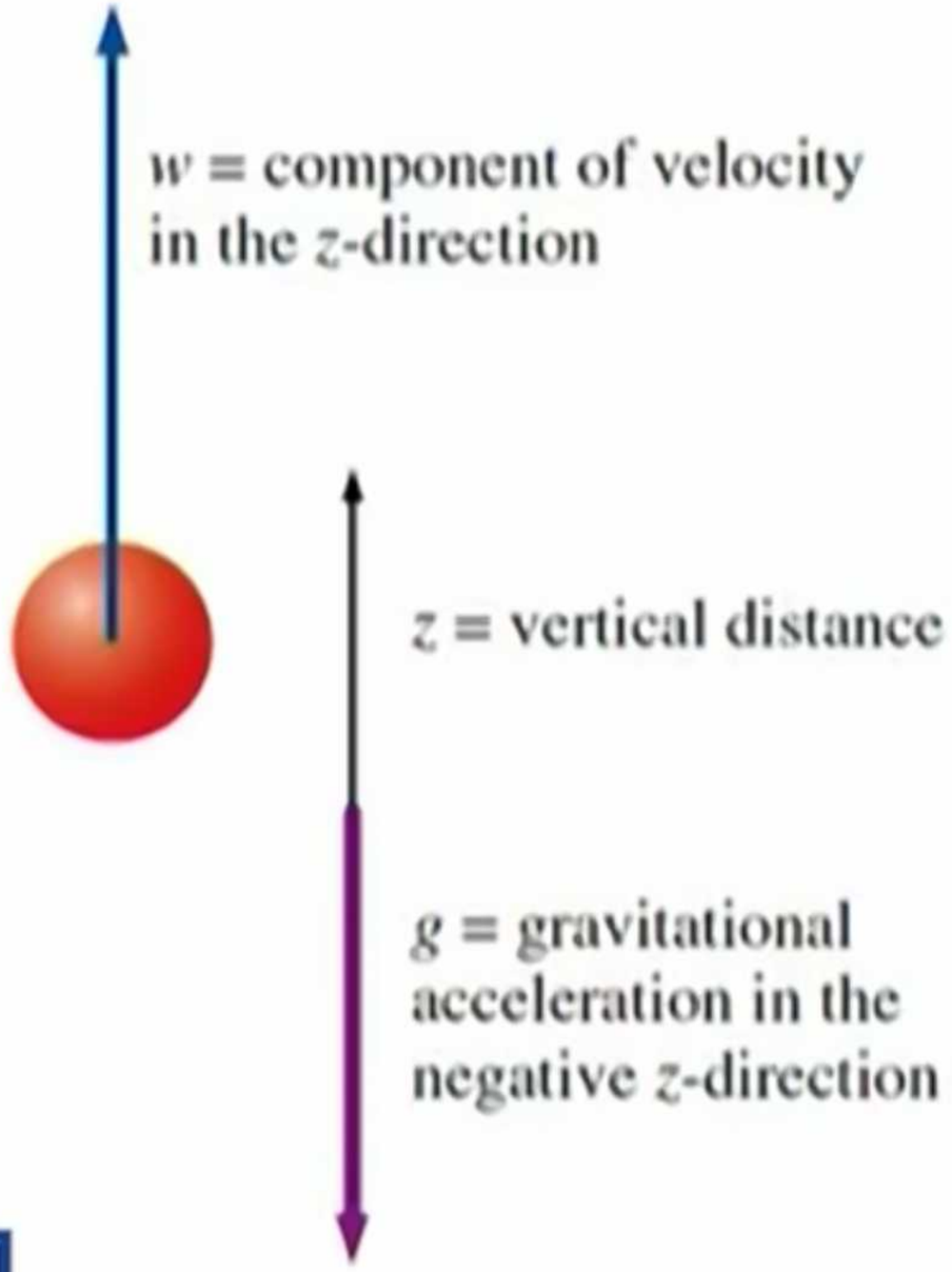
Non Dimensionalisation Of Equations



- **Object falling in a vacuum. Vertical velocity is drawn positively, so $w < 0$ for a falling object.**

LECTURE 102

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Non Dimensionalisation of Equations

Dimensional variables:

Dimensional quantities that change or vary in the problem. Examples: *z* (dimension of length) and *t* (dimension of time).

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Non Dimensionalisation of Equations

Nondimensional (or dimensionless) variables:

- Quantities that change or vary in the problem, but have no dimensions. Example: Angle of rotation, measured in degrees or radians, dimensionless units.

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Non Dimensionalisation of Equations

Nondimensional (or dimensionless) variables:

- **Dimensional constant: Gravitational constant *g*, while dimensional, remains constant.**
- **Parameters: Refer to the combined set of dimensional variables,**

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Non Dimensionalisation of Equations

Nondimensional (or dimensionless) variables:

- **and dimensional constants in the problem.**
- **Pure constants: The constant 1/2 and the exponent 2 in equation. Other common examples of pure constants are π and e.**

LECTURE 103

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Non Dimensionalisation of Equations

Nondimensional (or dimensionless) variables:

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Non Dimensionalisation Of Equations

To nondimensionalize an equation, we need to select scaling parameters, based on the primary dimensions contained in the original equation.

Primary dimensions of all parameters:

$$\{z\} = \{\mathrm{L}\} \qquad \{t\} = \{\mathrm{t}\} \qquad \{z_0\} = \{\mathrm{L}\} \qquad \{w_0\} = \{\mathrm{L/t}\} \qquad \{g\} = \{\mathrm{L/t^2}\}$$

Nondimensionalized variables: $z^* = \dfrac{z}{z_0} \qquad t^* = \dfrac{w_0 t}{z_0}$

$$\frac{d^2z}{dt^2} = \frac{d^2(z_0 z^*)}{d(z_0 t^*/w_0)^2} = \frac{w_0^2}{z_0}\frac{d^2z^*}{dt^{*2}} = -g \qquad \rightarrow \qquad \frac{w_0^2}{gz_0}\frac{d^2z^*}{dt^{*2}} = -1$$

$$\mathrm{Fr} = \frac{w_0}{\sqrt{gz_0}}$$

Froude number

VU

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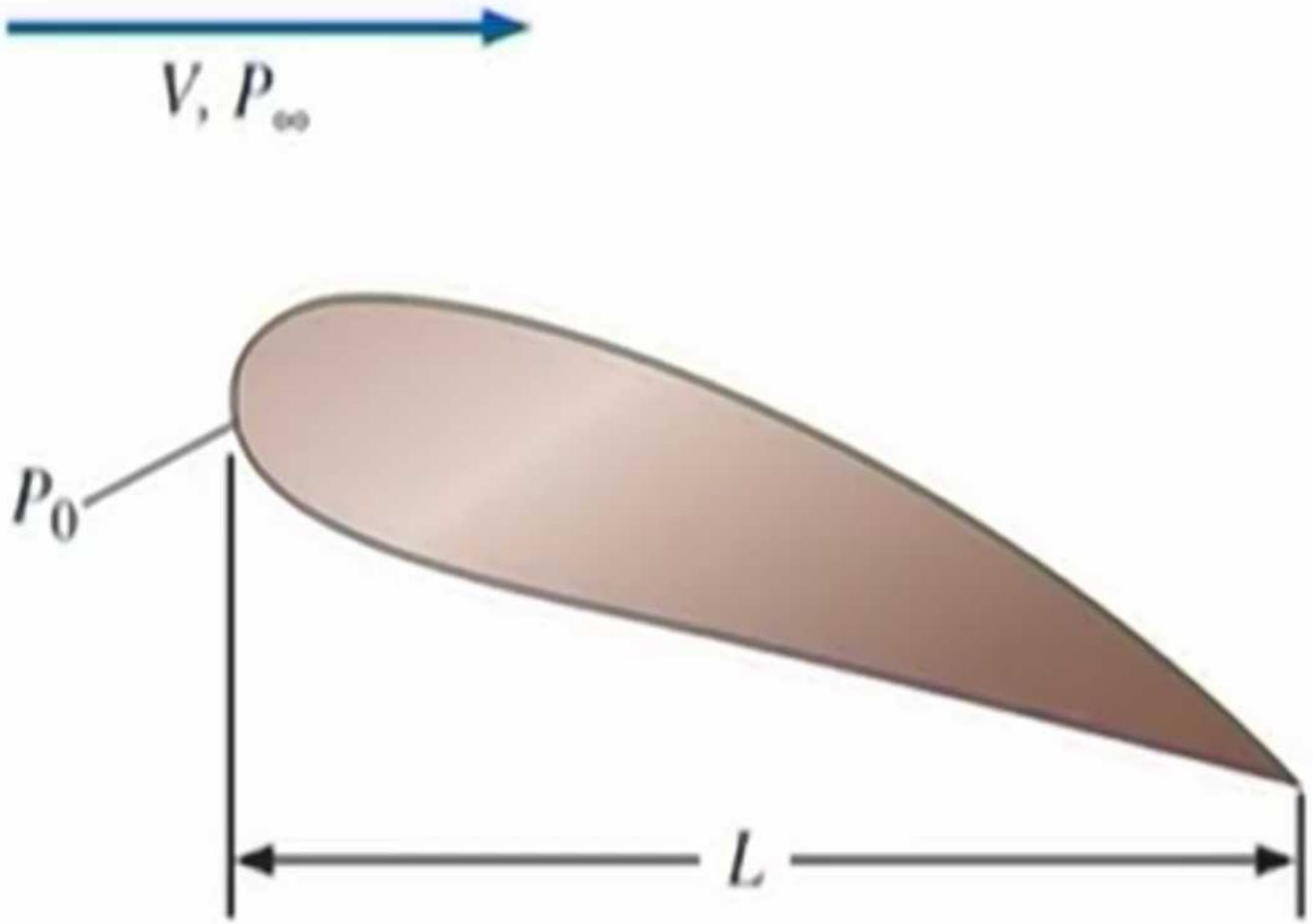
Non Dimensionalisation Of Equations



In a typical fluid flow problem, the scaling parameters usually include a characteristic length *L*, a characteristic velocity *V*, and a reference pressure difference $P_0 - P_\infty$. Other parameters and fluid properties such as density, viscosity, and gravitational acceleration enter the problem as well.

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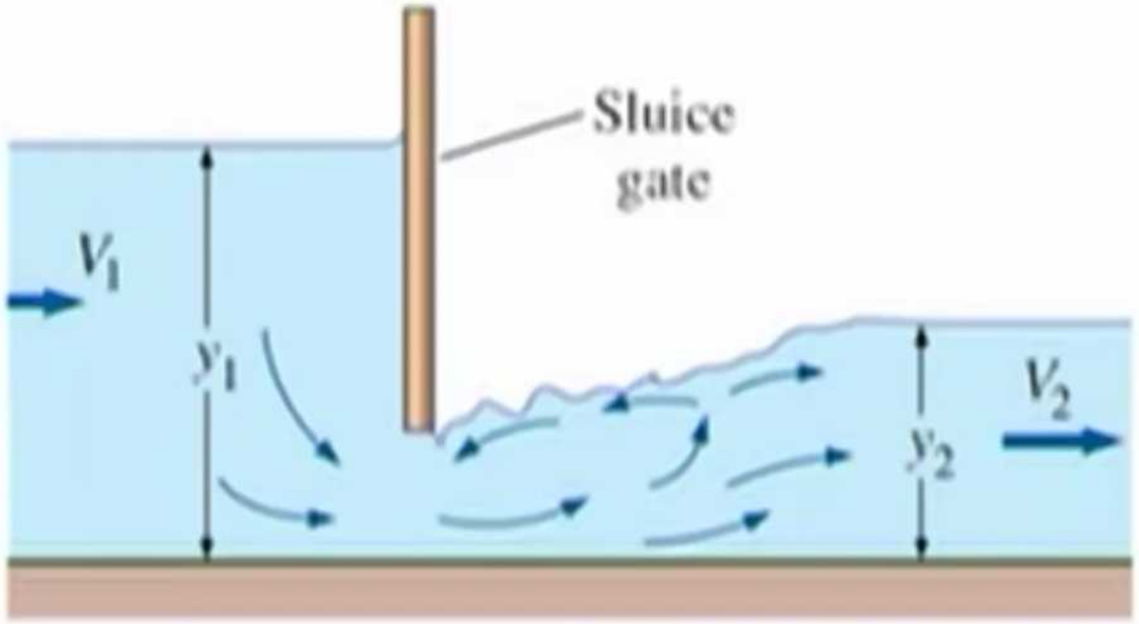
Non Dimensionalisation Of Equations



The *Froude number* is important in free-surface flows such as flow in open channels. Shown here is flow through a sluice gate. The Froude number upstream of the sluice gate is $\mathrm{Fr}_1 = V_1/\sqrt{gy_1}$, and it is $\mathrm{Fr}_2 = V_2/\sqrt{gy_2}$ downstream of the sluice gate.

Nondimensionalized equation of motion:

$$\frac{d^2z^*}{dt^{*2}} = -\frac{1}{\mathrm{Fr}^2}$$

Nondimensional result:

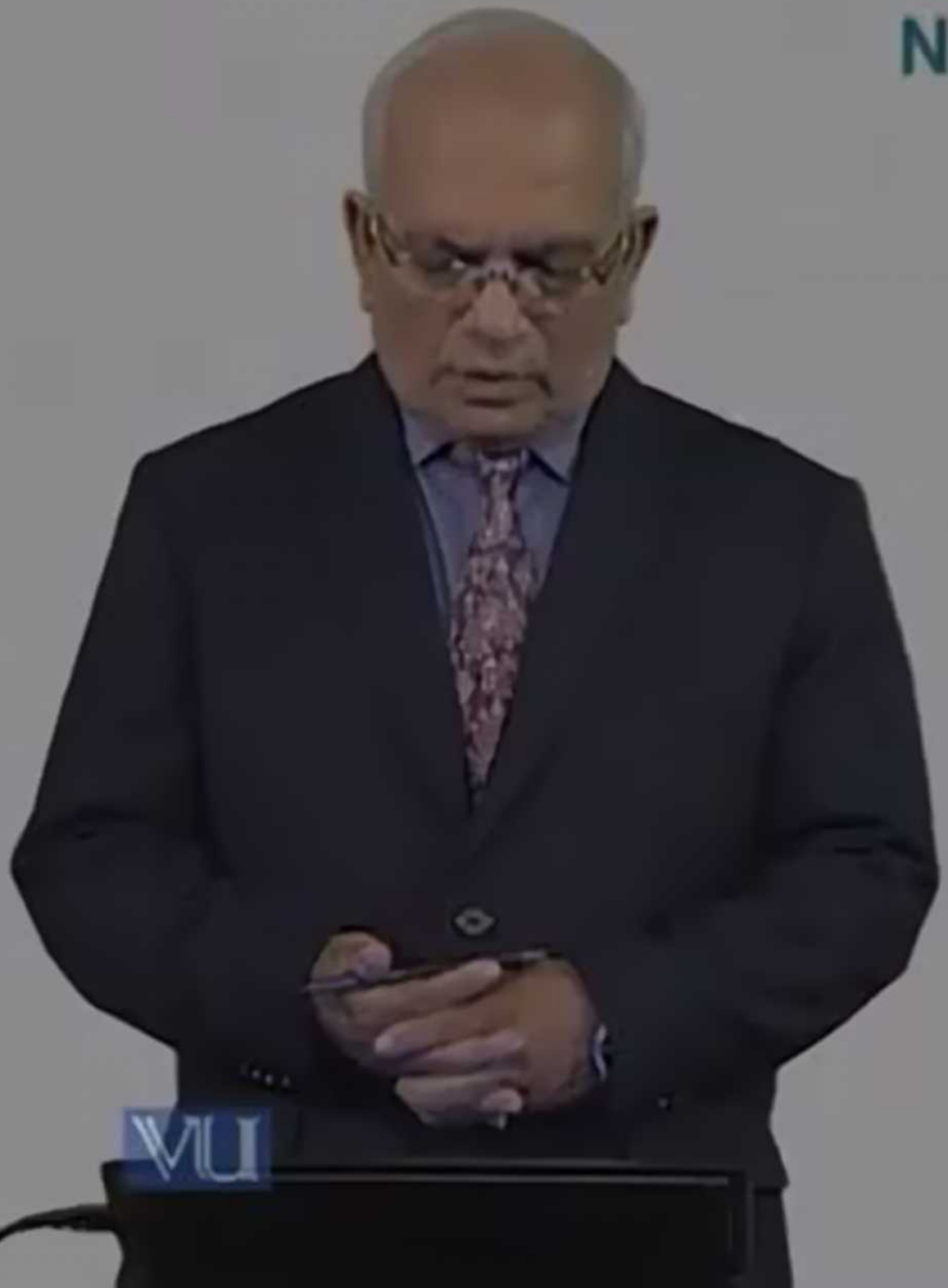
$$z^* = 1 + t^* - \frac{1}{2\mathrm{Fr}^2}t^{*2}$$

LECTURE 104

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Non Dimensionalisation of Equations



Advanced Fluid Dynamics

Non Dimensionalisation Of Equations

Relationships between key parameters in the problem are identified.

The number of parameters in a nondimensionalized equation is less than the number of parameters in the original equation.

The two key advantages of nondimensionalization of an equation.

EXAMPLE 7–3 **Illustration of the Advantages of Nondimensionalization**

Your little brother's high school physics class conducts experiments in a large vertical pipe whose inside is kept under vacuum conditions. The students are able to remotely release a steel ball at initial height z_0 between 0 and 15 m (measured from the bottom of the pipe), and with initial vertical speed w_0 between 0 and 10 m/s. A computer coupled to a network of photosensors along the pipe enables students to plot the trajectory of the steel ball (height z plotted as a function of time t) for each test. The students are unfamiliar with dimensional analysis or nondimensionalization techniques, and therefore conduct several "brute force" experiments to determine how the trajectory is affected by initial conditions z_0 and w_0. First they hold w_0 fixed at 4 m/s and conduct experiments at five different values of z_0: 3, 6, 9, 12, and 15 m. The experimental results are shown in Fig. 7–12*a*. Next, they hold z_0 fixed at 10 m and conduct experiments at five different values of w_0: 2, 4, 6, 8, and 10 m/s. These results are shown in Fig. 7–12*b*. Later that evening, your brother shows you the data and the trajectory plots and tells you that they plan to conduct more experiments at different values of z_0 and w_0. You explain to him that by first nondimensionalizing the data, the problem can be reduced to just *one* parameter, and no further experiments required. Prepare a nondimensional plot to prove your point and discuss.

VU

Analysis Equation 7–4 is valid for this problem, as is the nondimensionalization that resulted in Eq. 7–9. As previously discussed, this problem combines three of the original dimensional parameters (g, z_0, and w_0) into *one* nondimensional parameter, the Froude number. After converting to the dimensionless variables of Eq. 7–6, the 10 trajectories of Fig. 7–12*a* and *b* are replotted in dimensionless format in Fig. 7–13. It is clear that all the trajectories are of the same family, with the Froude number as the only remaining parameter. Fr^2 varies from about 0.041 to about 1.0 in these experiments. If any more experiments are to be conducted, they should include combinations of z_0 and w_0 that produce Froude numbers outside of this range. A large number of additional experiments would be unnecessary, since all the trajectories would be of the same family as those plotted in Fig. 7–13.

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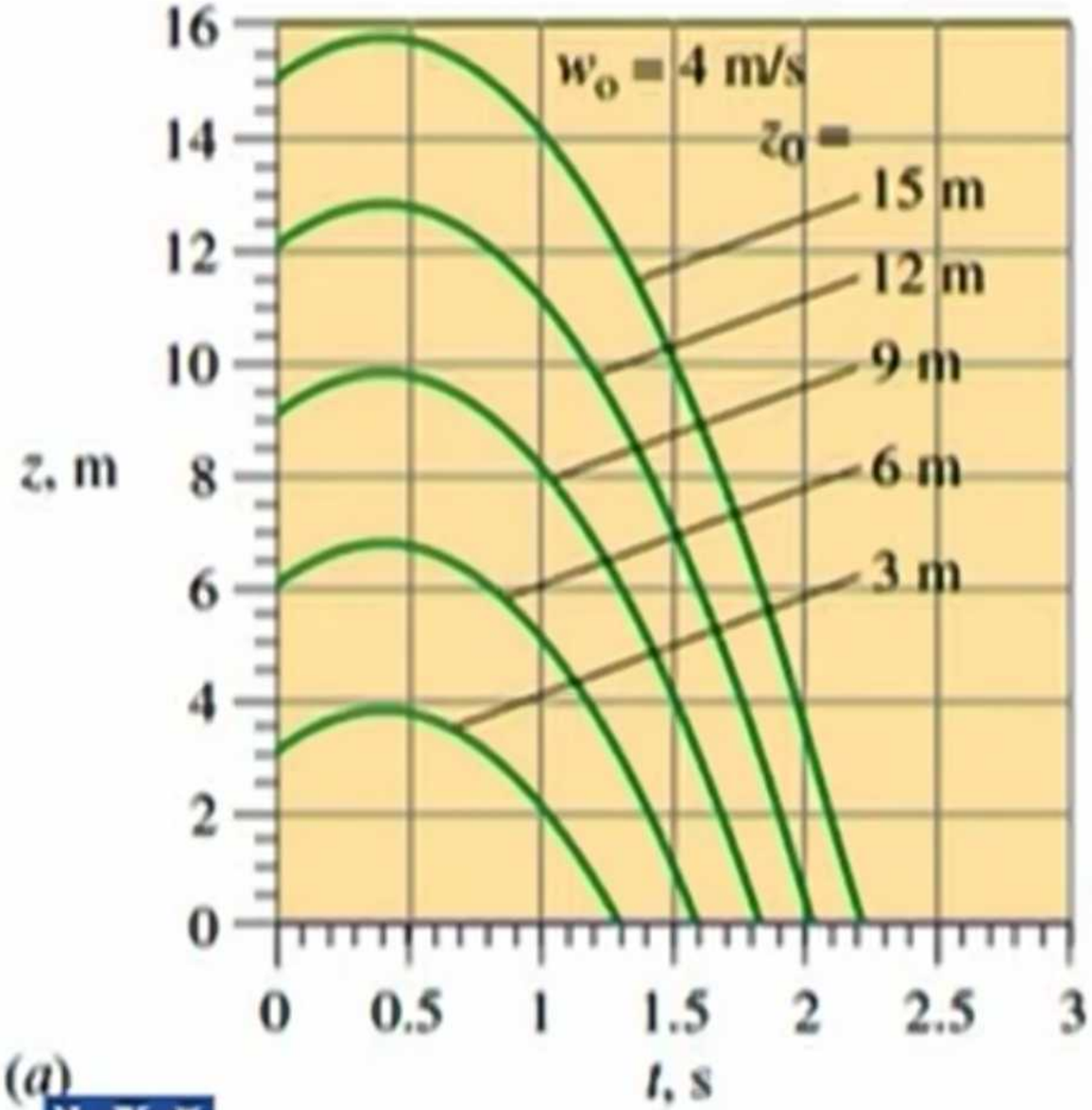
Non Dimensionalisation Of Equations



Trajectories of a steel ball falling in a vacuum: (*a*) w_0 fixed at 4 m/s, and (*b*) z_0 fixed at 10 m (Example 7–3).

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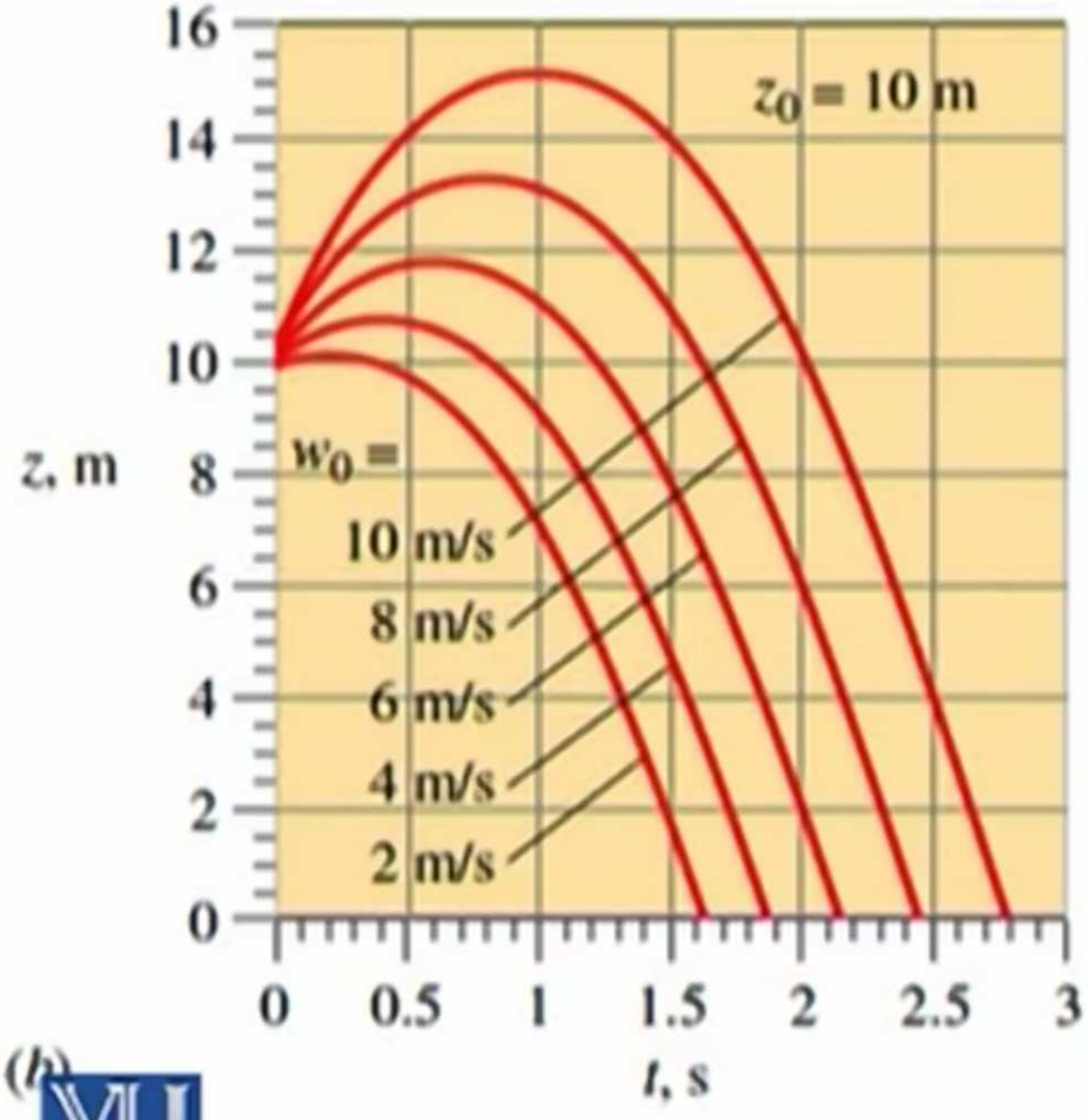
Non Dimensionalisation Of Equations



(*b*)

Trajectories of a steel ball falling in a vacuum: (*a*) w_0 fixed at 4 m/s, and (*b*) z_0 fixed at 10 m (Example 7–3).

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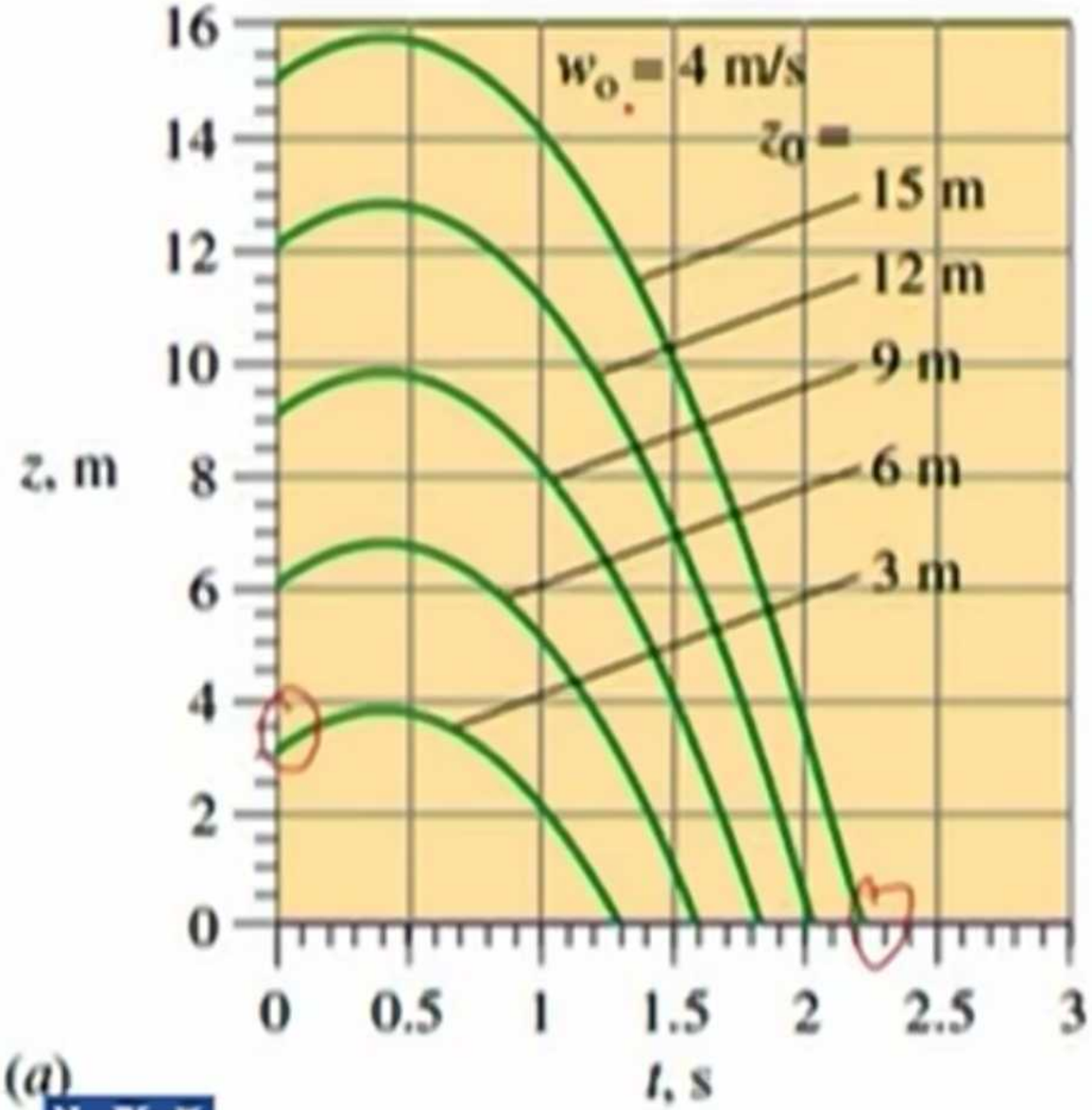
Non Dimensionalisation Of Equations



Trajectories of a steel ball falling in a vacuum: (*a*) w_0 fixed at 4 m/s, and (*b*) z_0 fixed at 10 m (Example 7–3).

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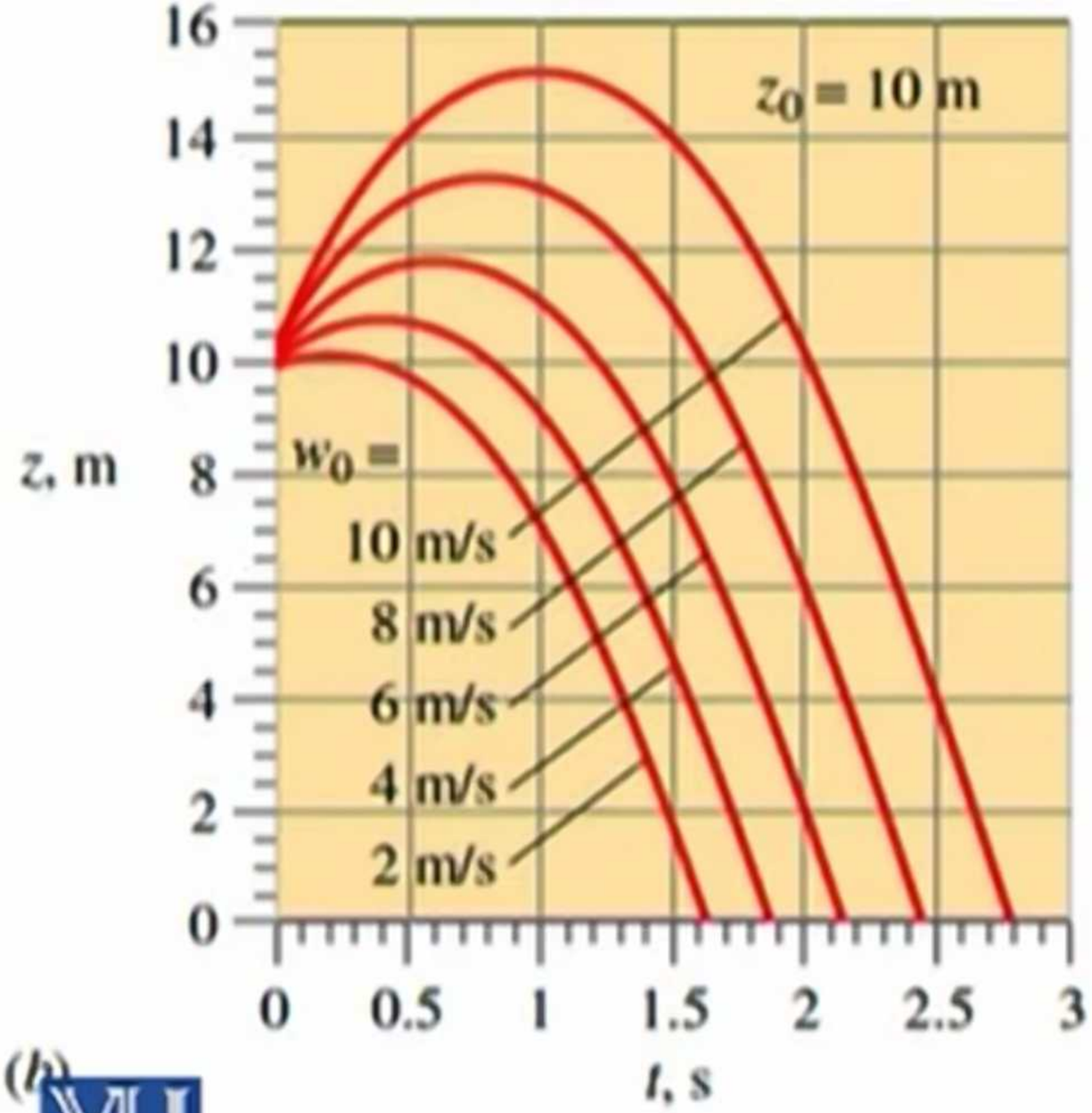
Non Dimensionalisation Of Equations



(b)

Trajectories of a steel ball falling in a vacuum: (*a*) w_0 fixed at 4 m/s, and (*b*) z_0 fixed at 10 m (Example 7–3).

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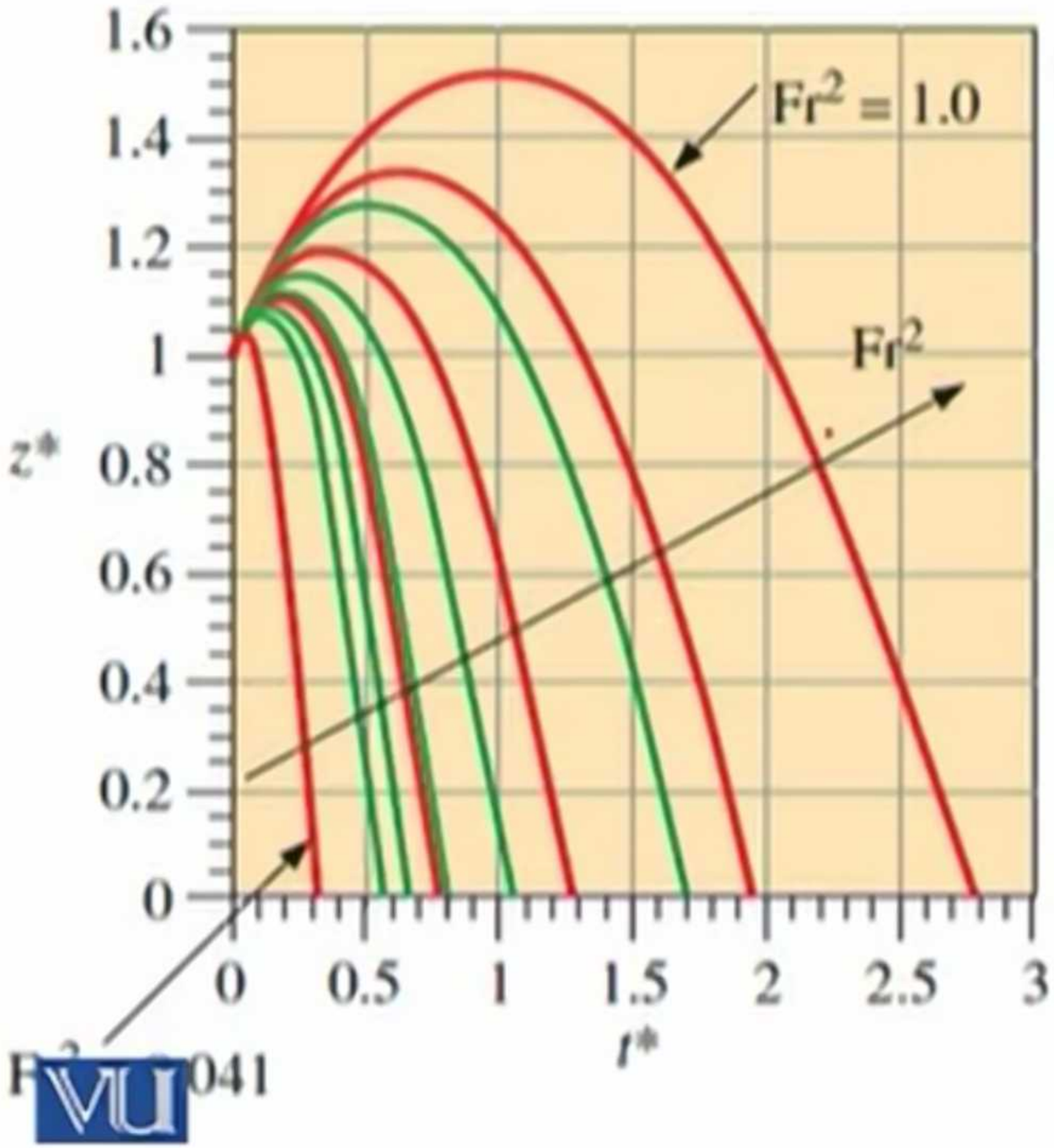
Non Dimensionalisation Of Equations



Trajectories of a steel ball falling in a vacuum. Data of Fig. 7–12*a* and *b* are nondimensionalized and combined onto one plot.

LECTURE 105

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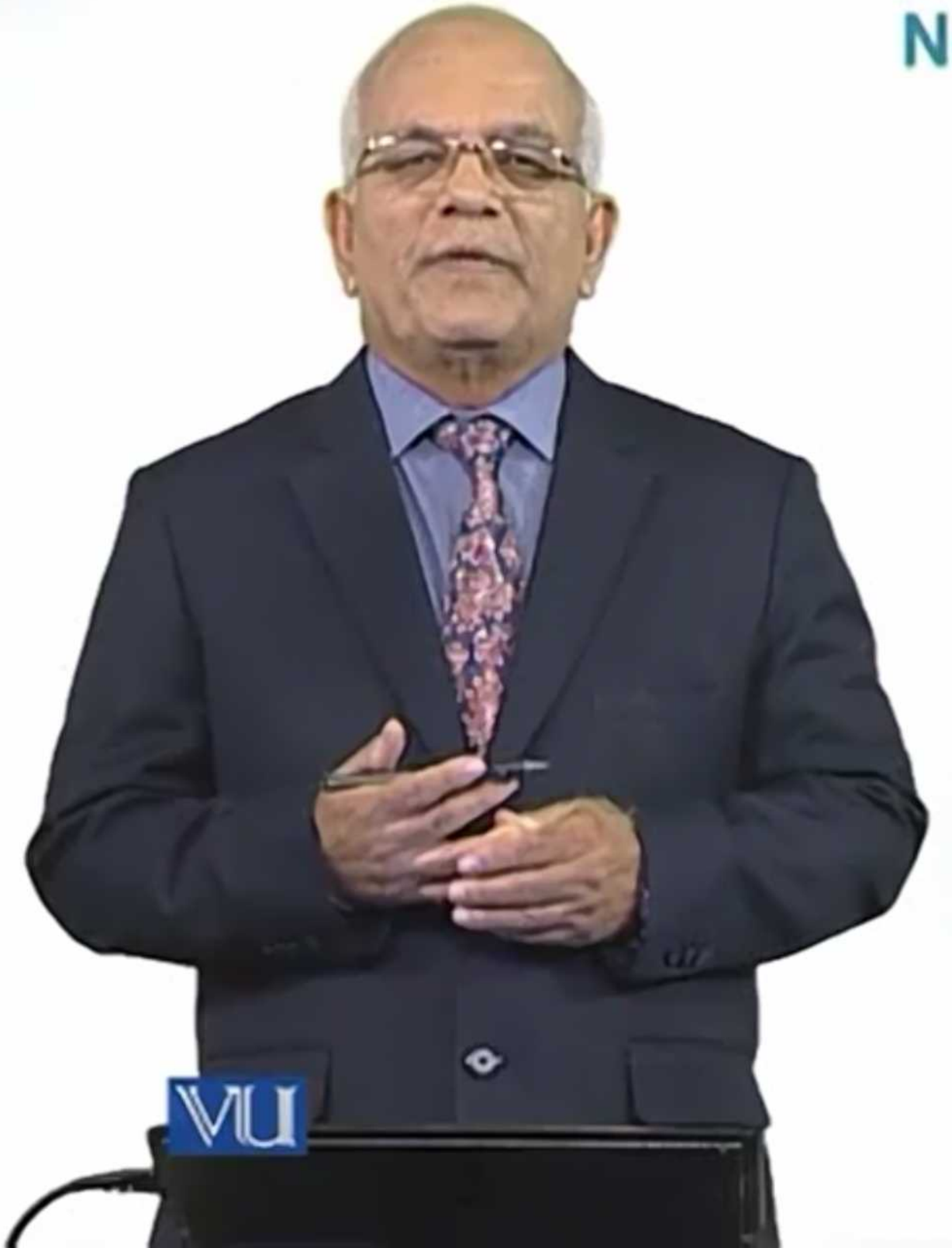
Non Dimensionalisation of Equations



Advanced Fluid Dynamics

Non Dimensionalisation Of Equations

EXAMPLE 7–4 **Extrapolation of Nondimensionalized Data**

The gravitational constant at the surface of the moon is only about one-sixth of that on earth. An astronaut on the moon throws a baseball at an initial speed of 21.0 m/s at a 5° angle above the horizon and at 2.0 m above the moon's surface (Fig. 7–14). (*a*) Using the dimensionless data of Example 7–3 shown in Fig. 7–13, predict how long it takes for the baseball to fall to the ground. (*b*) Do an *exact* calculation and compare the result to that of part (*a*).

Properties The gravitational constant on the moon is $g_{moon} \cong 9.81/6 = 1.63\ \text{m/s}^2$.

Analysis (*a*) The Froude number is calculated based on the value of g_{moon} and the vertical component of initial speed,

$$w_0 = (21.0\ \text{m/s})\sin(5°) = 1.830\ \text{m/s}$$

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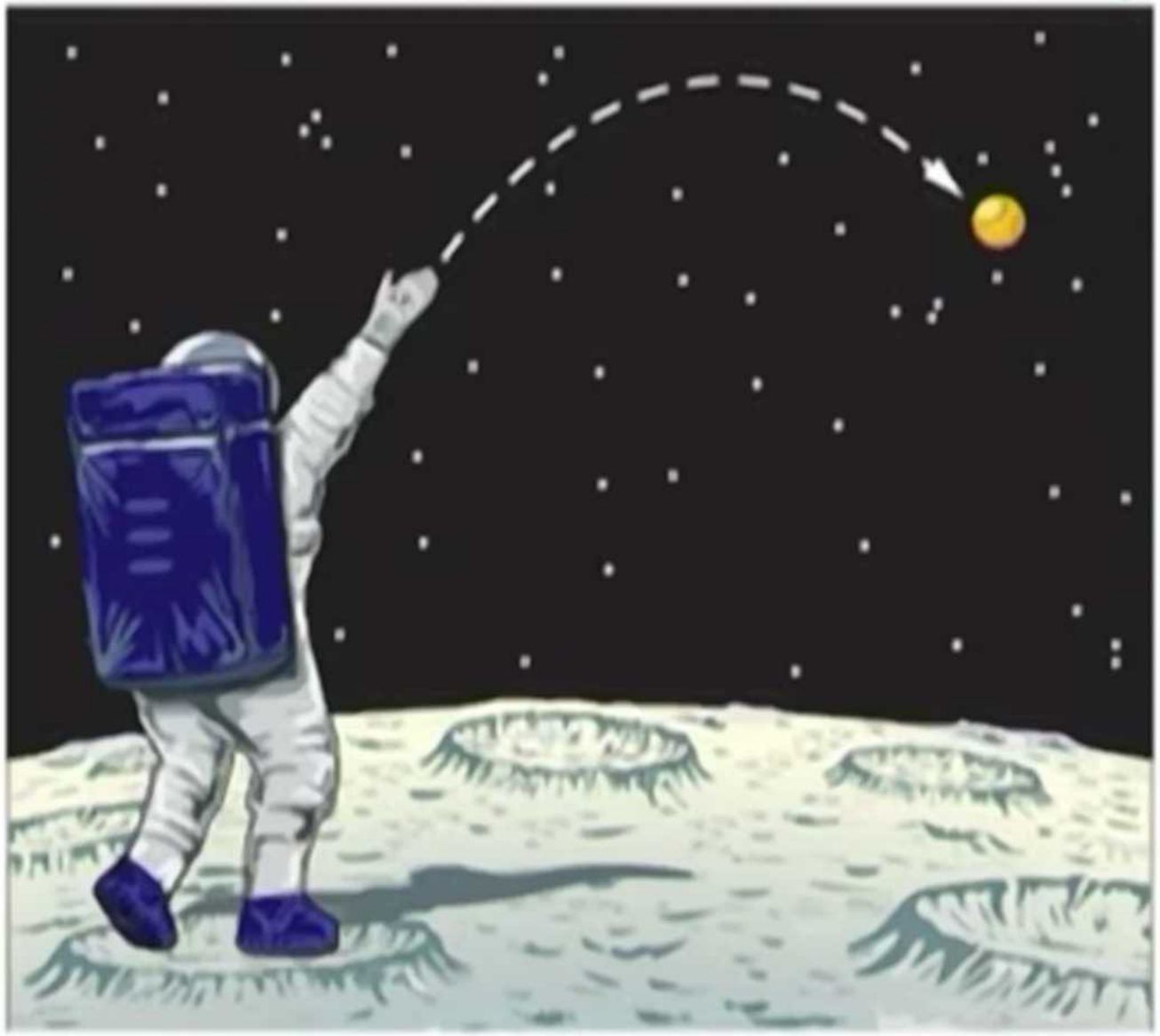
Non Dimensionalisation Of Equations



Throwing a baseball on the moon

Advanced Fluid Dynamics

Non Dimensionalisation Of Equations

$$\mathrm{Fr}^2 = \frac{w_0^2}{g_{\text{moon}} z_0} = \frac{(1.830\ \text{m/s})^2}{(1.63\ \text{m/s}^2)(2.0\ \text{m})} = 1.03$$

This value of Fr^2 is nearly the same as the largest value plotted in Fig. 7–13. Thus, in terms of dimensionless variables, the baseball strikes the ground at $t^* \cong 2.75$, as determined from Fig. 7–13. Converting back to dimensional variables using Eq. 7–6,

Estimated time to strike the ground: $$t = \frac{t^* z_0}{w_0} = \frac{2.75(2.0\ \text{m})}{1.830\ \text{m/s}} = 3.01\ \text{s}$$

(*b*) An exact calculation is obtained by setting $\dot{z}$ equal to zero in Eq. 7–5 and solving for time t (using the quadratic formula),

Advanced Fluid Dynamics

Non Dimensionalisation Of Equations

Exact time to strike the ground:

$$t = \frac{w_0 + \sqrt{w_0^2 + 2z_0 g}}{g}$$

$$= \frac{1.830 \text{ m/s} + \sqrt{(1.830 \text{ m/s})^2 + 2(2.0 \text{ m})(1.63 \text{ m/s}^2)}}{1.63 \text{ m/s}^2} = 3.05 \text{ s}$$

Discussion If the Froude number had landed between two of the trajectories of Fig. 7–13, interpolation would have been required. Since some of the numbers are precise to only two significant digits, the small difference between the results of part (*a*) and part (*b*) is of no concern. The final result is $t = 3.0$ s to two significant digits.

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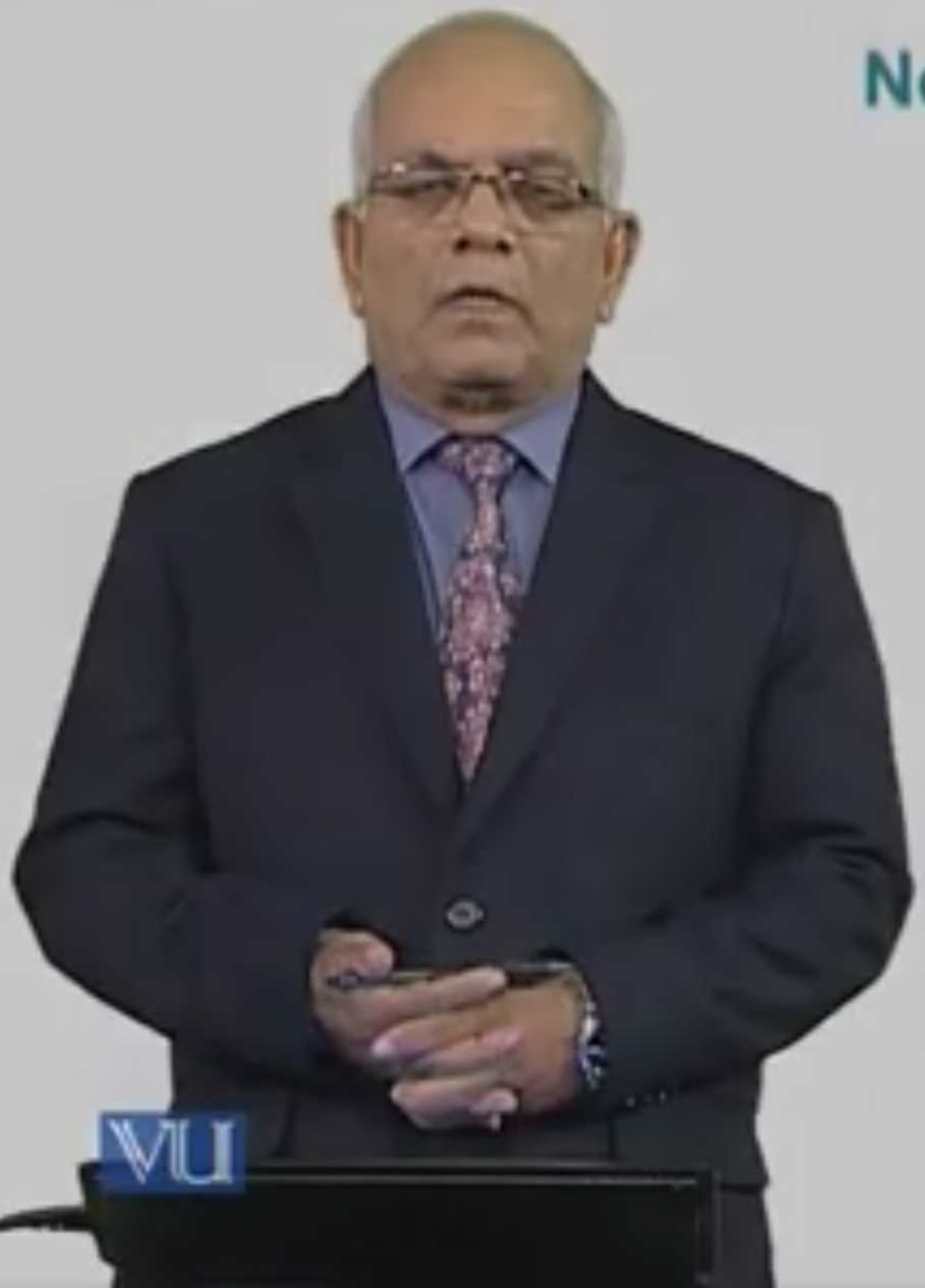
Non Dimensionalisation of Equations



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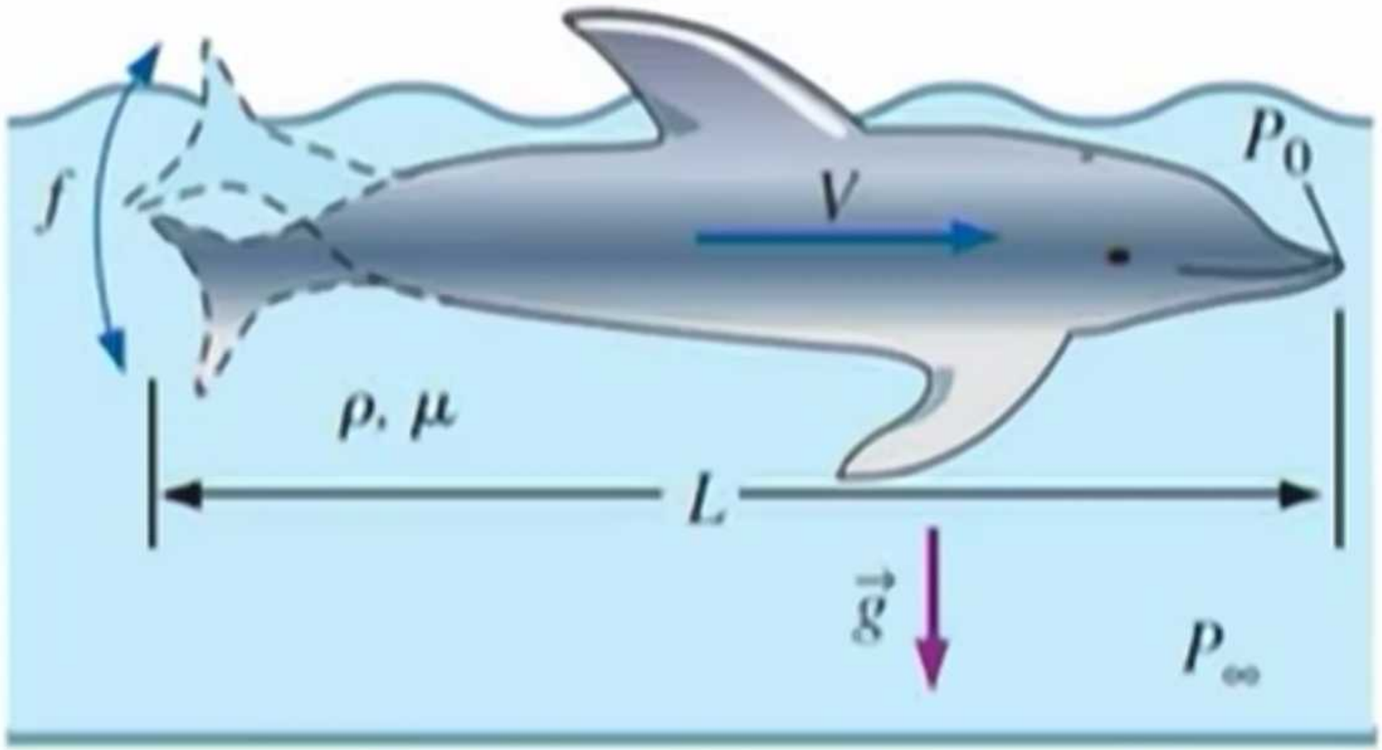
Non Dimensionalisation Of Equations



$$\mathrm{Re} = \frac{\rho V L}{\mu}$$

$$\mathrm{Fr} = \frac{V}{\sqrt{gL}}$$

$$\mathrm{St} = \frac{fL}{V}$$

$$\mathrm{Eu} = \frac{P_0 - P_\infty}{\rho V^2}$$

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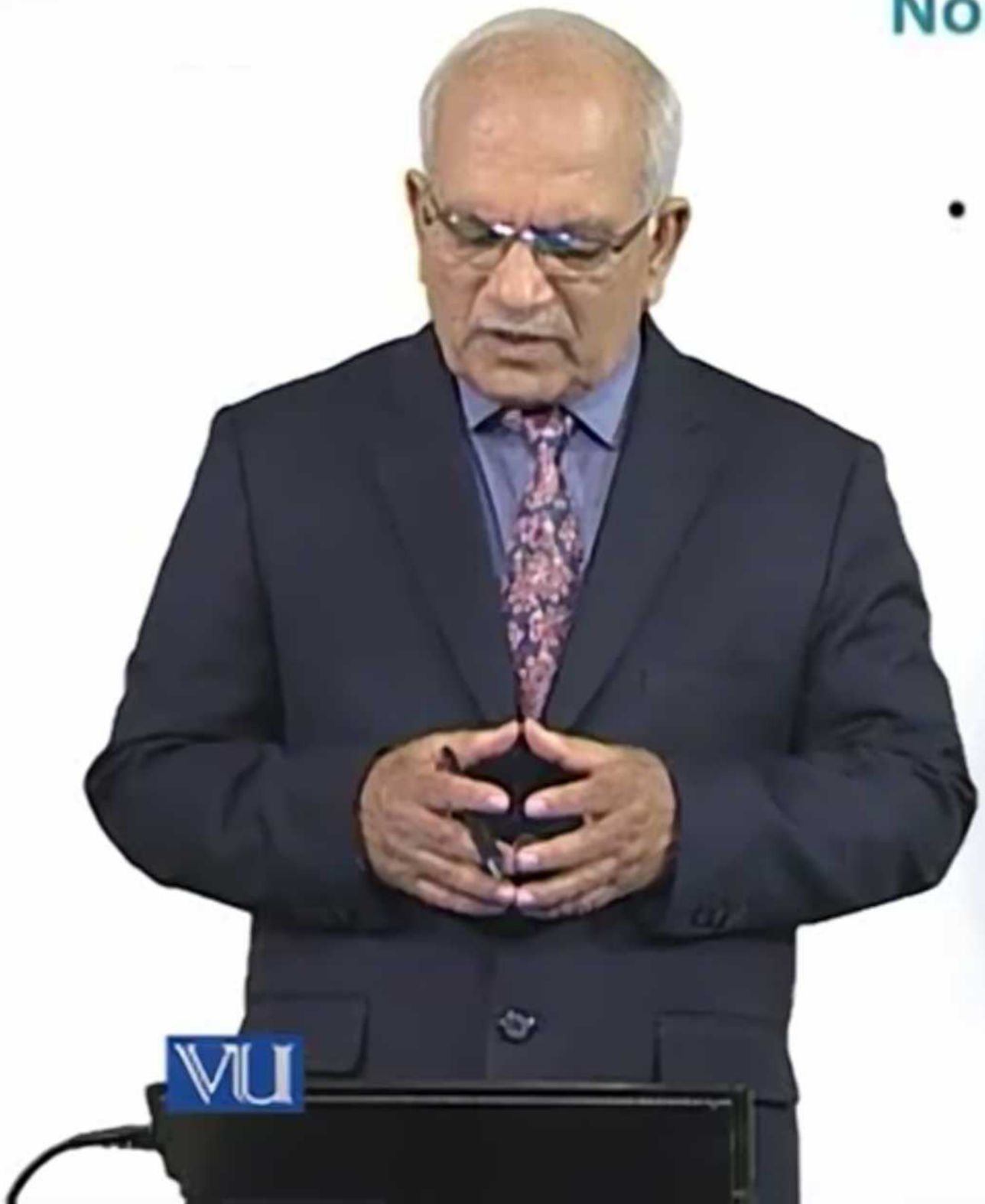

Non Dimensionalisation of Equations

- In a general unsteady fluid flow problem with a free surface, the scaling parameters include a characteristic length L, a characteristic velocity V, a characteristic frequency f, and a reference pressure difference $P_0 - P_\infty$.

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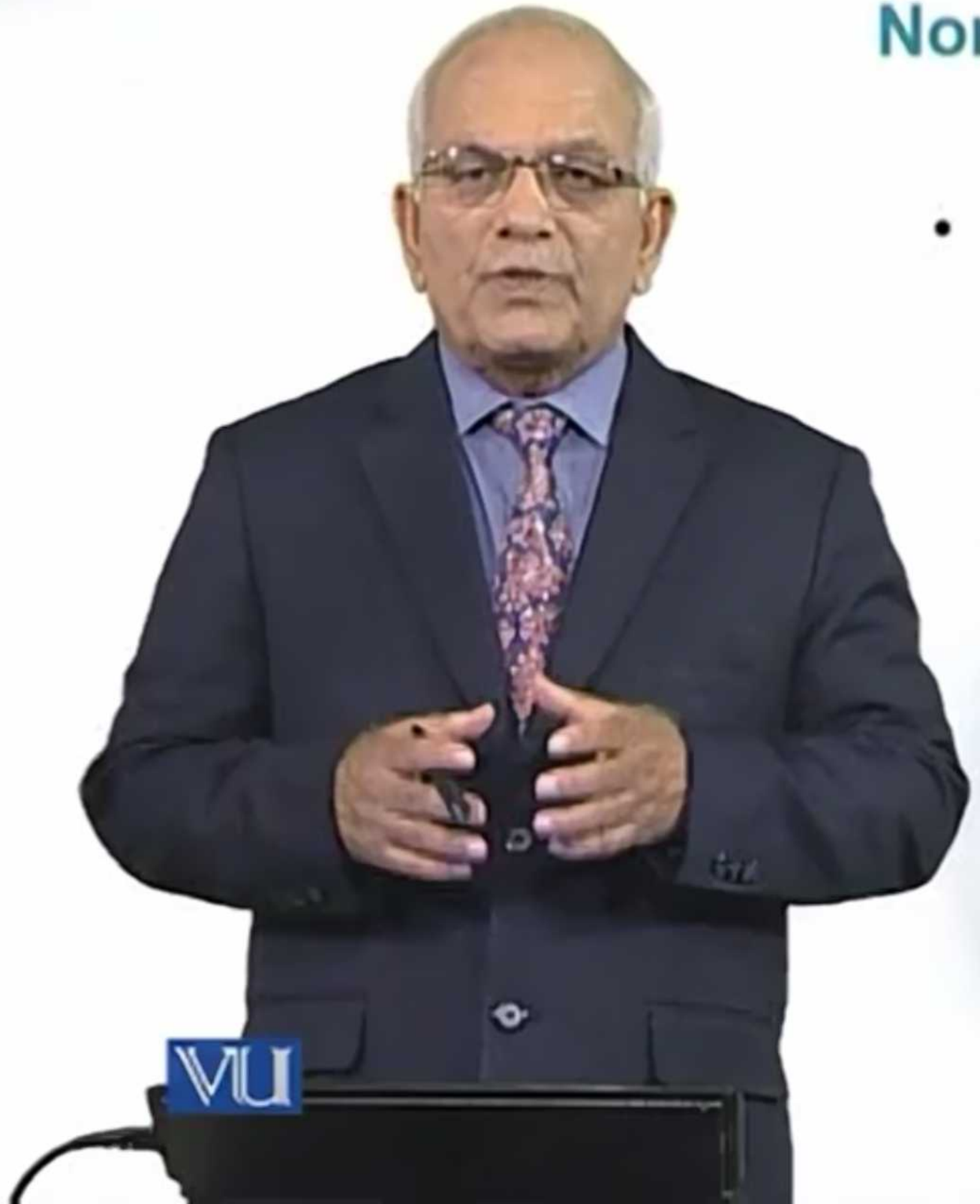

Non Dimensionalisation of Equations

- **Nondimensionaliza-tion of the differential equations of fluid flow produces four dimensionless parameters: the Reynolds number, Froude number, Strouhal number, and Euler number**

Advanced Fluid Dynamics

DIMENSIONAL ANALYSIS AND SIMILARITY

- In most experiments, to save time and money, tests are performed on a geometrically scaled model, rather than on the full-scale prototype.
- In such cases, care must be taken to properly scale the results. We introduce here a powerful technique called dimensional analysis.

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DIMENSIONAL ANALYSIS AND SIMILARITY

- The three primary purposes of dimensional analysis are
- To generate nondimensional parameters that help in the design of experiments (physical and/or numerical) and in the reporting of experimental results

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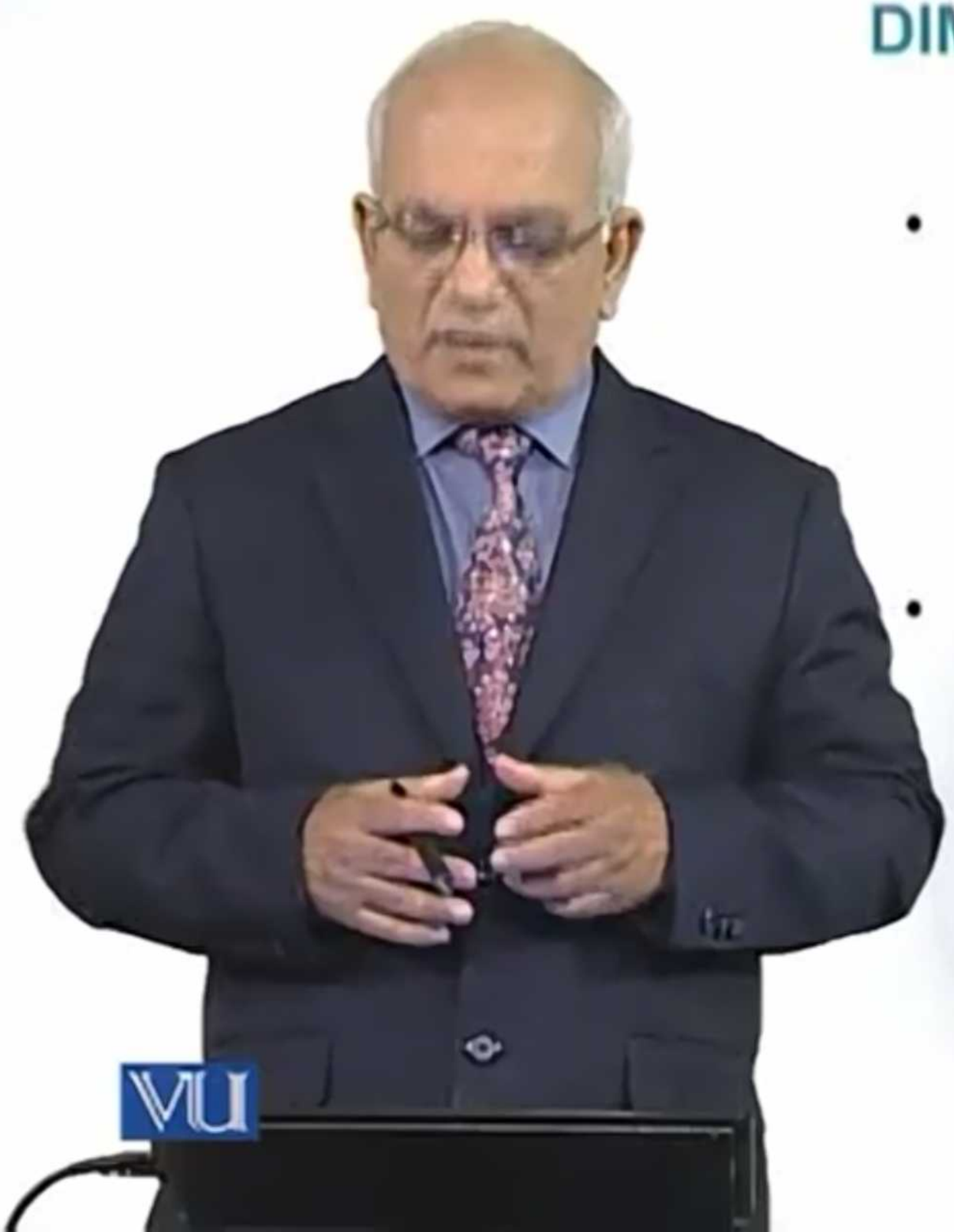

DIMENSIONAL ANALYSIS AND SIMILARITY

- To obtain scaling laws so that prototype performance can be predicted from model performance
- To (sometimes) predict trends in the relationship between parameters
- The principle of similarity

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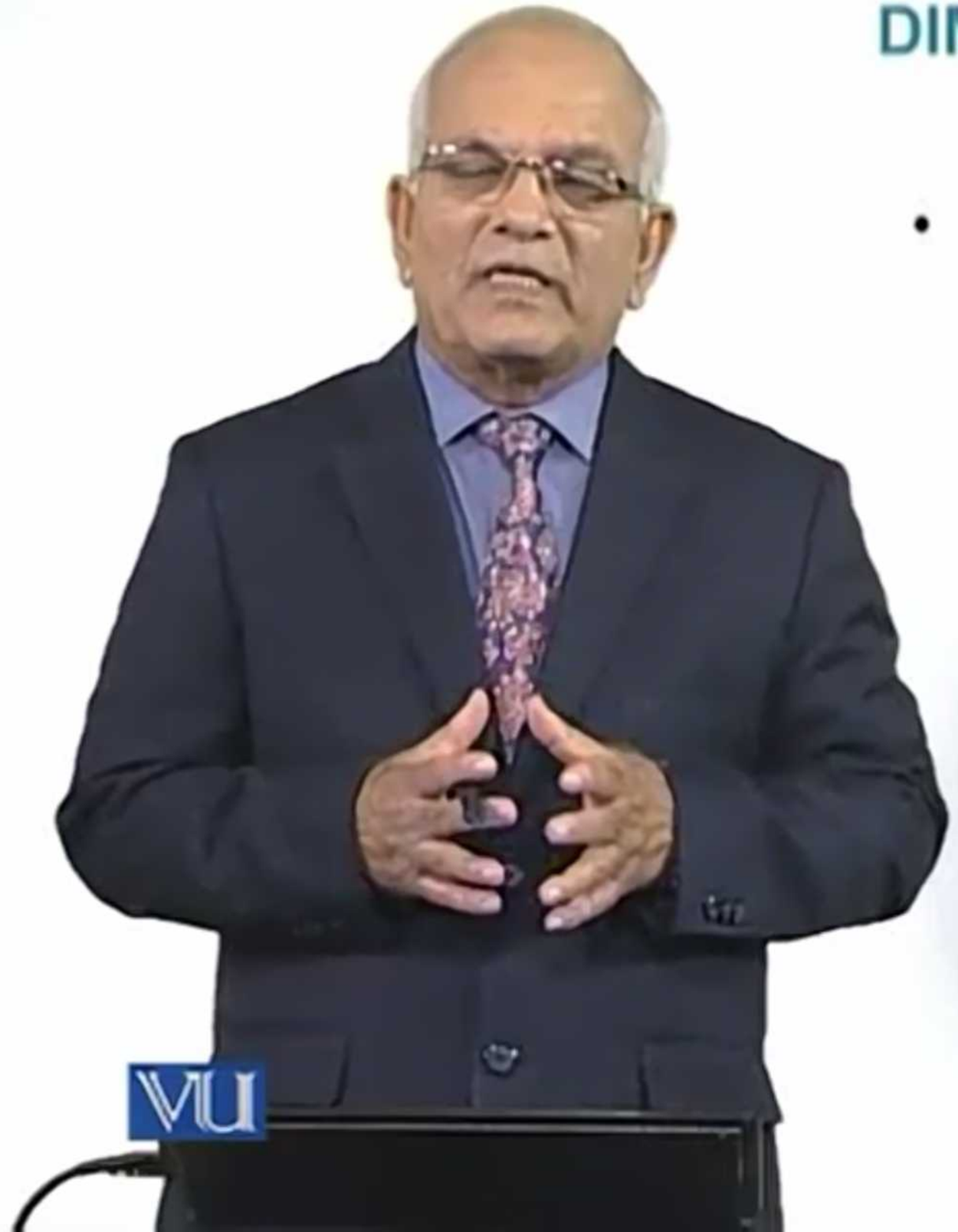

DIMENSIONAL ANALYSIS AND SIMILARITY

- **(2) Kinematic similarity—the velocity at any point in the model flow must be proportional (by a constant scale factor) to the velocity at the corresponding point in the prototype flow.**

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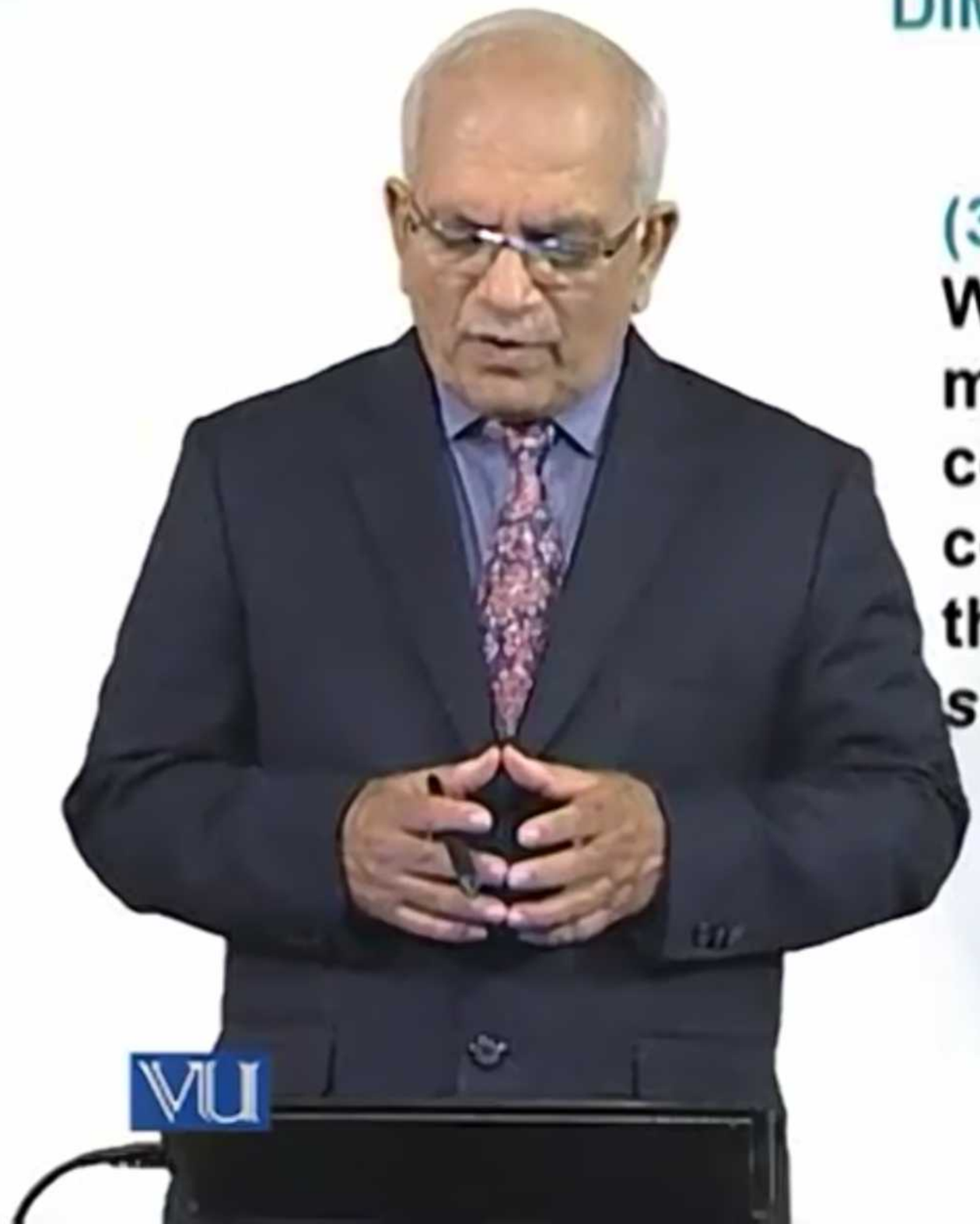

DIMENSIONAL ANALYSIS AND SIMILARITY

(3) dynamic similarity—When all *forces* in the model flow scale by a constant factor to corresponding forces in the prototype flow (*force-scale* equivalence).

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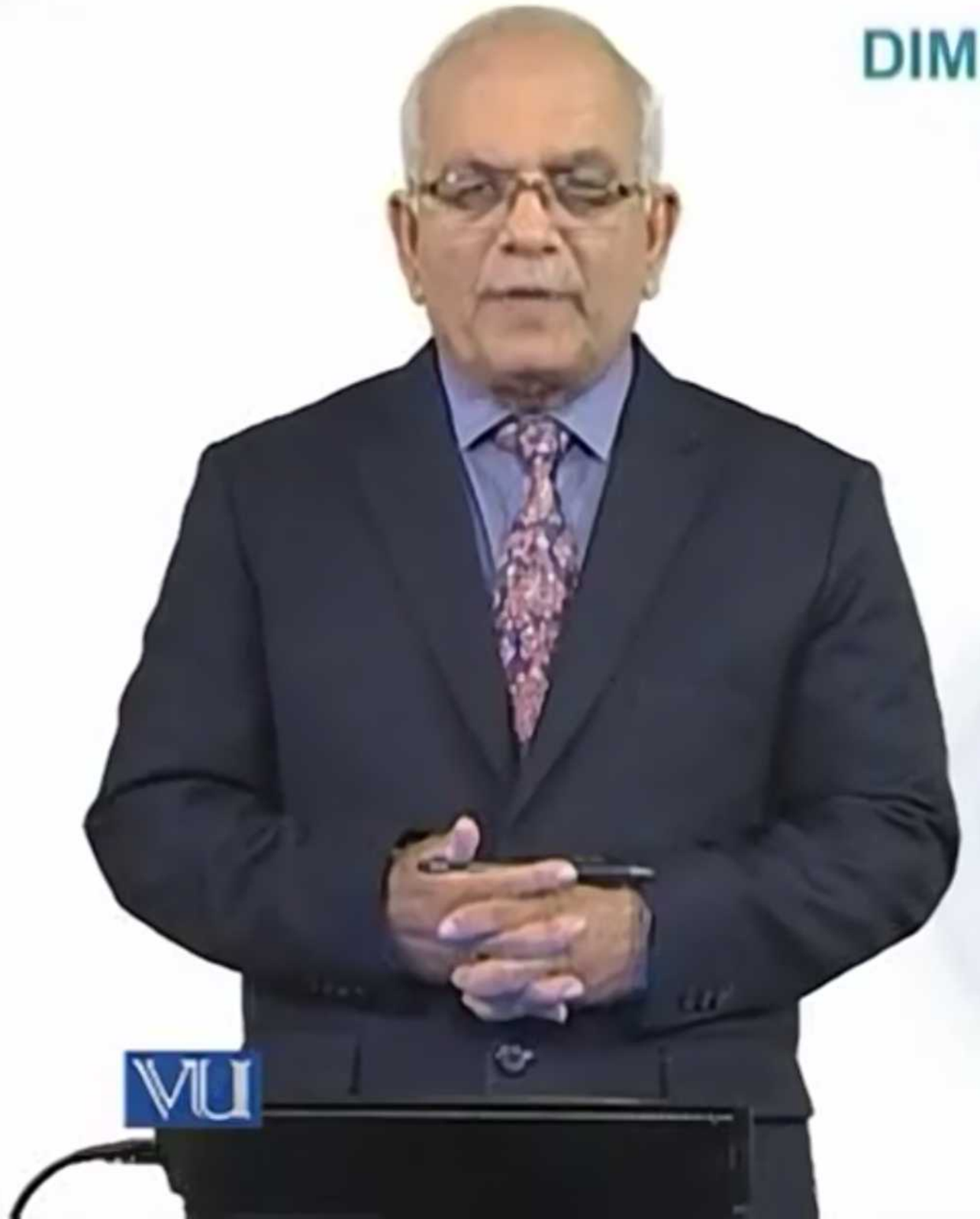
DIMENSIONAL ANALYSIS AND SIMILARITY



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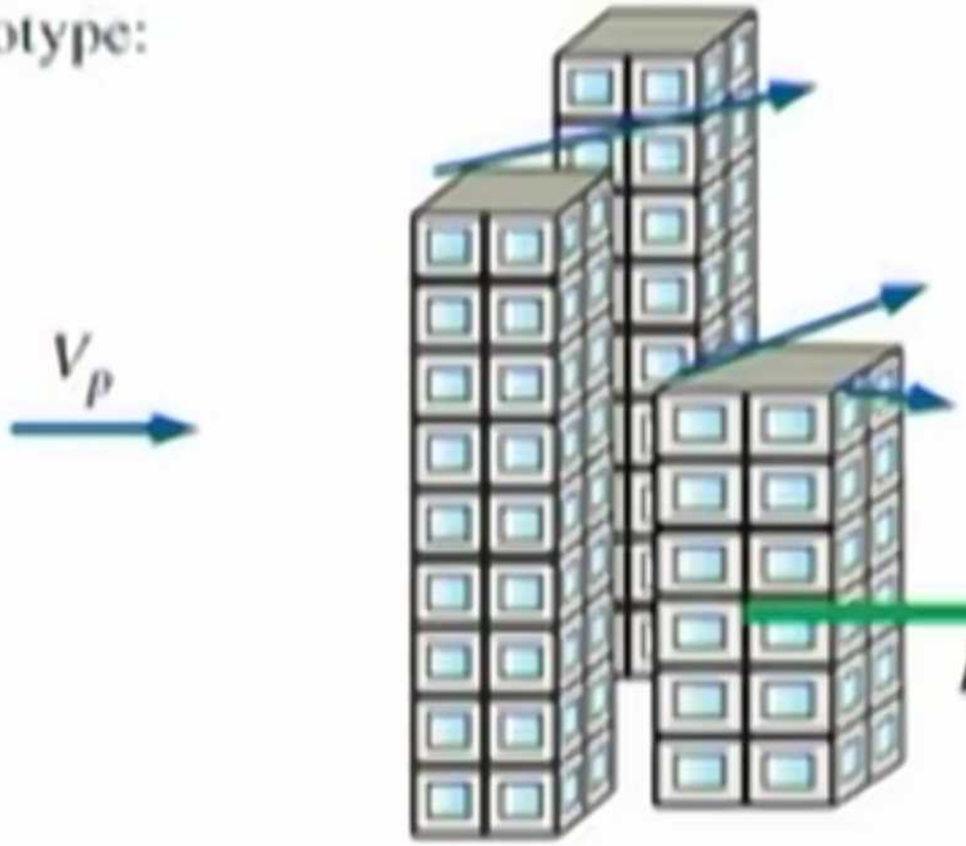
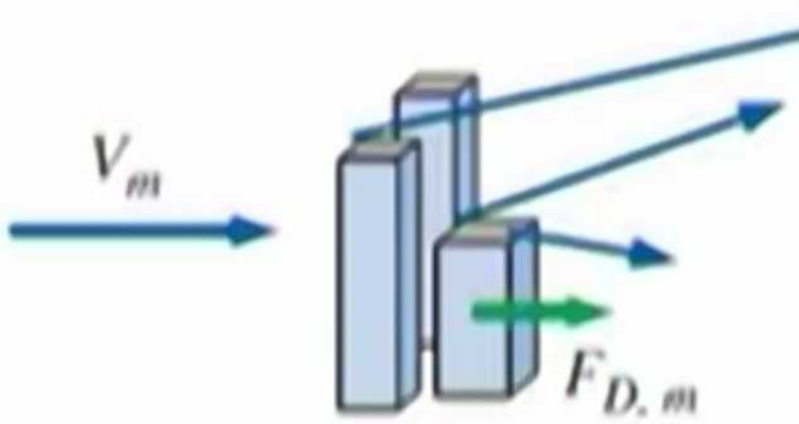
DIMENSIONAL ANALYSIS AND SIMILARITY

Prototype:



Model:



Kinematic similarity

is achieved when, at all locations, the speed in the model flow is proportional to that at corresponding locations in the prototype flow, and points in the same direction.

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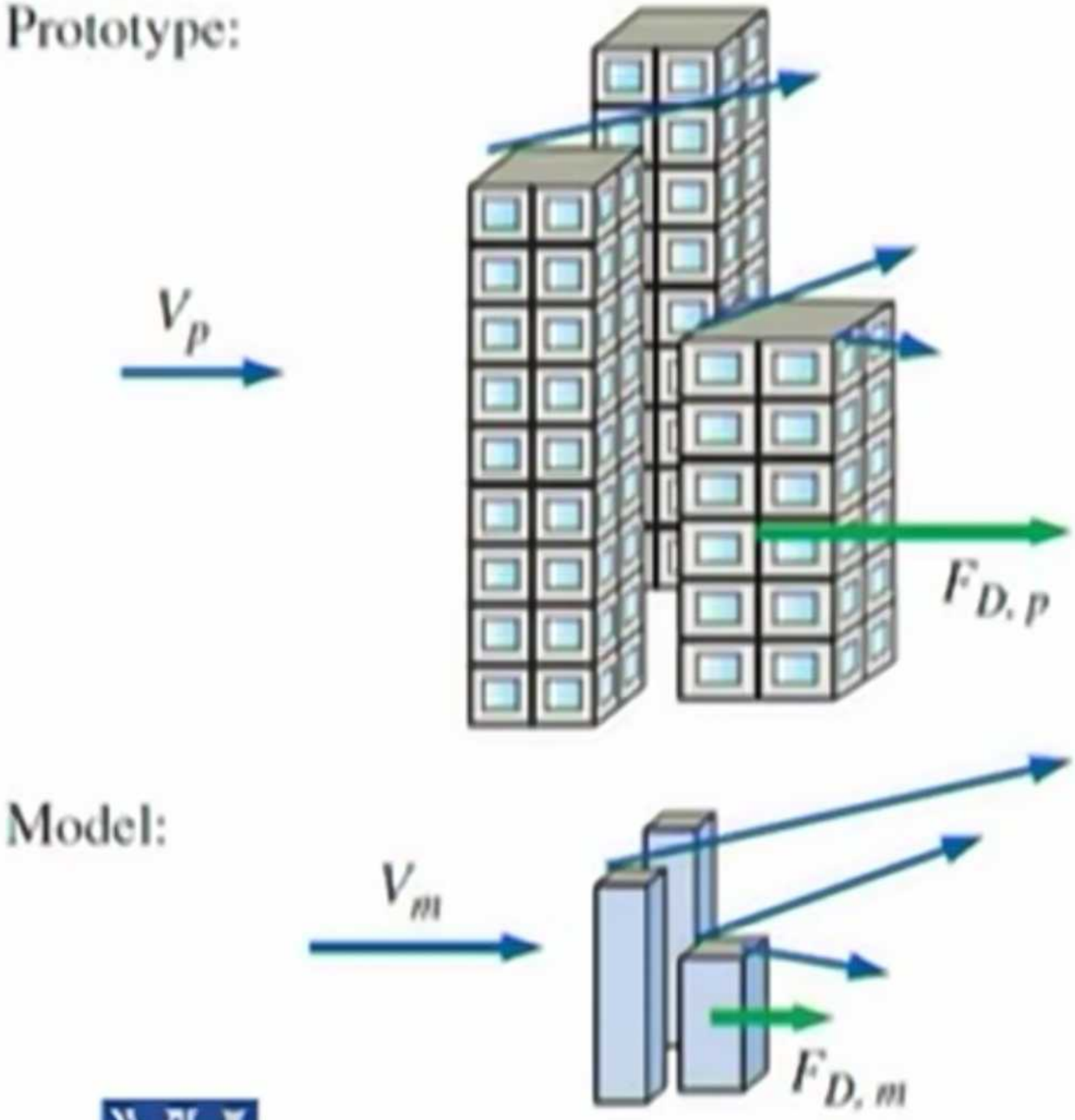
DIMENSIONAL ANALYSIS AND SIMILARITY



- **In a general flow field, complete similarity between a model and prototype is achieved only when there is geometric, kinematic, and dynamic similarity.**

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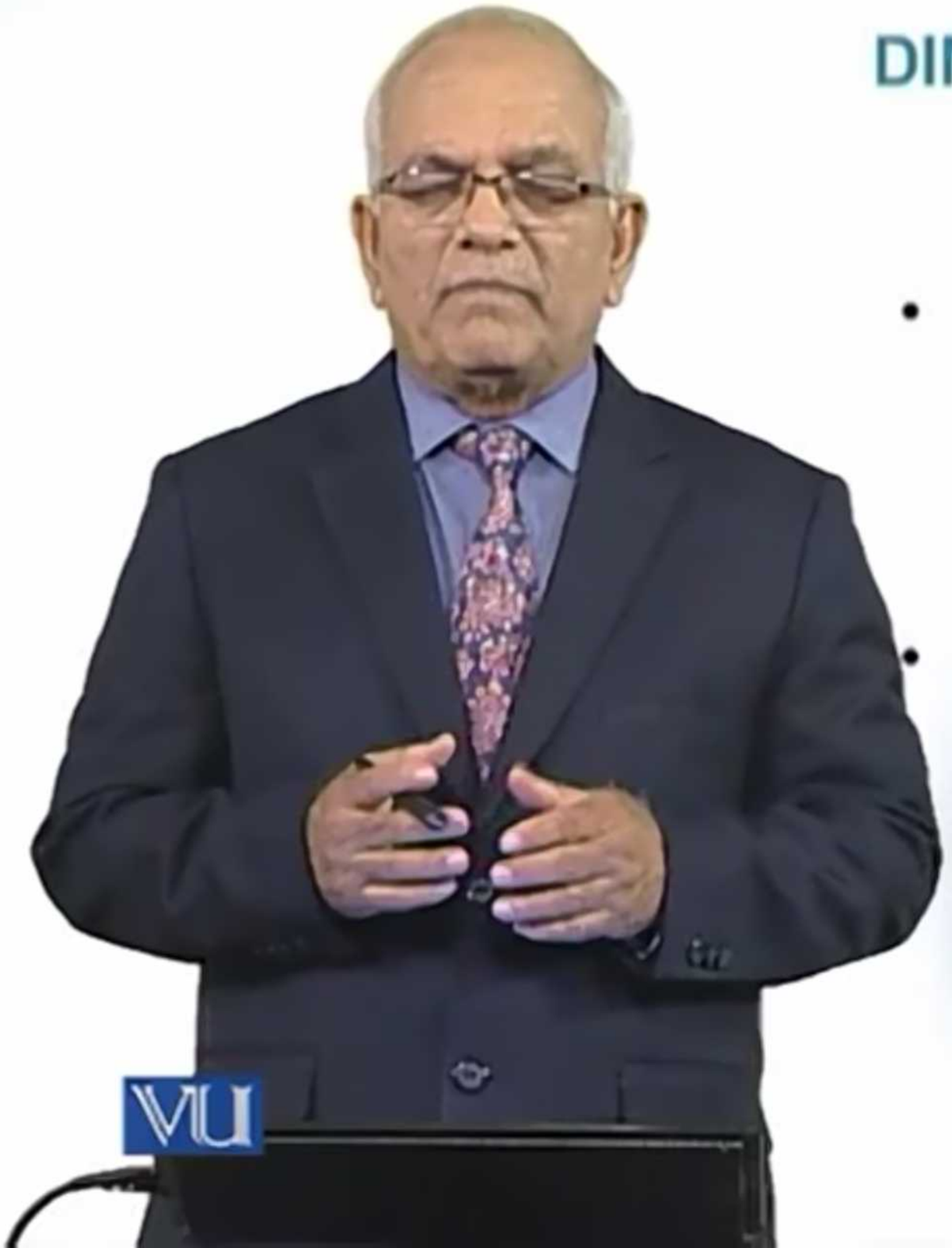

DIMENSIONAL ANALYSIS AND SIMILARITY

- We let uppercase Greek letter Pi (Π) denote a nondimensional parameter.
- In a general dimensional analysis problem, there is one Π that we call the dependent Π, giving it the notation Π_1.

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DIMENSIONAL ANALYSIS AND SIMILARITY

- **The parameter Π_1 is in general a function of several other Π's, which we call independent Π's.**

Functional relationship between Π*'s:* $$\Pi_1 = f(\Pi_2, \Pi_3, \ldots, \Pi_k)$$

To ensure complete similarity, the model and prototype must be geometrically similar, and all independent groups must match between model and prototype.

To achieve similarity

$$\text{If} \quad \Pi_{2,m} = \Pi_{2,p} \quad \text{and} \quad \Pi_{3,m} = \Pi_{3,p} \ldots \quad \text{and} \quad \Pi_{k,m} = \Pi_{k,p},$$
$$\text{then} \quad \Pi_{1,m} = \Pi_{1,p} \qquad (7\text{–}12)$$

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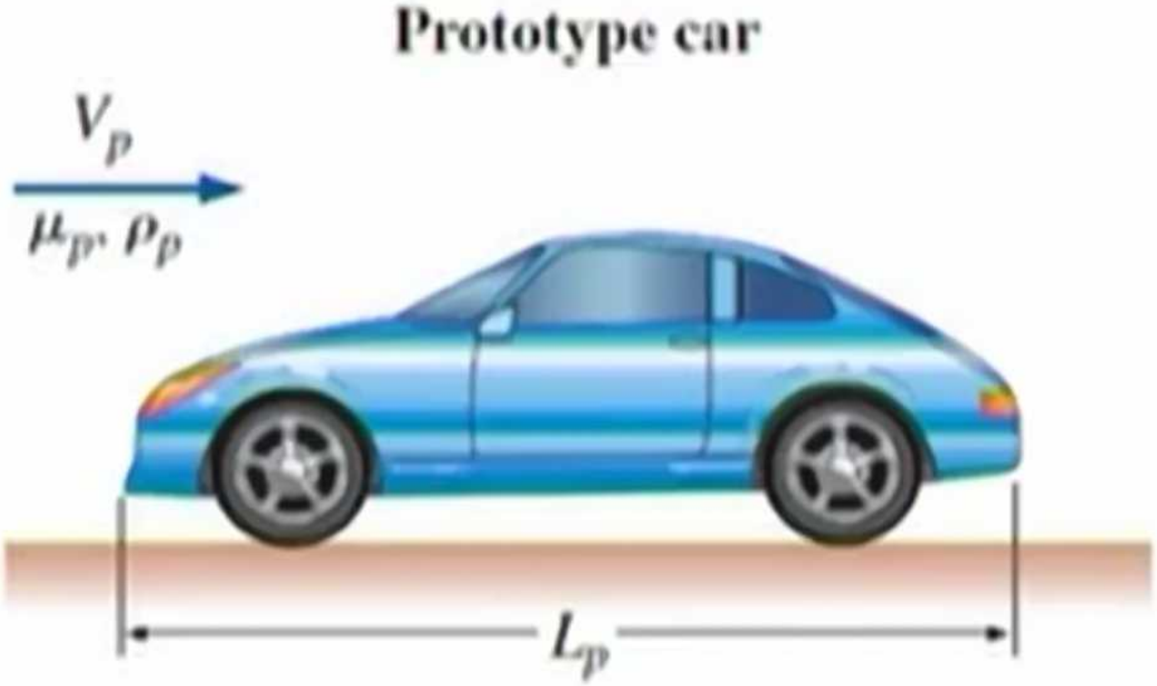
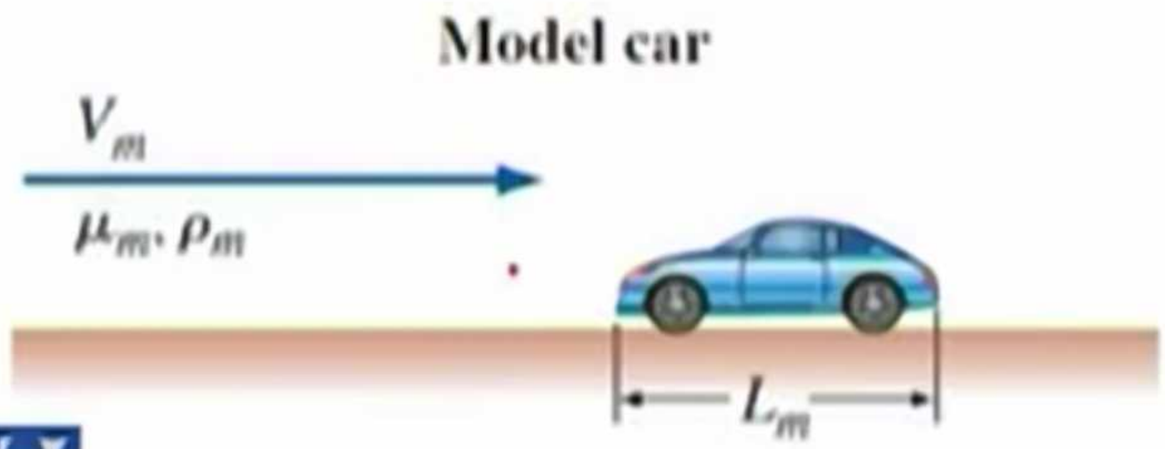
DIMENSIONAL ANALYSIS AND SIMILARITY





Geometric similarity **between a prototype car of length L_p and a model car of length L_m.**

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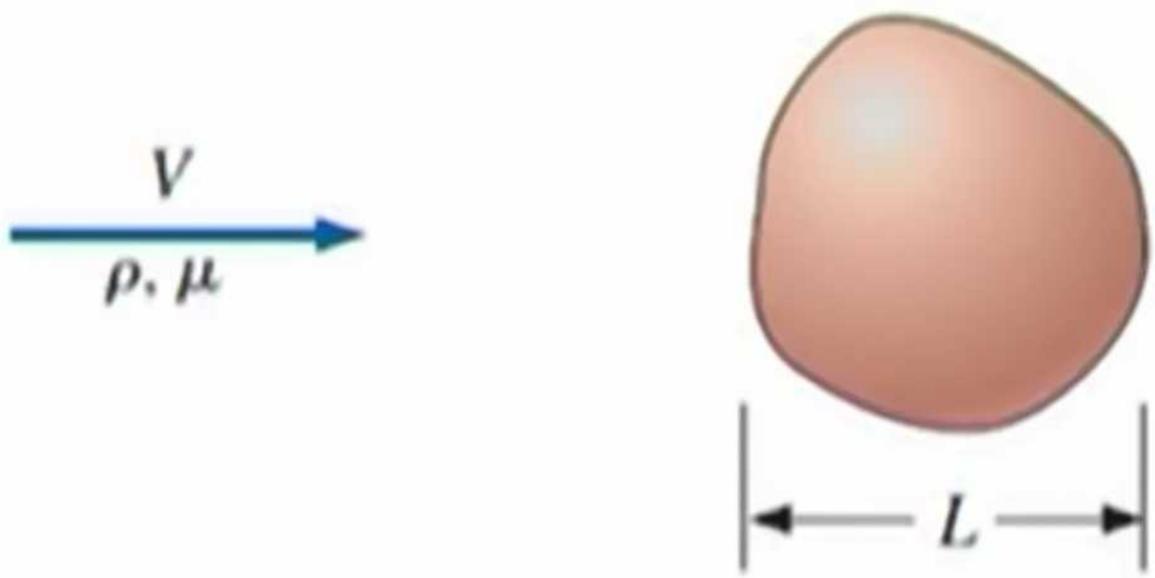
DIMENSIONAL ANALYSIS AND SIMILARITY



$$\mathrm{Re} = \frac{\rho V L}{\mu} = \frac{V L}{\nu}$$

The *Reynolds number* Re is formed by the ratio of density, characteristic speed, and characteristic length to viscosity. Alternatively, it is the ratio of characteristic speed and length to *kinematic viscosity*, defined as $\nu = \mu/\rho$.

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DIMENSIONAL ANALYSIS AND SIMILARITY

$$\Pi_1 = f(\Pi_2) \qquad \text{where} \qquad \Pi_1 = \frac{F_D}{\rho V^2 L^2} \qquad \text{and} \qquad \Pi_2 = \frac{\rho V L}{\mu}$$

The Reynolds number is the most well known and useful dimensionless parameter in all of fluid mechanics.

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DIMENSIONAL ANALYSIS AND SIMILARITY

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DIMENSIONAL ANALYSIS AND SIMILARITY

EXAMPLE 7–5 Similarity between Model and Prototype Cars

The aerodynamic drag of a new sports car is to be predicted at a speed of 50.0 mi/h at an air temperature of 25°C. Automotive engineers build a one-fifth scale model of the car to test in a wind tunnel. It is winter and the wind tunnel is located in an unheated building; the temperature of the wind tunnel air is only about 5°C. Determine how fast the engineers should run the wind tunnel in order to achieve similarity between the model and the prototype.

Properties For air at atmospheric pressure and at $T = 25°C$, $\rho = 1.184$ kg/m^3 and $\mu = 1.849 \times 10^{-5}$ kg/m·s. Similarly, at $T = 5°C$, $\rho = 1.269$ kg/m^3 and $\mu = 1.754 \times 10^{-5}$ kg/m·s.

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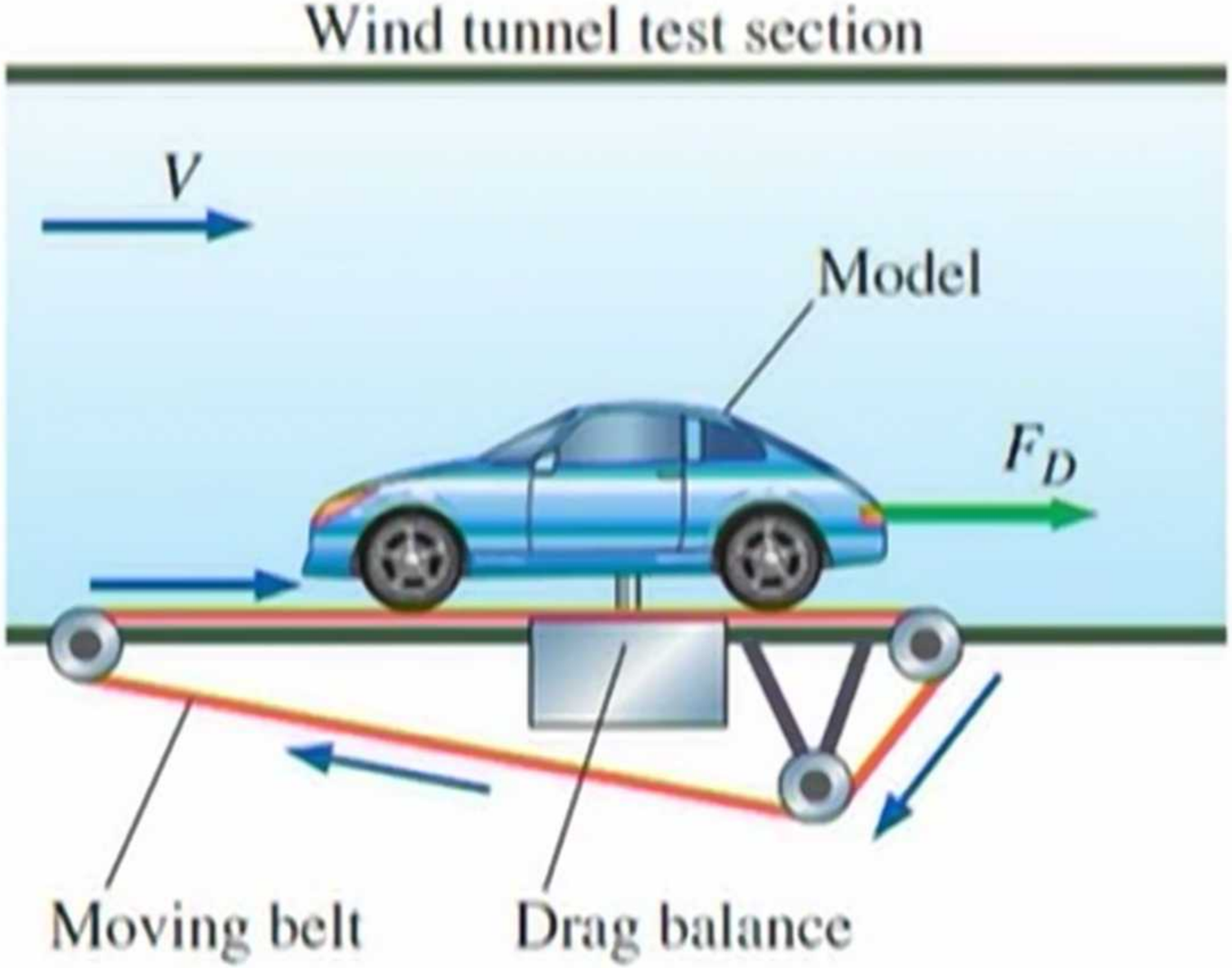

A *drag balance* is a device used in a wind tunnel to measure the aerodynamic drag of a body. When testing automobile models, a *moving belt* is often added to the floor of the wind tunnel to simulate the moving gr (from the car's frame of reference).

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Analysis Since there is only one independent Π in this problem, the similarity equation (Eq. 7–12) holds if $\Pi_{2,m} = \Pi_{2,p}$, where Π_2 is given by Eq. 7–13, and we call it the Reynolds number. Thus, we write

$$\Pi_{2,m} = \mathrm{Re}_m = \frac{\rho_m V_m L_m}{\mu_m} = \Pi_{2,p} = \mathrm{Re}_p = \frac{\rho_p V_p L_p}{\mu_p}$$

DIMENSIONAL ANALYSIS AND SIMILARITY

which we solve for the unknown wind tunnel speed for the model tests, V_m,

$$V_m = V_p\left(\frac{\mu_m}{\mu_p}\right)\left(\frac{\rho_p}{\rho_m}\right)\left(\frac{L_p}{L_m}\right)$$

$$= (50.0\ \text{mi/h})\left(\frac{1.754 \times 10^{-5}\ \text{kg/m·s}}{1.849 \times 10^{-5}\ \text{kg/m·s}}\right)\left(\frac{1.184\ \text{kg/m}^3}{1.269\ \text{kg/m}^3}\right)(5) = \mathbf{221\ mi/h}$$

Thus, to ensure similarity, the wind tunnel should be run at 221 mi/h (to three significant digits). Note that we were never given the actual length of either car, but the ratio of L_p to L_m is known because the prototype is five times larger than the scale model. When the dimensional parameters are rearranged as nondimensional ratios (as done here), the unit system is irrelevant. Since the units in each numerator cancel those in each denominator, no unit conversions are necessary.

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DIMENSIONAL ANALYSIS AND SIMILARITY

Discussion By arranging the dimensional parameters as nondimensional ratios, the units cancel nicely even though they are a mixture of SI and English units. Because both velocity and length are squared in the equation for Π_1, the higher speed in the wind tunnel nearly compensates for the model's smaller size, and the drag force on the model is nearly the same as that on the prototype. In fact, if the density and viscosity of the air in the wind tunnel were *identical* to those of the air flowing over the prototype, the two drag forces would be identical as well (Fig. 7–20).

LECTURE 110

MTH642 FLUID MECHANICS

Advanced Fluid Dynamics

Dimensional Analysis and Similarity

- Nature of Dimensional Analysis
- Buckingham Pi Theorem
- Significant Dimensionless Groups in Fluid Mechanics
- Flow Similarity and Model Studies

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Dimensional Analysis and Similarity

- Understand dimensions, units, and dimensional homogeneity
- Understand benefits of dimensional analysis
- Know how to use the method of repeating variables

Understand the concept of similarity and how to apply it to experimental modeling

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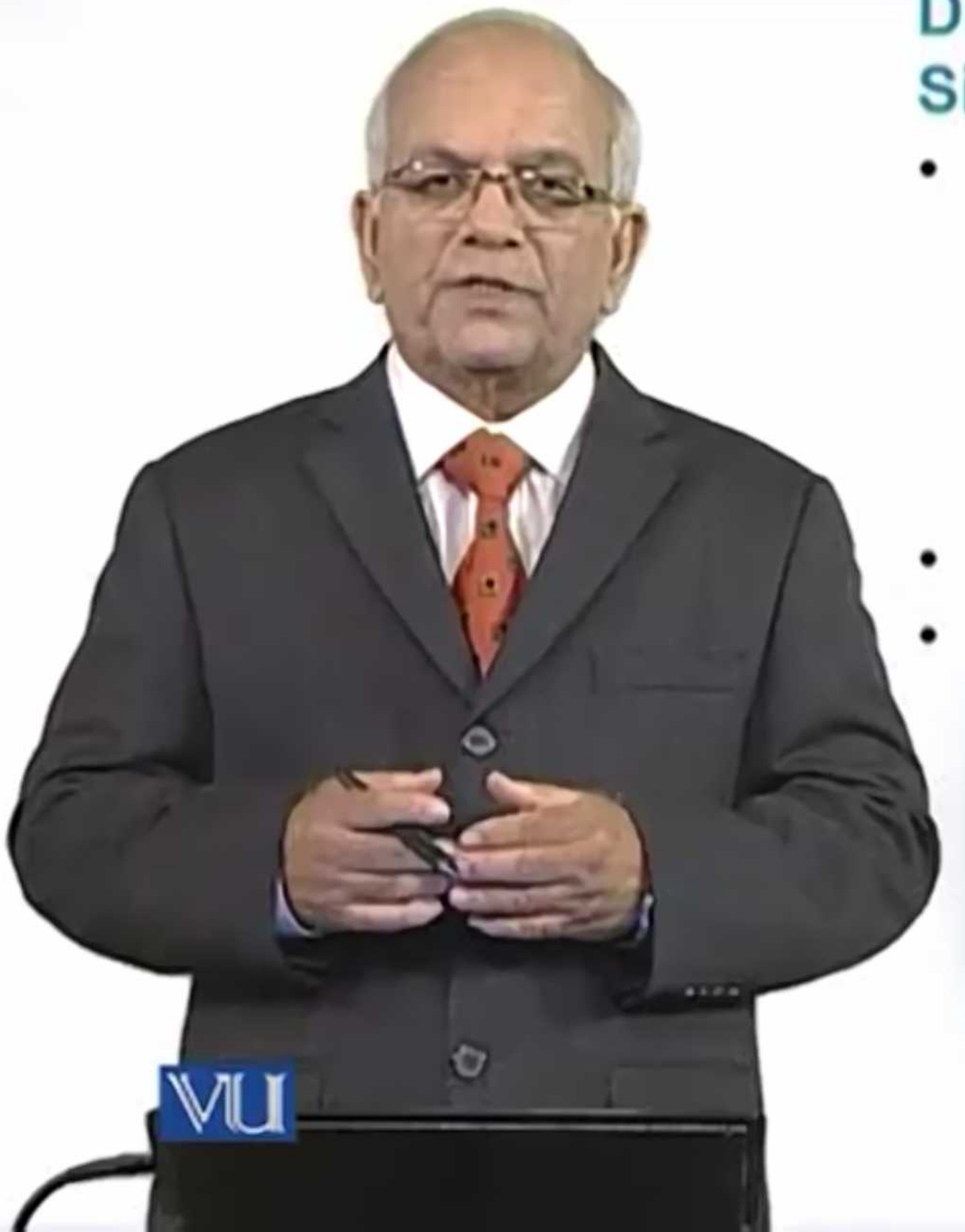

Dimensional Analysis and Similarity

- All non-primary dimensions can be formed by a combination of the 7 primary dimensions
- Examples
- {Velocity} m/sec = {Length/Time} = {L/t}

{Force} N = {Mass Length/Time} = {mL/t^2}

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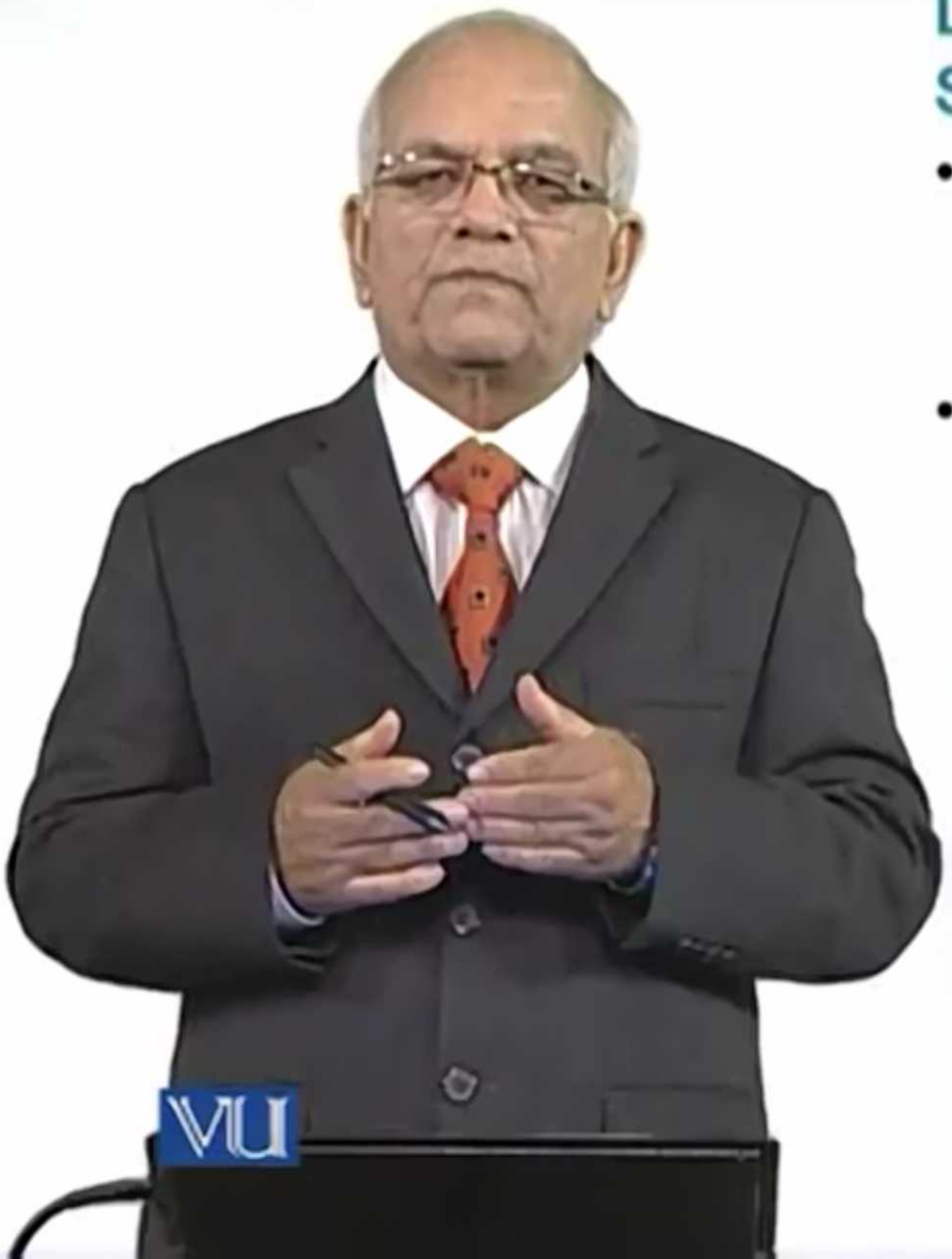

Dimensional Analysis and Similarity

- Every additive term in an equation must have the same dimensions
- Example: Bernoulli equation

$$p + \frac{1}{2}\rho V^2 + \rho g z = C$$

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Dimensional Analysis and Similarity

Nondimensionalization of Equations

$$p + \frac{1}{2}\rho V^2 + \rho g z = C$$

{p} = {force/area}={mass x length/time x 1/length^2} = {m/(t^{2}L)}

{1/2ρV^2} = {mass/length^3 x (length/time)^2} = {m/(t^{2}L)}

{ρgz} = {mass/length^3 x length/time^2 x length} ={m/(t^{2}L)}

VU

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Nondimensionality of Equations

- To nondimensionalize, for example, the Bernoulli equation, the first step is to list primary dimensions of all dimensional variables and constants

{time^2} =

{time^2 x length}

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Dimensional Analysis And Similarity

Nondimensionalization of Equations

For B equation

$$\{p\} = \{m/(t^2L)\}$$
$$\{\rho\} = \{m/L^3\}$$
$$\{V\} = \{L/t\}$$
$$\{g\} = \{L/t^2\}$$
$$\{z\} = \{L\}$$

Next, we need to select Scaling Parameters. For this example, select L, U_0, ρ_0

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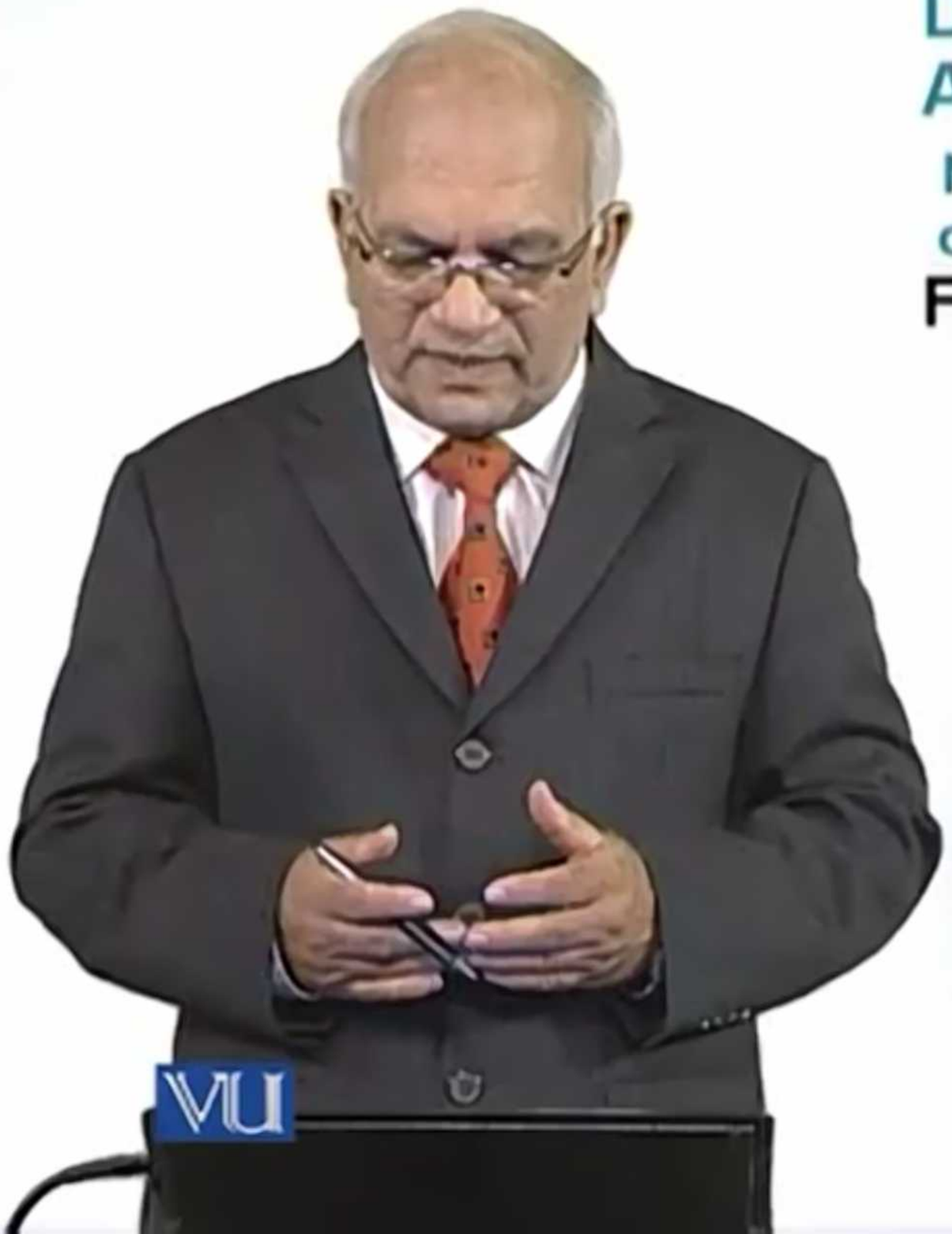

Dimensional Analysis And Similarity

Nondimensionalization of Equations

For B equation

$$\{p\} = \{m/(t^2L)\}$$
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$$\{z\} = \{L\}$$

Next, we need to select Scaling Parameters. For this example, select L, U_0, ρ_0

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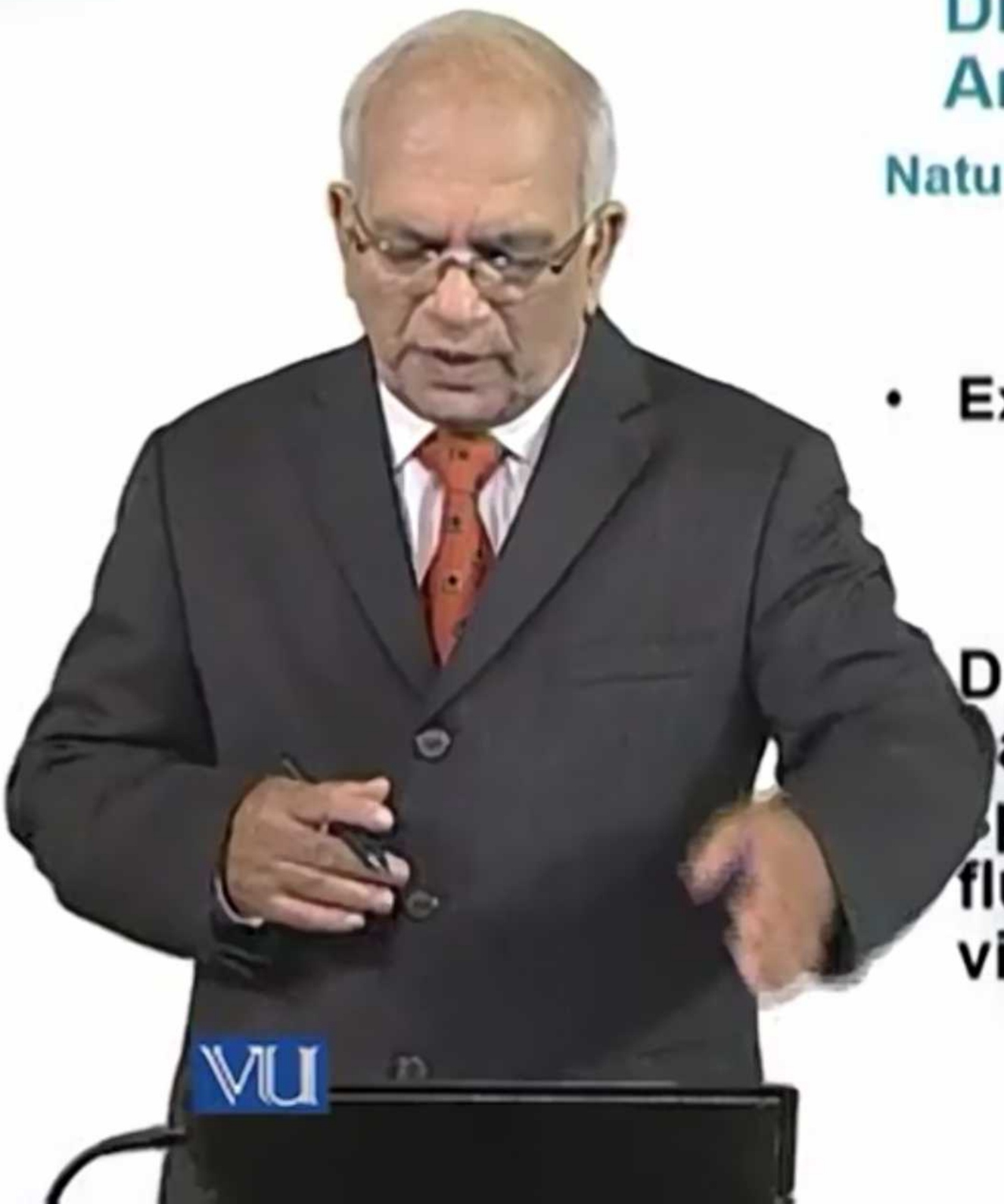
Dimensional Analysis And Similarity

Nature of Dimensional Analysis



- **Example: Drag on a Sphere**

$$F = f(D,\ V,\ \rho,\ \mu)$$

Drag depends on FOUR parameters:
Sphere size (*D*); speed (*V*); fluid density (ρ); fluid viscosity (μ)

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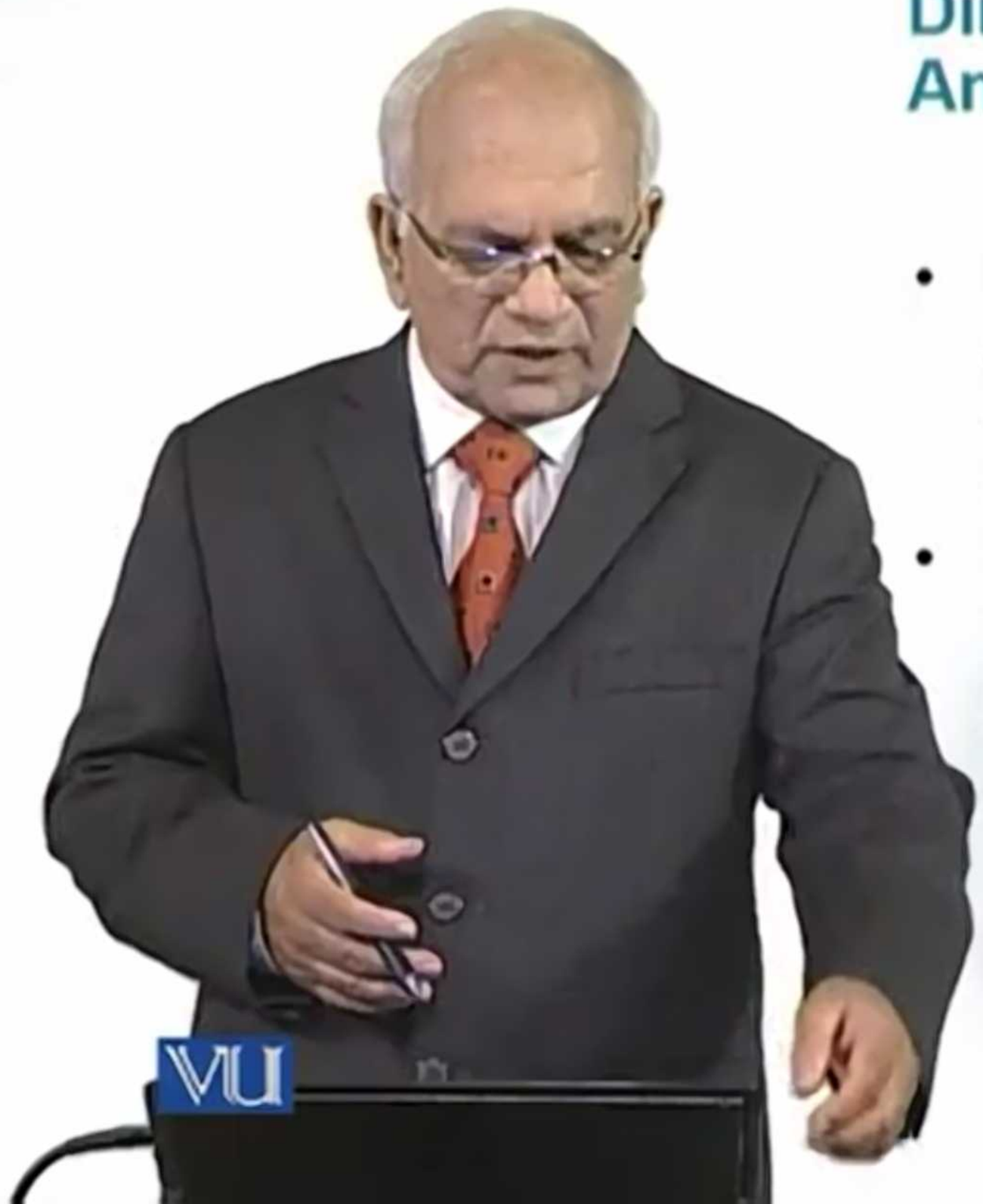
Dimensional Analysis And Similarity



- Difficult to know how to set up experiments to determine dependencies
- Difficult to know how to present results (four graphs?)

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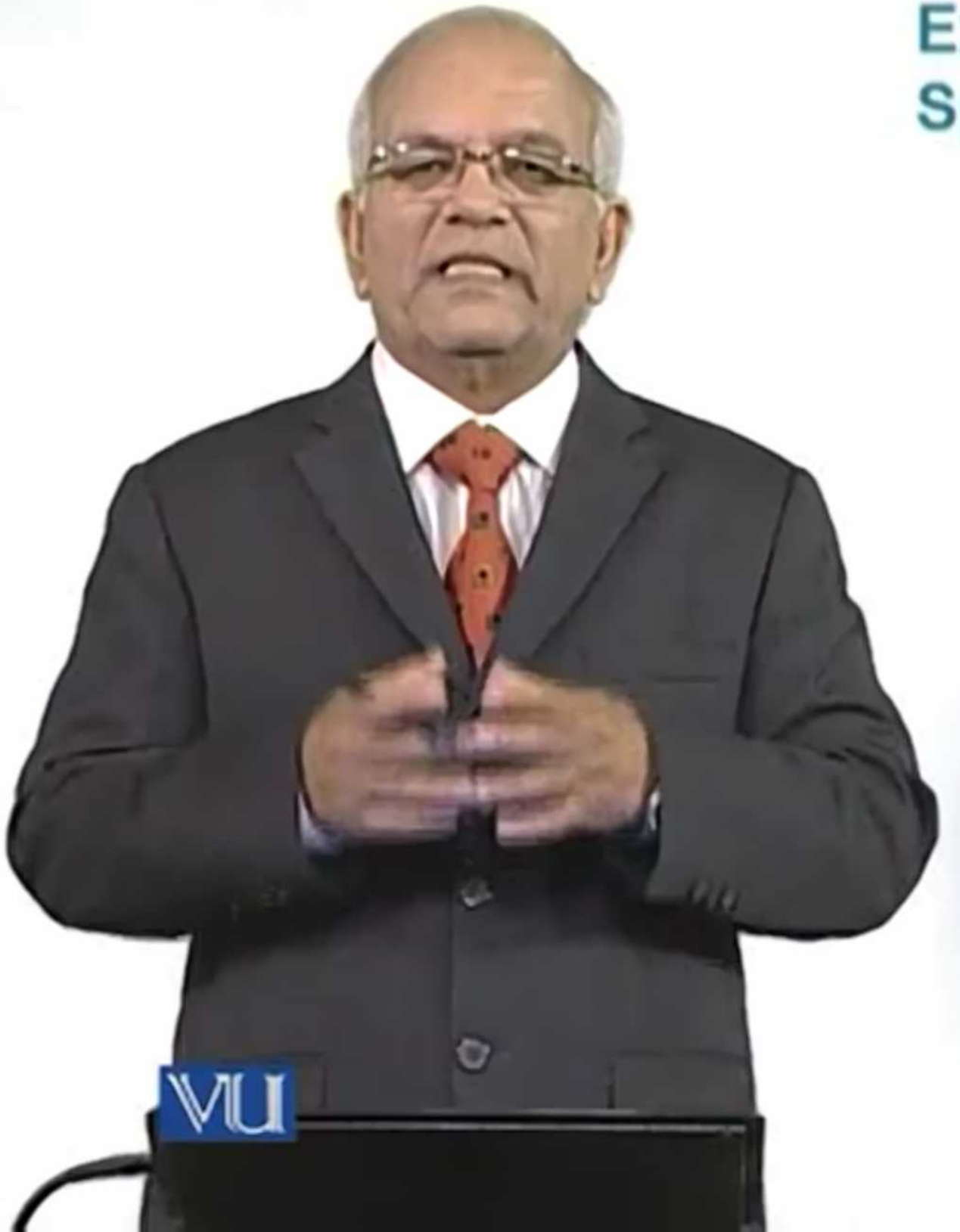

Example: Drag on a Sphere

$$\frac{F}{\rho V^2 D^2} = f\left(\frac{\rho V D}{\mu}\right)$$

- Only one dependent and one independent variable
- Easy to set up experiments to determine dependency
- Easy to present results (one graph)

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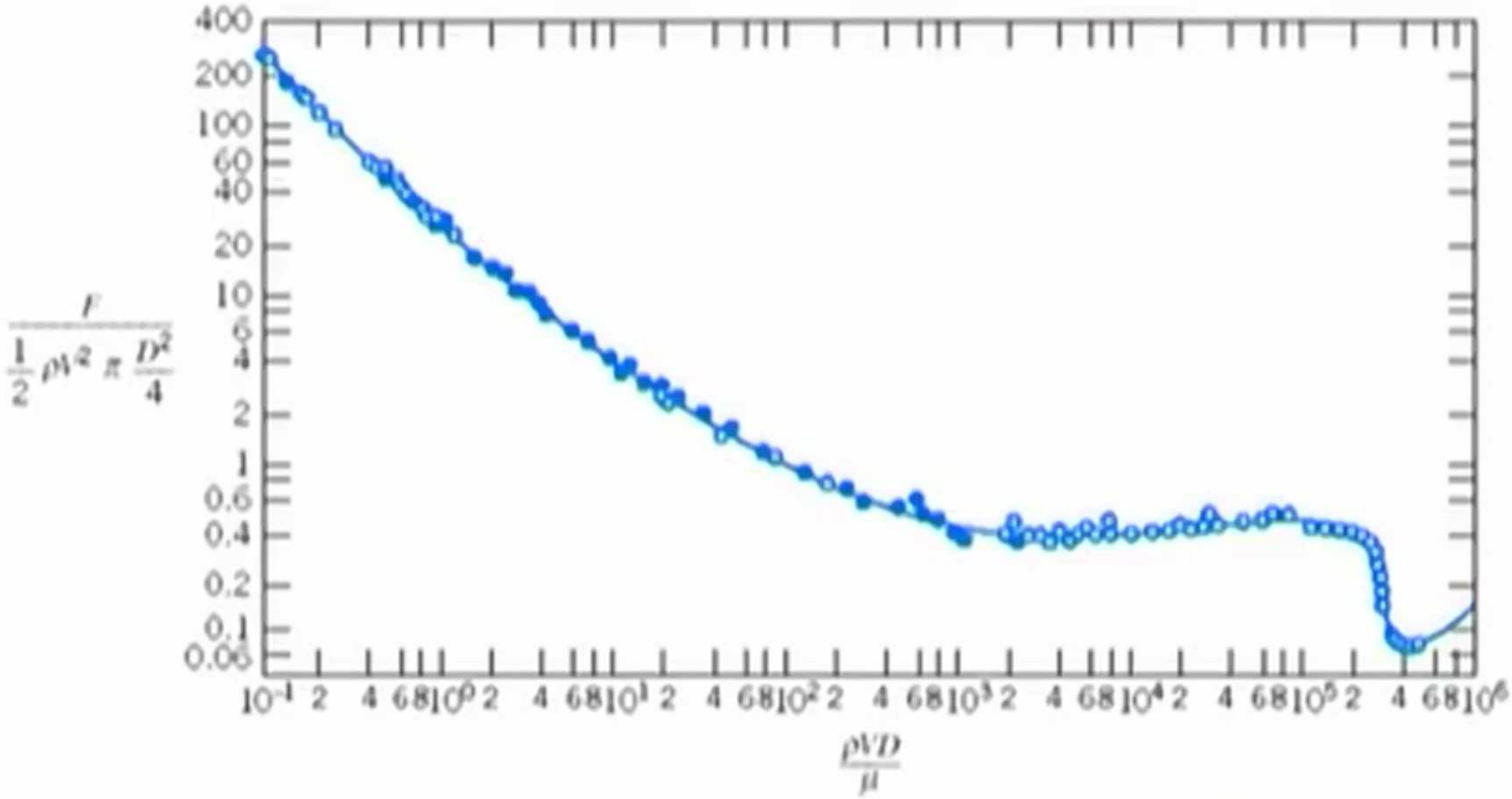
DIMENSIONAL ANALYSIS AND SIMILARITY

Nature of Dimensional Analysis



Fi Experimentally derived relation between the nondimensional parameters [4].